



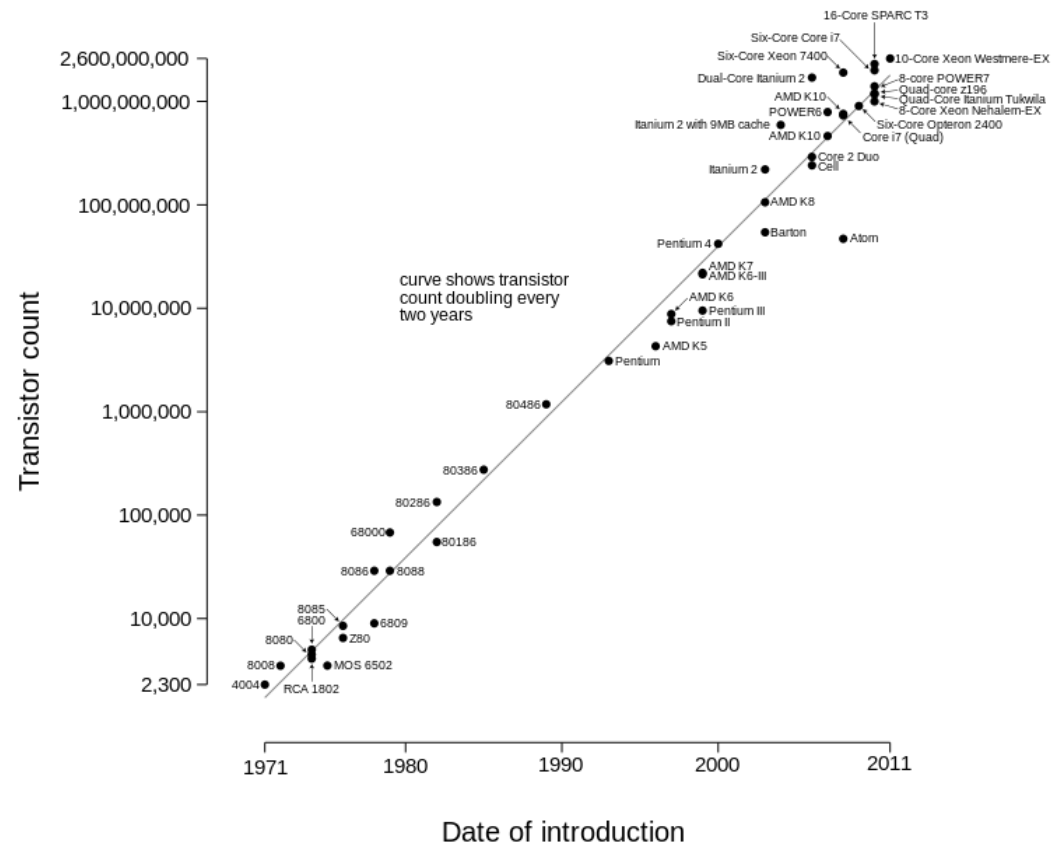
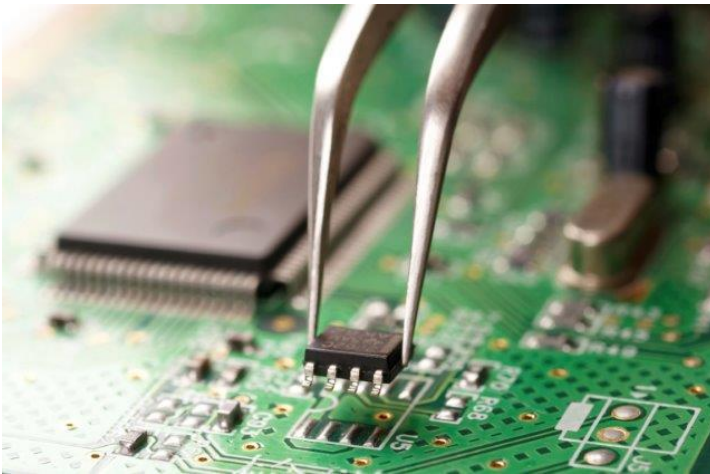
CHAPTER IV

Semiconducting Materials

Semiconductors-----the foundation of modern electronics.

Microprocessor Transistor Counts 1971-2011 & Moore's Law

Late 1940s, the first semiconductor device known as a transistor was discovered at Bell Labs.

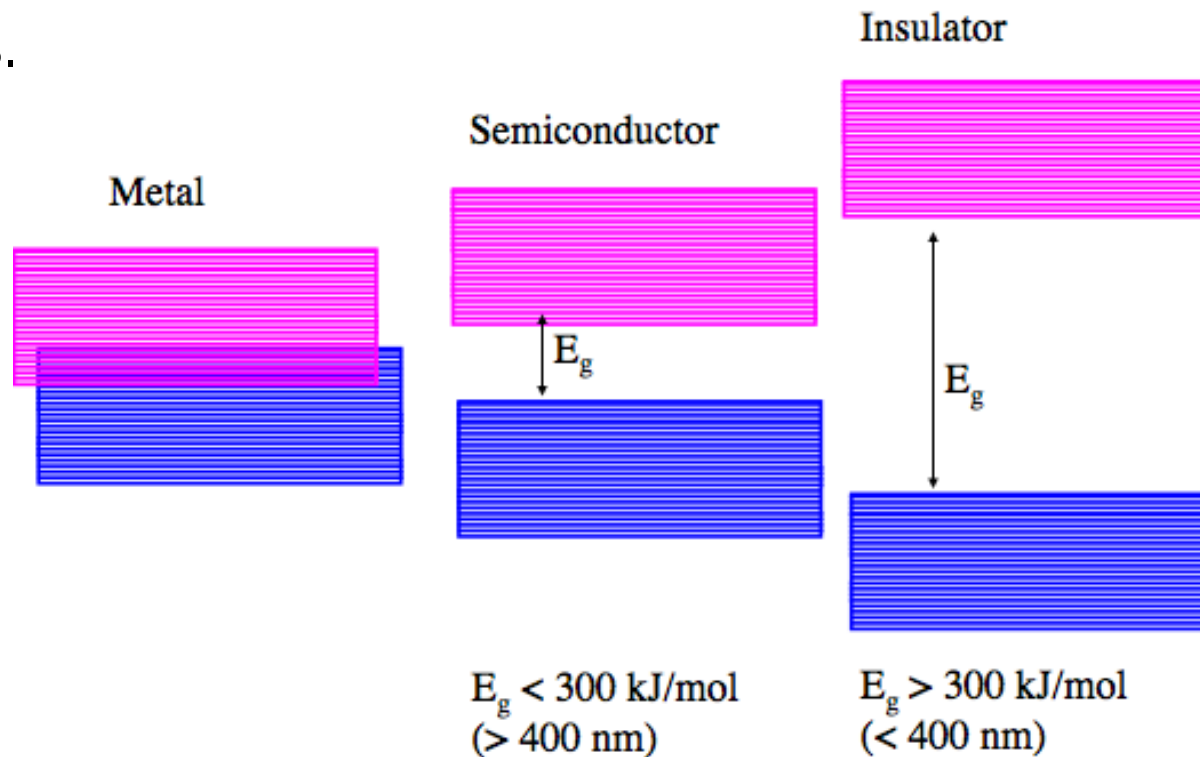


As of 2015, the highest transistor count in a commercially available CPU (in one chip) is over 5.5 billion transistors, in Intel's 18-core Xeon Haswell-EP



4.1 Properties and types of semiconductors

Semiconductors possess electrical conductivity intermediate between conductors such as metals and insulators such as ceramics.



The bandgap for semiconductors are typically in the range $150\text{--}290 \text{ kJ mol}^{-1}$ (*i.e.*, $1\text{--}4 \text{ eV}$).



Semiconducting Materials

The image displays a periodic table of elements, color-coded by category: Non-metals (yellow), Metals (green), and Metalloids (blue). A red circle highlights the elements Carbon (C), Nitrogen (N), and Oxygen (O), which are part of the 'greenhouse gases' mentioned in the text. To the right of the periodic table is a graph showing the energy spectrum of solar radiation. The y-axis is labeled 'energy [W/(m²·nm)]' and ranges from 0.4 to 1.6. The x-axis is labeled 'wavelength [nm]' and ranges from 0.4 to 1.6. The graph shows a yellow curve representing the solar spectrum, with a peak around 0.5 μm. The area under the curve is shaded yellow, and the area above the curve is shaded green.

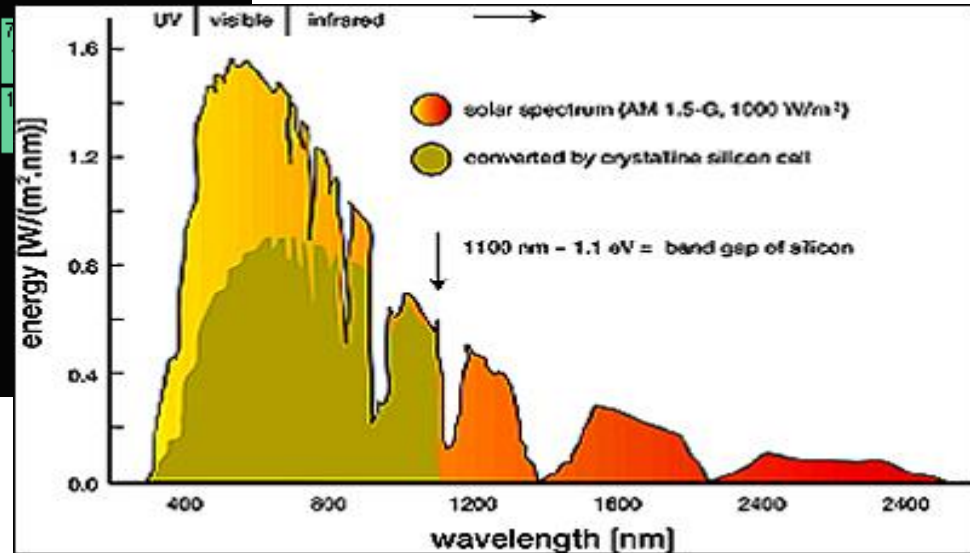
Periodic Table Legend:

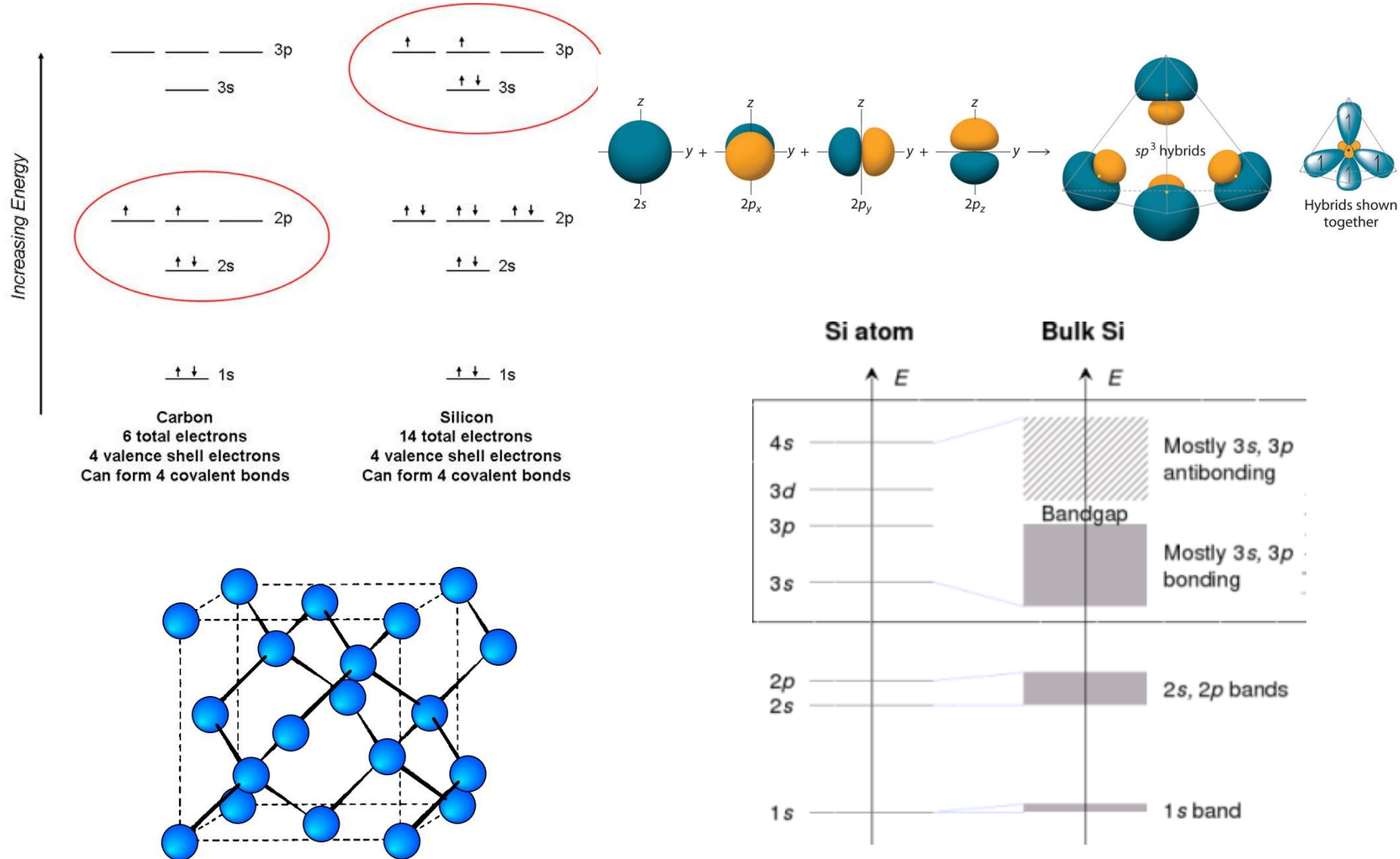
- Non-metals (Yellow)
- Metals (Green)
- Metalloids (Blue)

Highlighted Elements (Greenhouse Gases): Carbon (C), Nitrogen (N), Oxygen (O).

Graph Data (Approximate):

Wavelength [nm]	Energy [W/(m²·nm)]
0.4	0.4
0.5	1.6
0.6	1.2
0.7	0.8
0.8	0.6
0.9	0.5
1.0	0.4
1.6	0.4





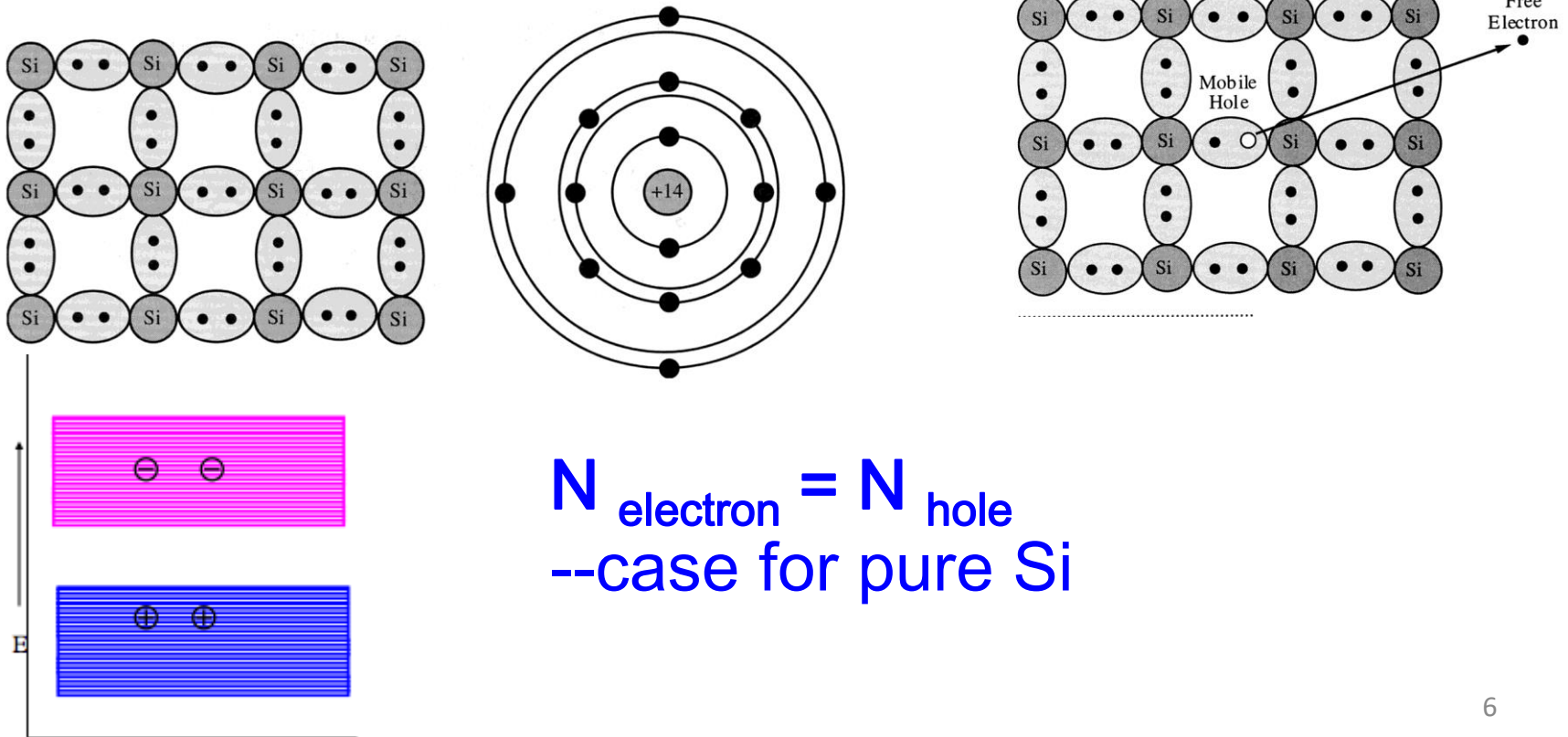
Single crystalline structure of silicon



Intrinsic and extrinsic semiconductors

• Intrinsic

Intrinsic semiconductors contain the same numbers of free bonding electrons, and “holes” created from the migration of electrons from the valence to conduction bands.





- **Fermi level**

The highest occupied state at absolute zero.

- **Fermi-distribution function $f(E)$**

The probability that a particle will have energy E

At absolute zero, fermions will fill up all available energy states below a level E_F called the Fermi energy with one (and only one) particle. They are constrained by the Pauli exclusion principle. At higher temperatures, some are elevated to levels above the Fermi level.

$$f(E) = \frac{1}{e^{(E - E_F)/kT} + 1}$$

Fermi-Dirac

See the Maxwell-Boltzmann distribution for a general discussion of the exponential term.

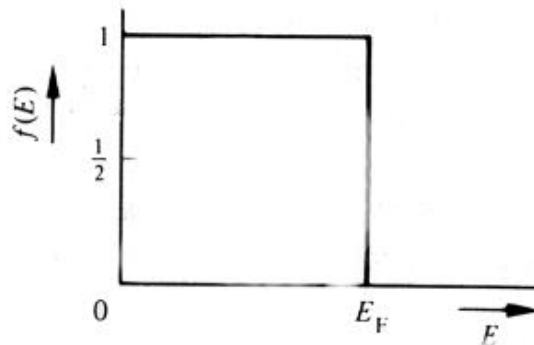
For low temperatures, those energy states below the Fermi energy E_F have a probability of essentially 1, and those above the Fermi energy essentially zero.

The quantum difference which arises from the fact that the particles are indistinguishable.

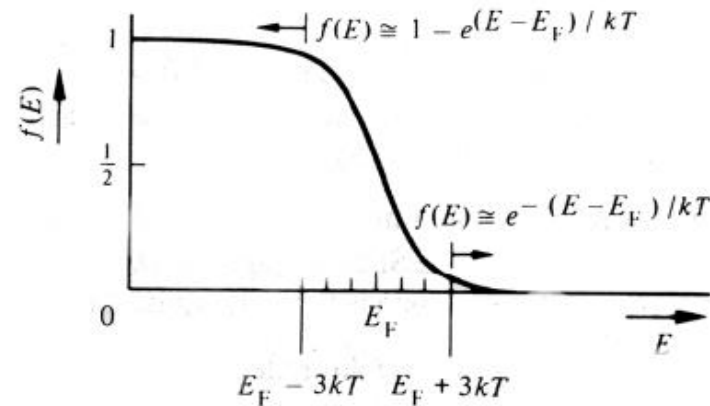


$f(E)$ is the probability that a state at energy E is occupied

$1-f(E)$ is the probability that a state at energy E is unoccupied

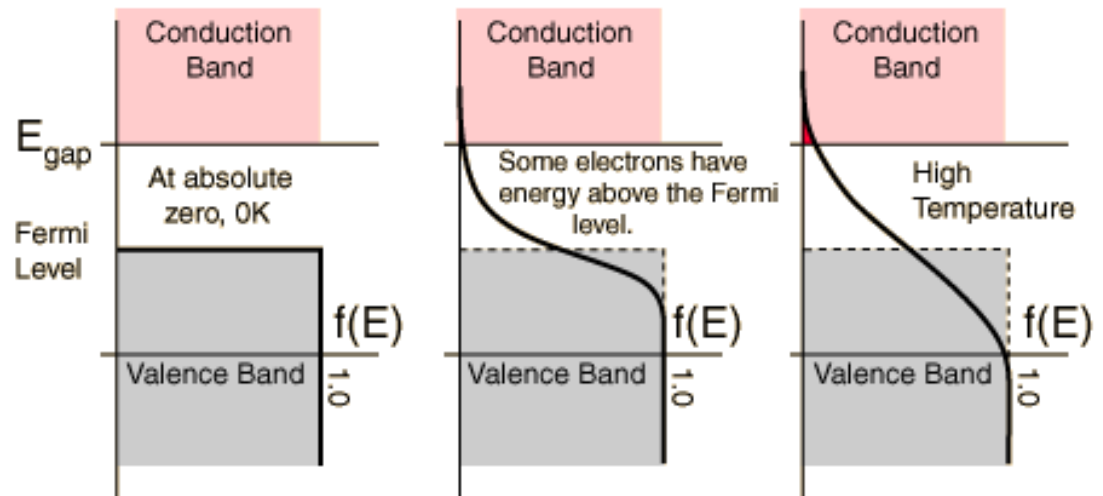
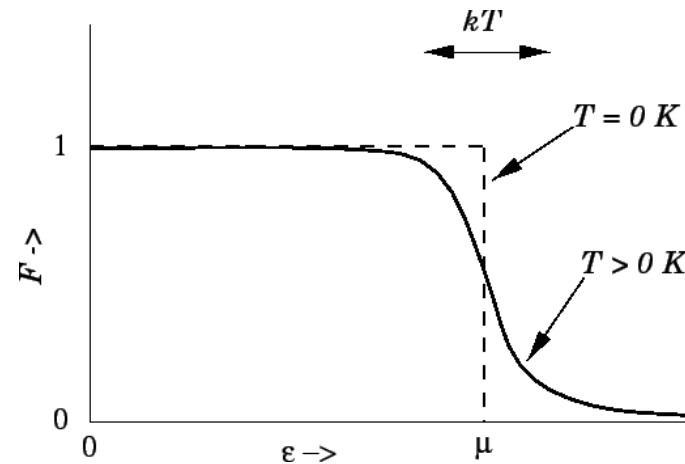
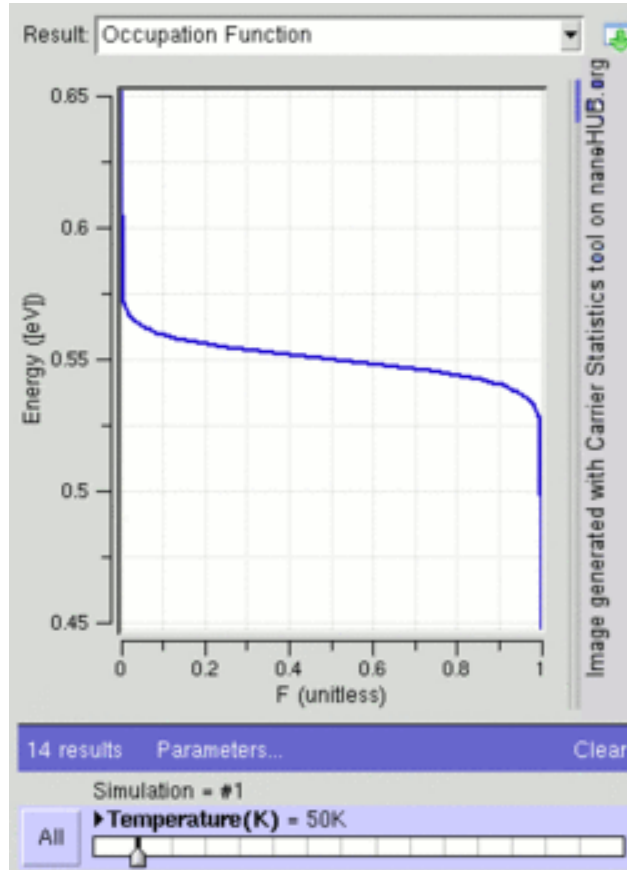


(a) $T \rightarrow 0 \text{ K}$



(b) $T > 0 \text{ K}$

- At $T=0\text{K}$, occupancy is “digital”: No occupation of states above E_F and complete occupation of states below E_F
- At $T>0\text{K}$, occupation probability is reduced with increasing energy.
- $f(E=E_F) = 1/2$ regardless of temperature.

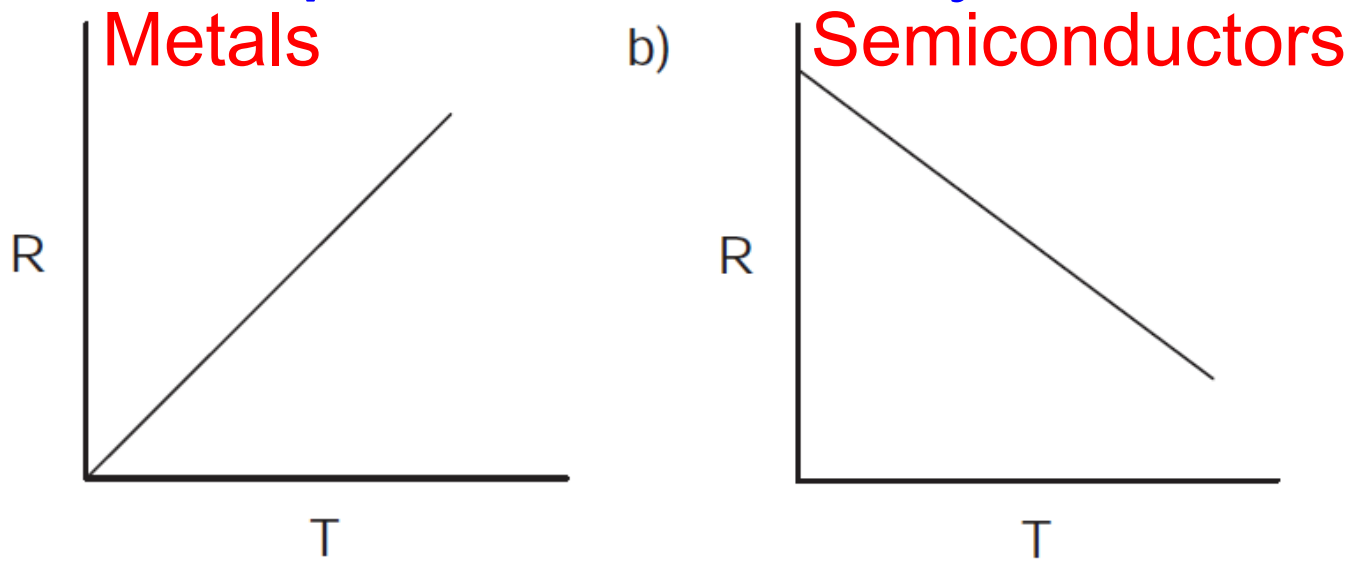


No electrons can be above the valence band at 0K, since none have energy above the Fermi level and there are no available energy states in the band gap.

At high temperatures, some electrons can reach the conduction band and contribute to electric current.



Temperature dependent conductivity



As the temperature is increased:

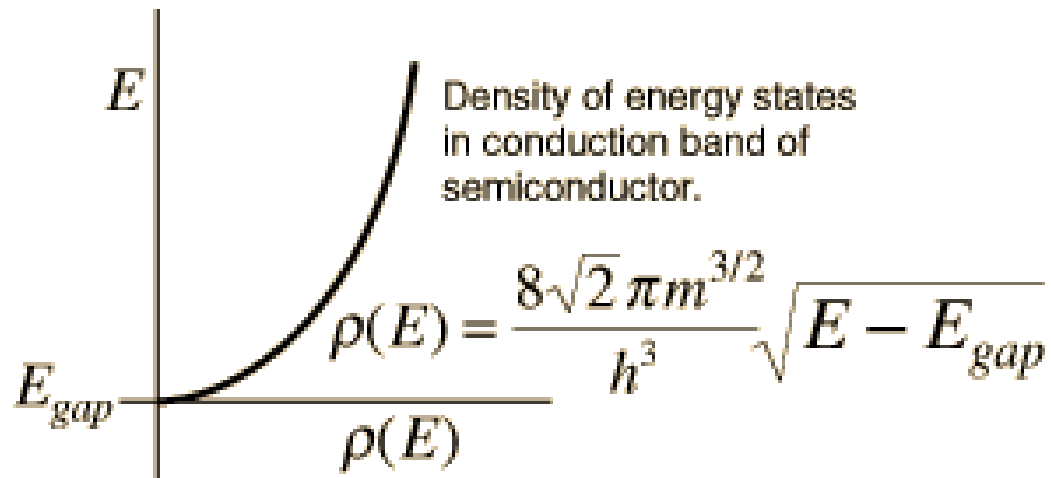
- Metals show a decreased conductivity.
- Semiconductors show an increased conductivity.

The thermal motion of metal atoms causes less efficient electron mobility through the lattice, whereas a temperature increase causes the bandgap to narrow for semiconductors, resulting in more effective electrical conductivity.

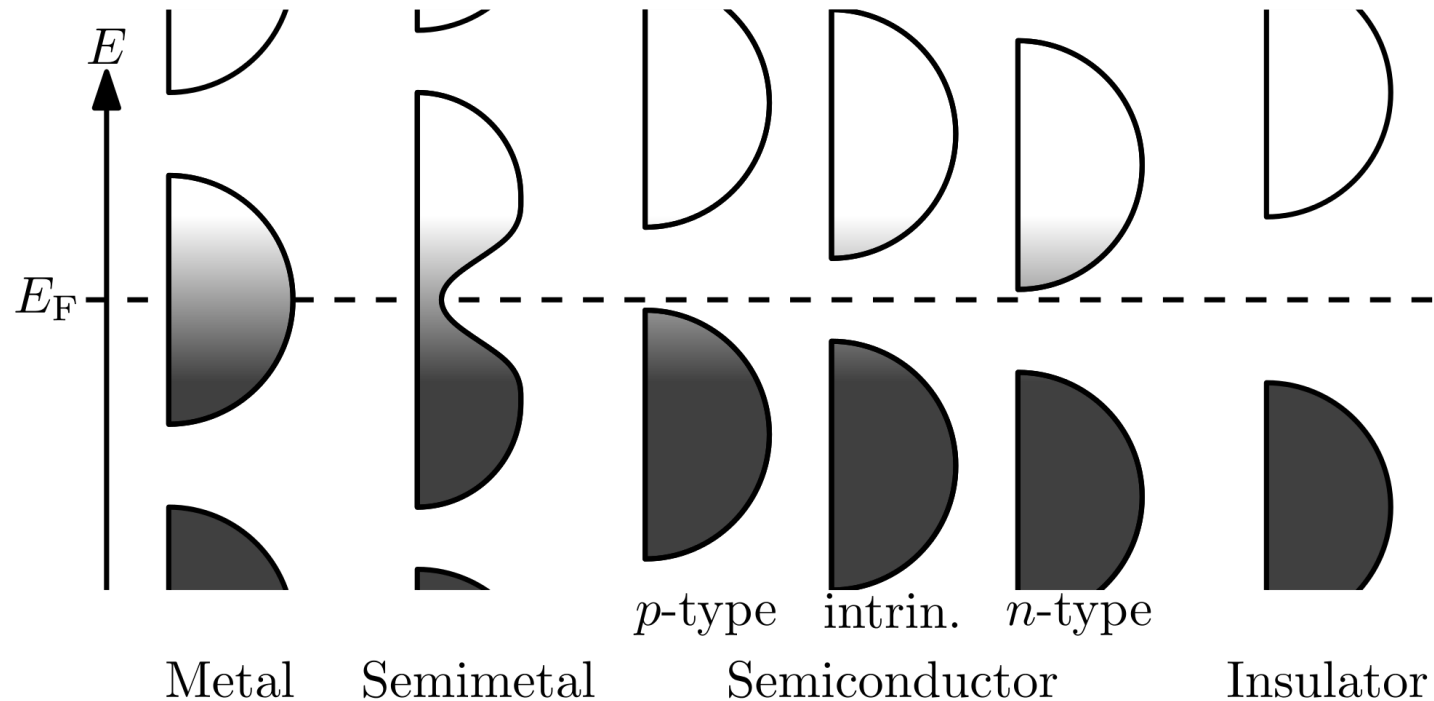


- The density of states (DOS)**

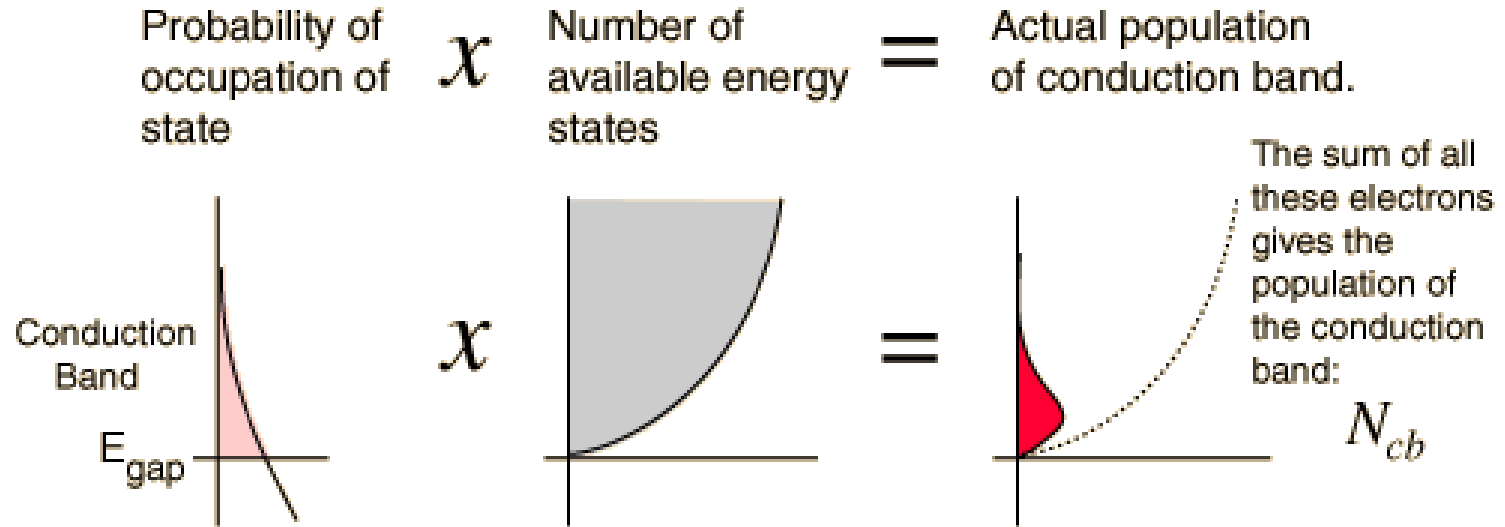
The density of states gives the number of allowed electron (or hole) states per volume at a given energy. It can be derived from basic quantum mechanics.



the number of available energy levels

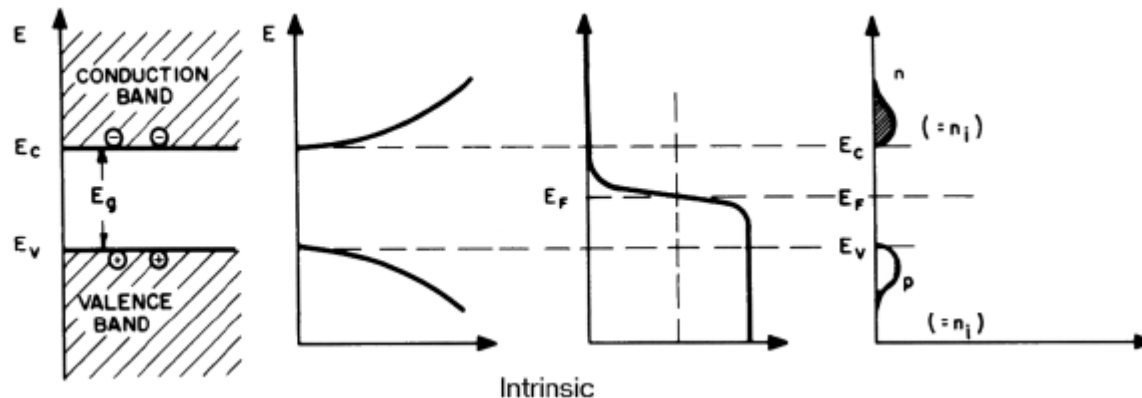


Here, height is energy while width is the density of available states for a certain energy in the material listed. The shade follows the Fermi–Dirac distribution (black = all states filled, white = no state filled). In metals and semimetals the Fermi level E_F lies inside at least one band. In insulators and semiconductors the Fermi level is inside a band gap; however, in semiconductors the bands are near enough to the Fermi level to be thermally populated with electrons or holes.



Intrinsic semiconductors

Intrinsic:
Equal number
of electrons
and holes

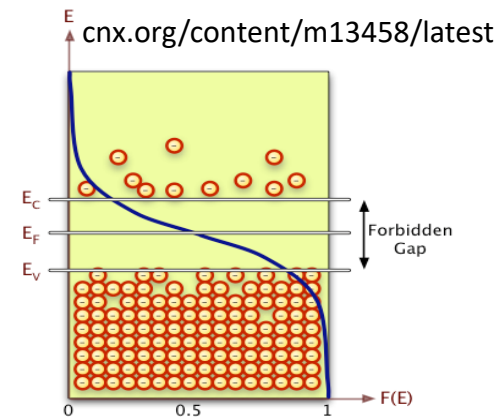
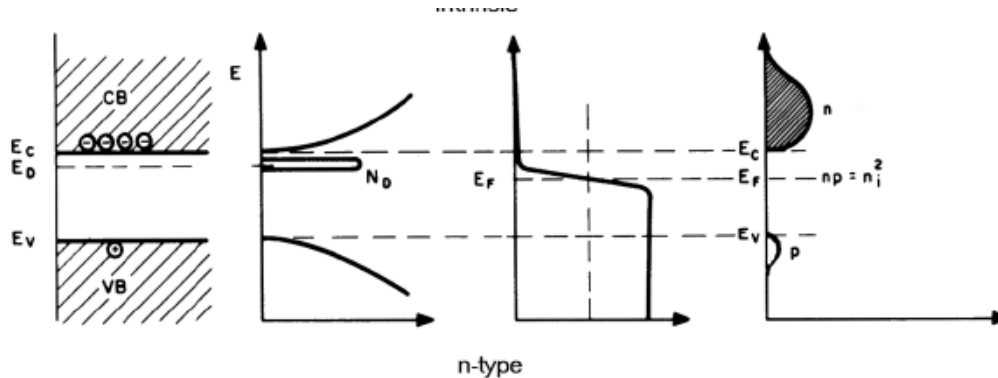




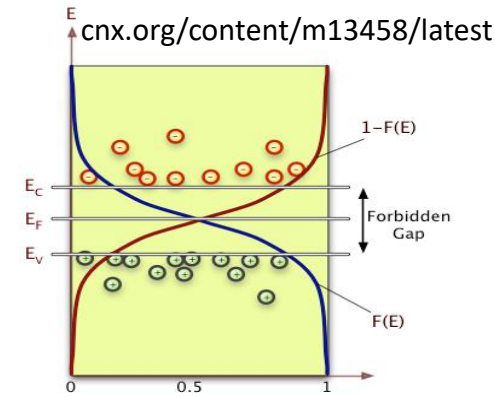
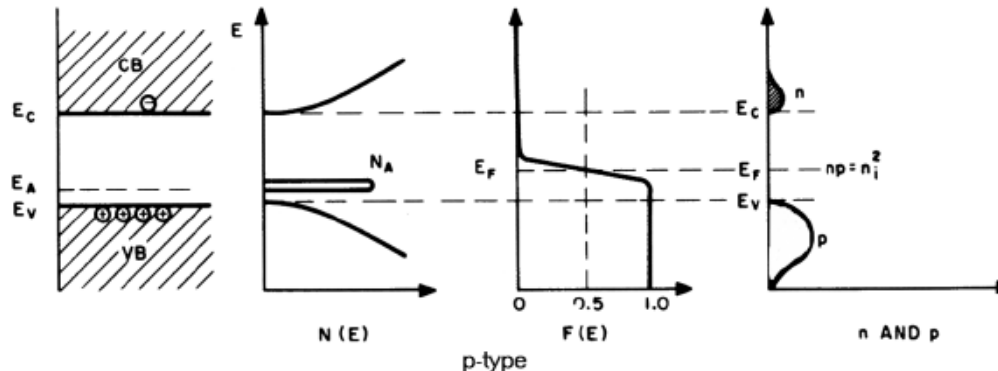
• Extrinsic

For **extrinsic semiconductors**, the Fermi level corresponds to a level slightly above or below conduction or valence bands, depending on the dopants introduced into the lattice – either an excess of electrons or holes

n-type: more electrons than holes

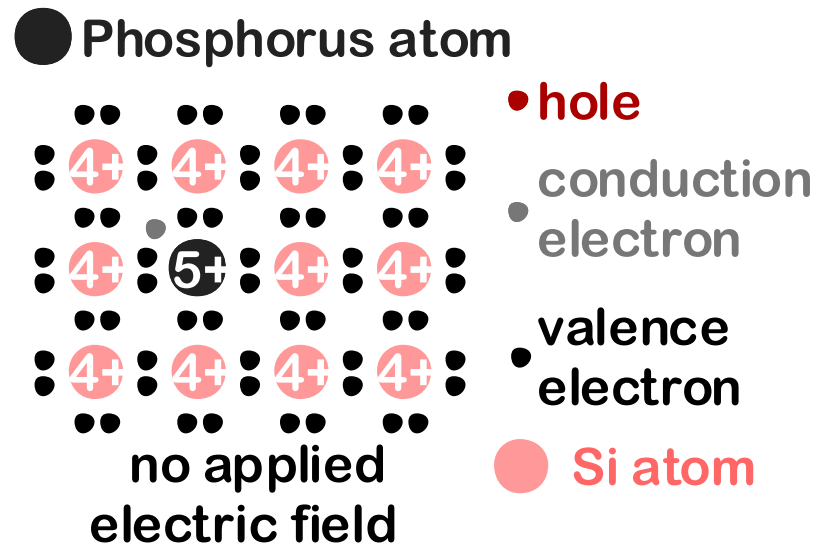


p-type: more holes than electrons





N-type: An N-type semiconductor material has an excess of electrons. In this way, free electrons are available within the lattices and their overall movement in one direction under the influence of a potential difference results in an electric current flow. This in an N-type semiconductor, the charge carriers are electrons.



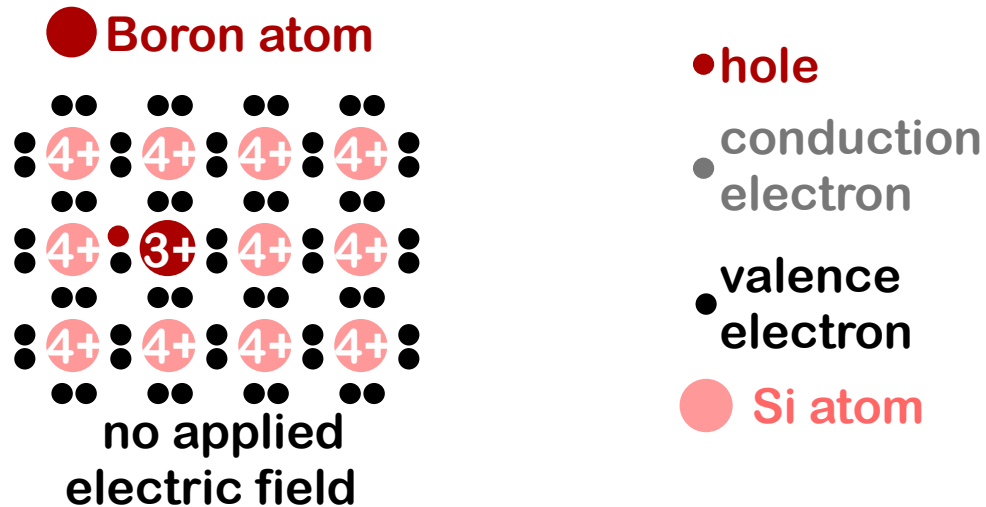
Dopants: Valence > 4, electron-rich dopants (e.g., N, P);



P-type: In a P-type semiconductor material there is a shortage of electrons, i.e. there are 'holes' in the crystal lattice. Electrons may move from one empty position to another and in this case it can be considered that the holes are moving. This can happen under the influence of a potential difference and the holes can be seen to flow in one direction resulting in an electric current flow. It is actually harder for holes to move than for free electrons to move and therefore the mobility of holes is less than that of free electrons. Holes are positively charged carriers.



P-type



Dopants:

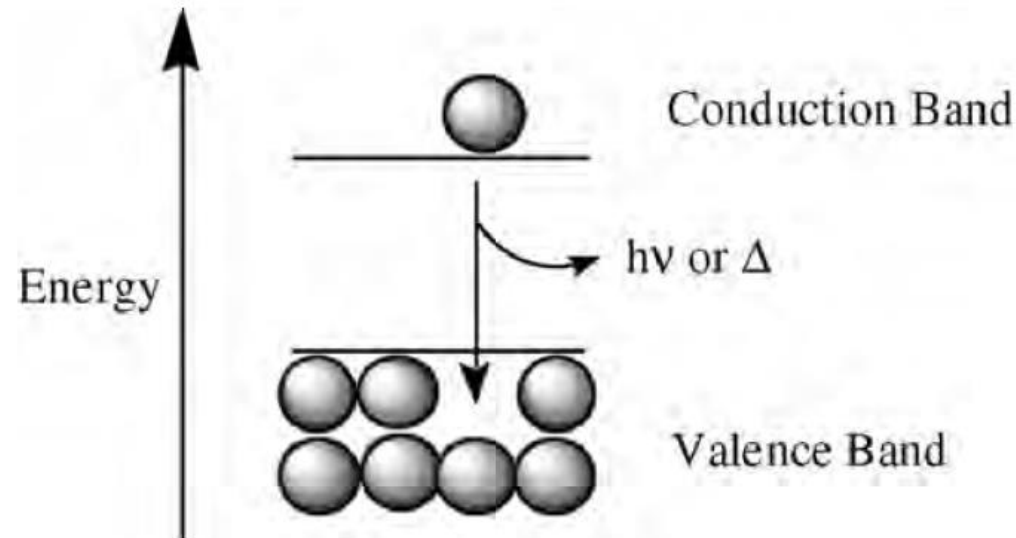
Valence < 4, electron-deficient dopants (e.g., B, Al);



Charge carriers: “electron” and “hole”

When an electric field is applied to a metal, negatively charged electrons are accelerated and carry the resulting current. In a semiconductor the charge is not carried exclusively by electrons. Positively charged **holes** also carry charge. These may be viewed either as vacancies in the otherwise filled valence band, or equivalently as positively charged particles.

Recombination



Since an electron that has been promoted to the conduction band will have a greater energy than those left in the valence band, there is a possibility for the electron to lose this excess energy. The spontaneous return of electrons in the conduction band to the valence band is known as **recombination**, and is usually accompanied by light emission and heat.



Wave-particle duality

- De Broglie Wave Equation

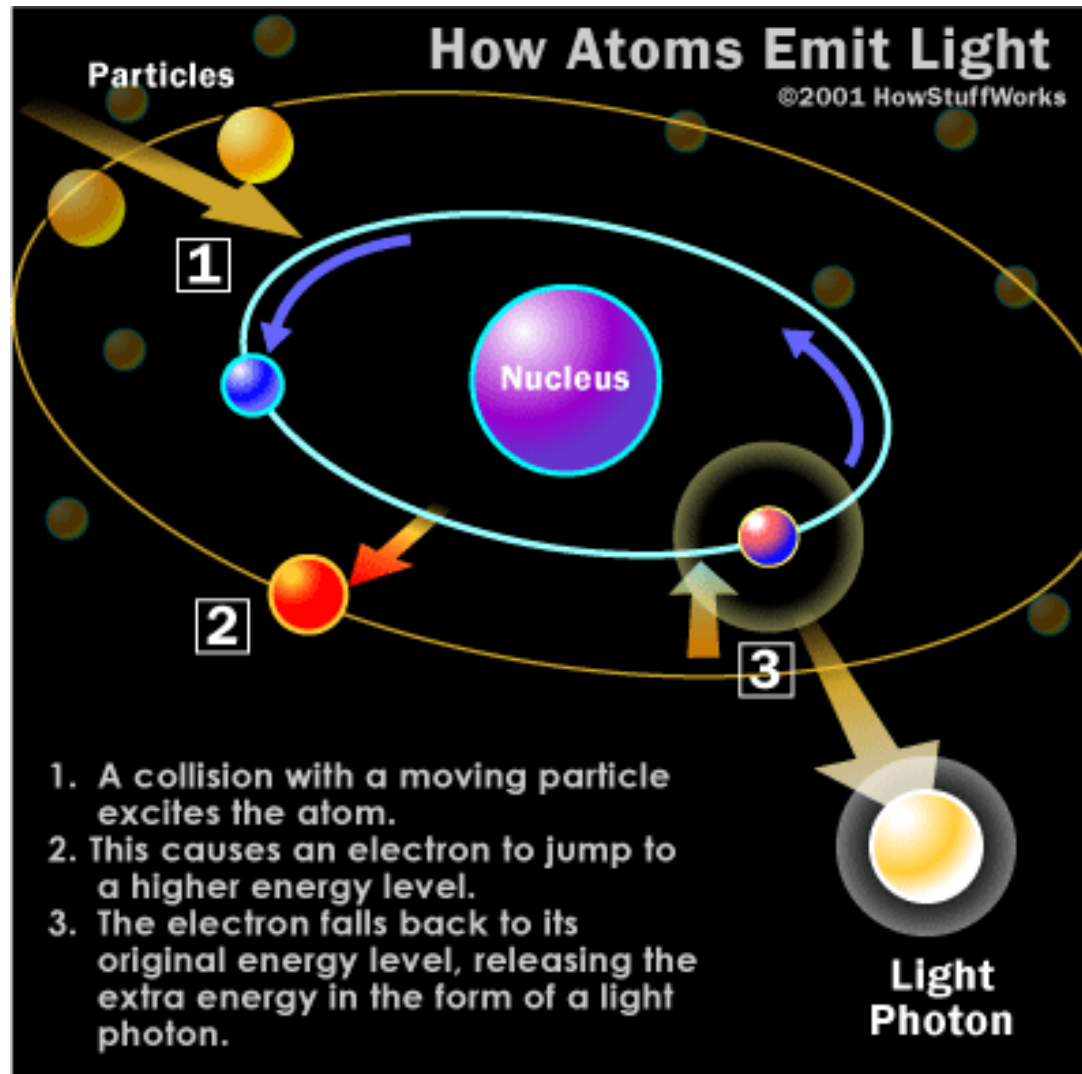
$$\lambda = \frac{h}{p}$$

λ : wavelength

h : Planck's constant

p : momentum

Every particle has wave nature as well, but it is only truly evident when a particle is very light, such as an electron ($m = 9.11 \times 10^{-28} g$).





Photon vs Electron		
More Information Online WWW.DIFFERENCEBETWEEN.COM		
	Photon	Electron
DEFINITION	Photon is a type of elementary particle that acts as a carrier of energy.	Electron is a subatomic particle that present in all the atoms.
CHARGE	No charge	Negative charge; $-1.602 \times 10^{-19} \text{ C}$
REST MASS	No rest mass	Very small mass; $9.11 \times 10^{-31} \text{ kilograms}$
SPEED	Can go at the speed of light	Practically cannot obtain the speed of light
PROPERTIES	Displays more wave properties.	Displays more particle properties.

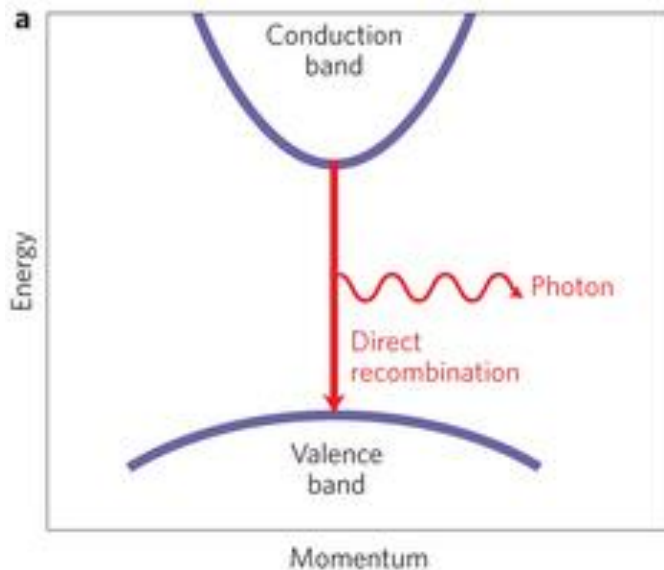


Bloch wavefunctions

- Each wavefunction represents the energy of an electron at a specific location in the lattice, referred to as k-space.
- An **E-k diagram** may be constructed, with the potential energy of the electron on the y-axis, and the wavevector, k , on the x-axis.
- The wavevector represents lattice directions of the semiconductor crystal; changing values of k represents a change in momentum of the electron.

Direct and Indirect bandgaps

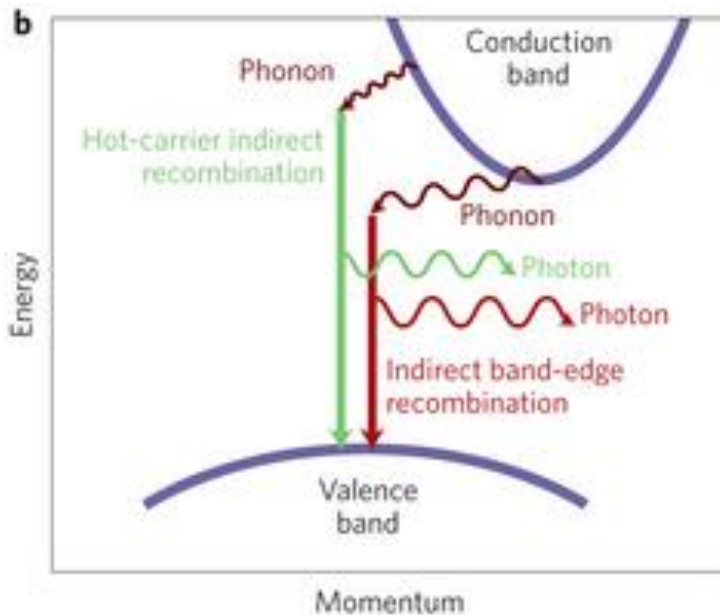
Direct-band gap semiconductors (e.g. GaAs, InP, AlGaAs)



- For a **direct-band gap material**, the minimum of the **conduction band (CB)** and maximum of the **valance band (VB)** lies at the same momentum, k , values.
- When an electron sitting at the bottom of the **CB** recombines with a hole sitting at the top of the **VB**, there will be no change in momentum values.
- Energy is conserved by means of emitting a photon, such transitions are called as **radiative transitions**.

Direct and Indirect bandgaps

Indirect-band gap semiconductors (e.g. Si and Ge)



- For an indirect-band gap material; the minimum of the CB and maximum of the VB lie at different k-values.
- When an e^- and hole recombine in an indirect-band gap s/c, phonons must be involved to conserve momentum.



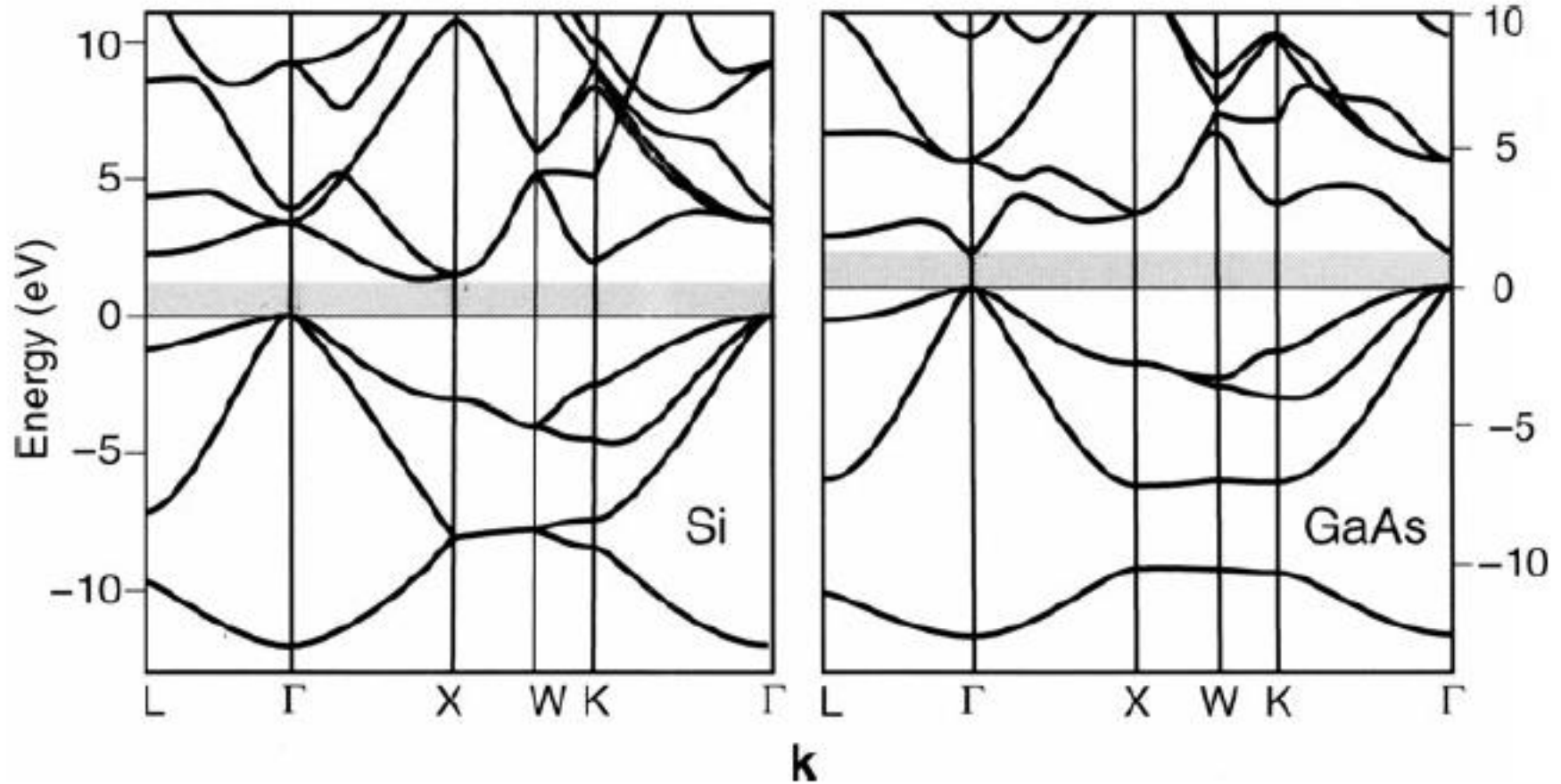
Phonons

- Atoms vibrate about their mean position at a finite temperature. These vibrations produce vibrational waves inside the crystal.
- Phonons are the quanta of these vibrational waves. Phonons travel with a velocity of sound.
- Their wavelength is determined by the crystal lattice constant. Phonons can only exist inside the crystal.



Direct bandgap vs Indirect bandgaps

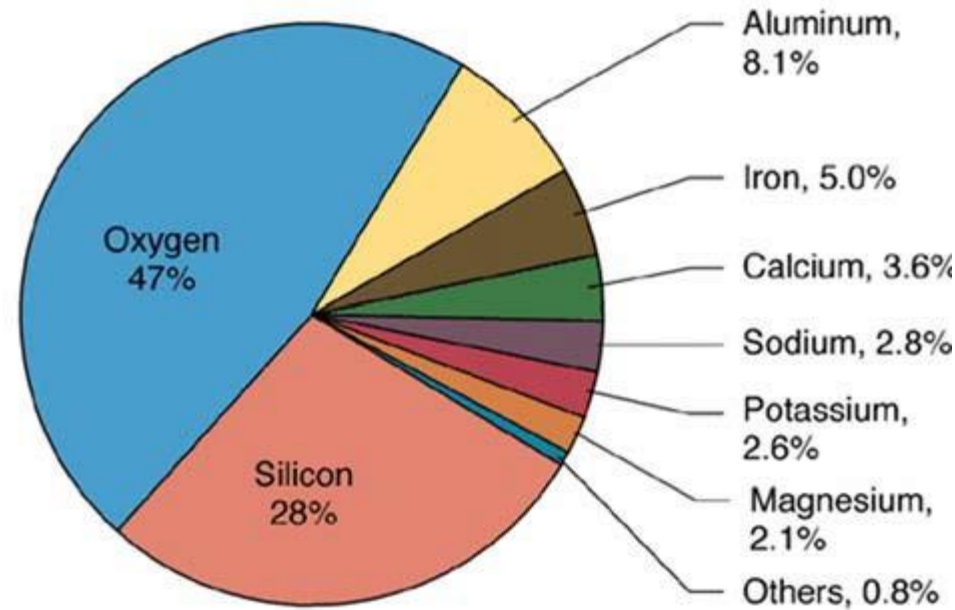
- The indirect process proceeds at a much slower rate, as it requires three entities to intersect in order to proceed: an electron, a photon and a phonon.
- The recombination process is much more efficient for a direct band gap semiconductor than for an indirect band gap semiconductor, where the process must be mediated by a phonon.
- As a result of such considerations, gallium arsenide and other direct band gap semiconductors are used to make optical devices such as LEDs and semiconductor lasers, whereas silicon, which is an indirect band gap semiconductor, is not.







4.2 Silicon-based applications



Silicon is the second most abundant element in the earth's crust, next to oxygen.



Where can you find silicon?



Sand



Quartz

Silicon oxide



Amethyst



Flint



Opal



Where can you find silicon?

Silicate



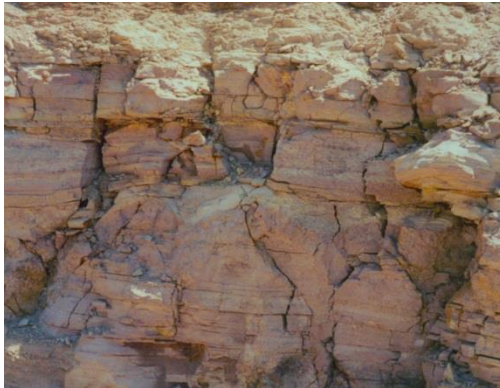
granite



asbestos



feldspar



Clay



Mica



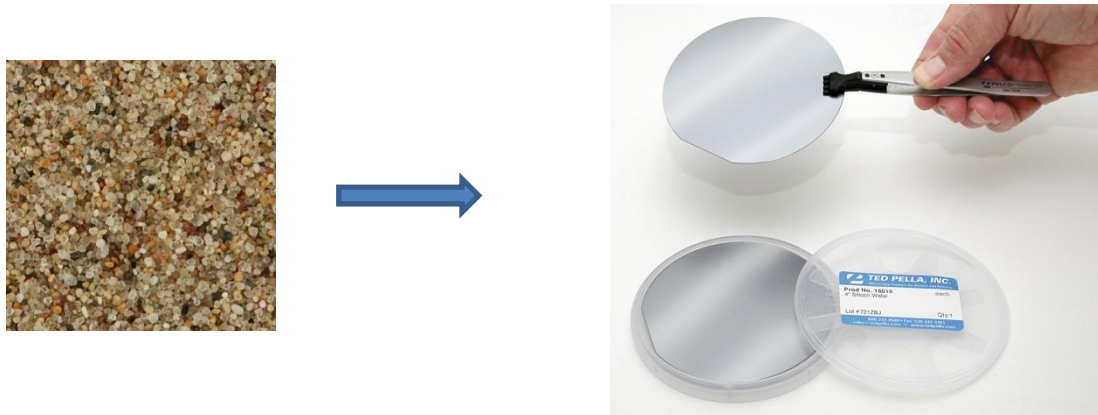
Silicon-based electronic devices





4.2.1 Silicon Wafer Production

All these devices are based on silicon wafer.



Silicon wafer is fabricated from sea sand.

metallurgical-grade silicon (MG-Si) >98%.

Finally, electronic-grade silicon (EG-Si) 99.9999999999%.



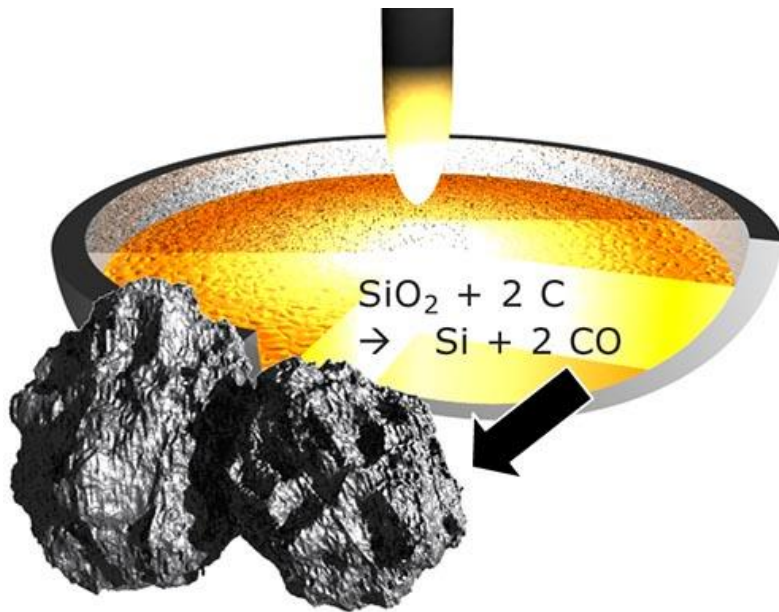
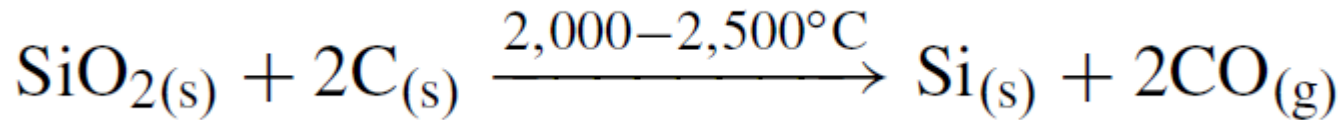
What do you mean a 99.999999999% purity?

only one out of every billion atoms in the solid is something other than silicon!

To put this into perspective, imagine stacking yellow tennis balls from the earth's surface to the moon; replacing only one of these with a blue ball would represent the level of impurities in EG-Si.

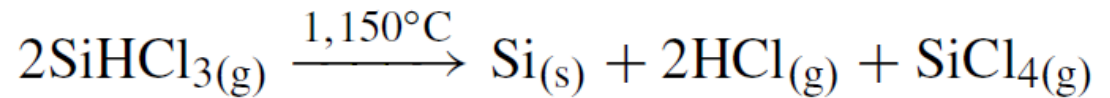
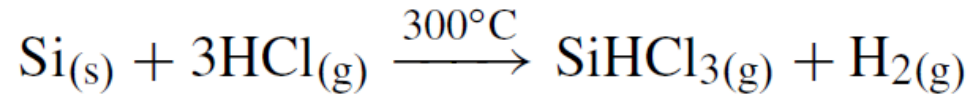
Every year, between 15,000 and 20,000 tones of EG-Si is manufactured throughout the world for an ever-increasing number of applications. **How?**

- MG-Si production



- Quartz sand is reduced with carbon (charcoal and coal) in an electric arc furnace at high temperatures (2000-2500°C) to metallurgical grade silicon (> 98 % pure).
- A slight excess of SiO₂ prevents silicon carbide (SiC) from forming.

- EG-Si production

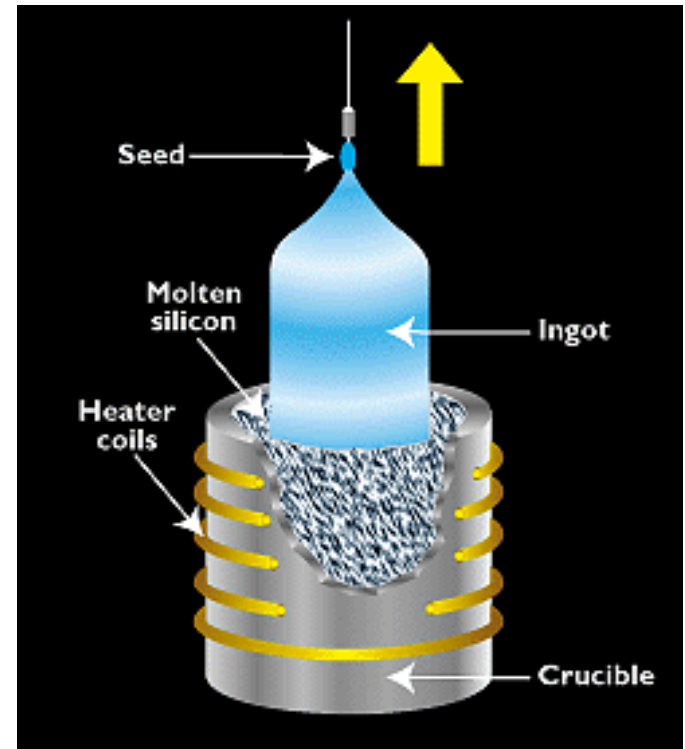


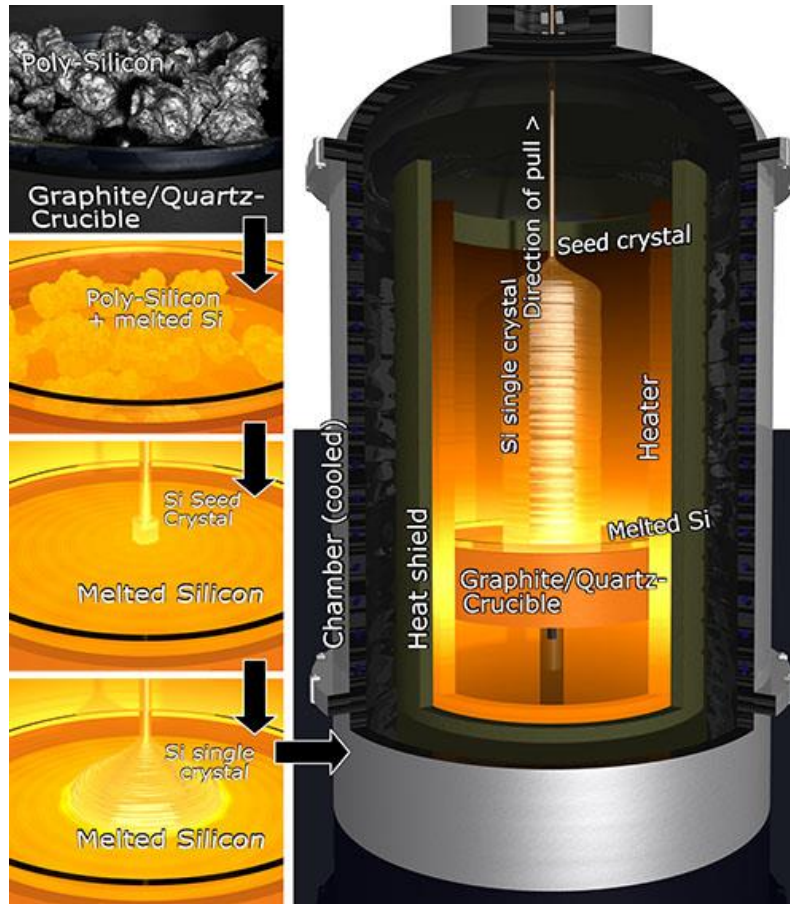
- Reaction of MG-Si with hydrogen chloride gas at a moderate temperature converts the silicon to trichlorosilane gas.
- When SiHCl_3 is heated to a temperature of *ca.* $1,150^{\circ}\text{C}$, it decomposes into high-purity silicon and gaseous by-products.

- Conversion polycrystalline EG-Si to a single crystal

Czochralski (CZ) method

- ✓ Silicon and any desired dopants are molten together in a crucible at a temperature above the melting point of silicon, $1,414^{\circ}\text{C}$.
- ✓ A rod fastened with a single crystal of silicon is positioned on the surface of the melt, and is pulled upward while rotating.





- In order to ensure high purity of the ingot, this process is normally performed in vacuo (ca. 10^{-6} Torr), or under an inert atmosphere (e.g., 99.999% Ar) using an inert chamber such as quartz.
- Even under these reaction conditions, the CZ method technique suffers from O and C impurities that arise predominantly from the crucible walls.
- It should be noted that small quantities of O impurities are actually desirable, as they may trap unwanted transition metal impurities, a process referred to as **gettering**.
- The CZ technique is especially useful to yield doped semiconductors.



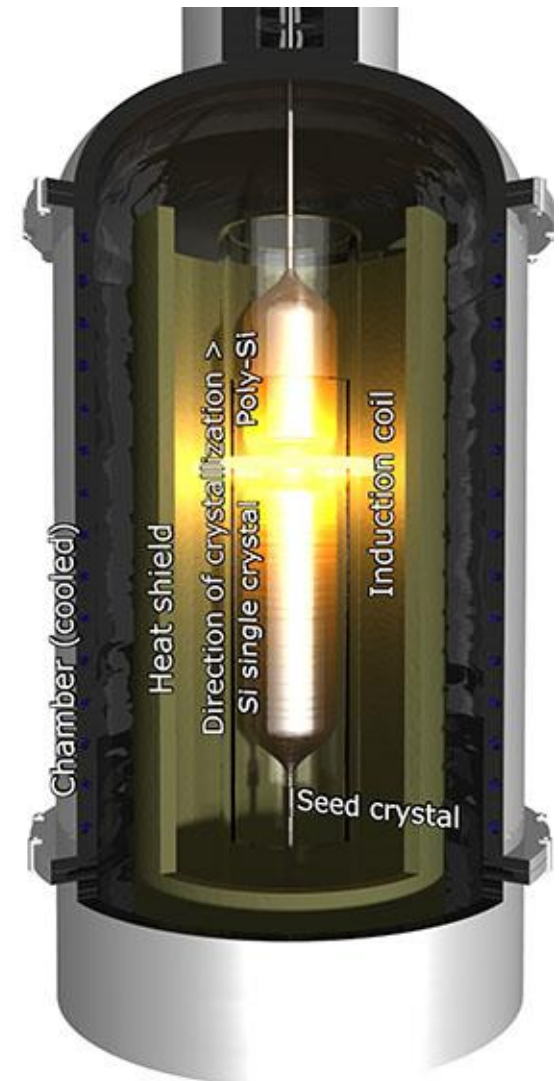
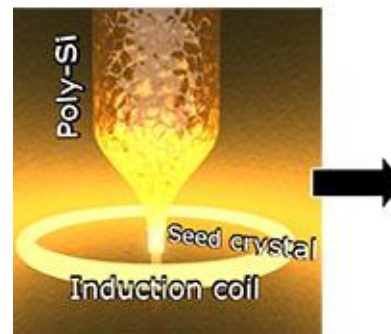
- The Silicon Cylinder is known as an Ingot
- Typical Ingot is About 1 or 2 meters in Length
- Can be Sliced into Hundreds of Smaller Circular Pieces called Wafers
- Each Wafer Yields Hundreds or Thousands of Integrated Circuits



- Conversion polycrystalline EG-Si to a single crystal

Float-zone (FZ) method

- The float-zone technique uses inductive heating to convert poly-Si into single crystal ingots of EG-Si.
- As a heating coil is slowly passed from one end to the other, the impurities become concentrated in the moving molten zone, becoming concentrated at the finishing end of the cylinder





FZ Technique: Advantages and Disadvantages

GOOD

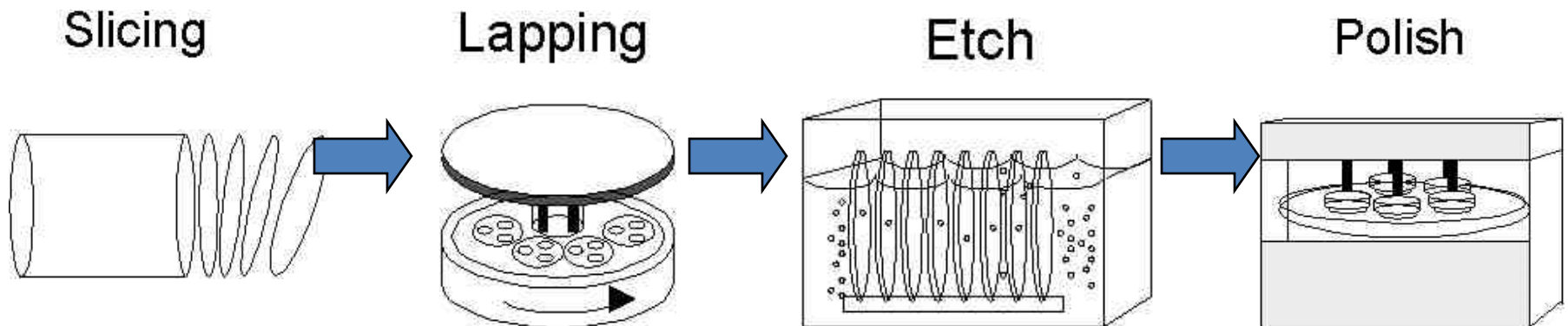
The main advantage of the float-zone technique is the very low impurity concentration in the silicon crystal. In particular the oxygen and carbon concentration are much lower as compared to CZ silicon, since the melt does not come into contact with a quartz crucible, and no hot graphite container is used.

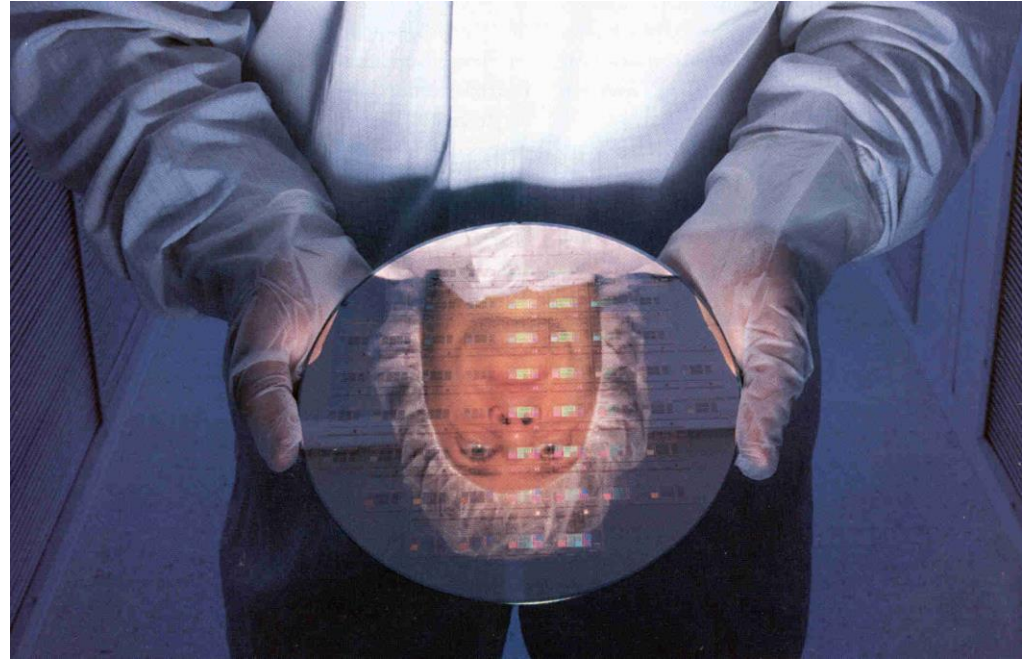
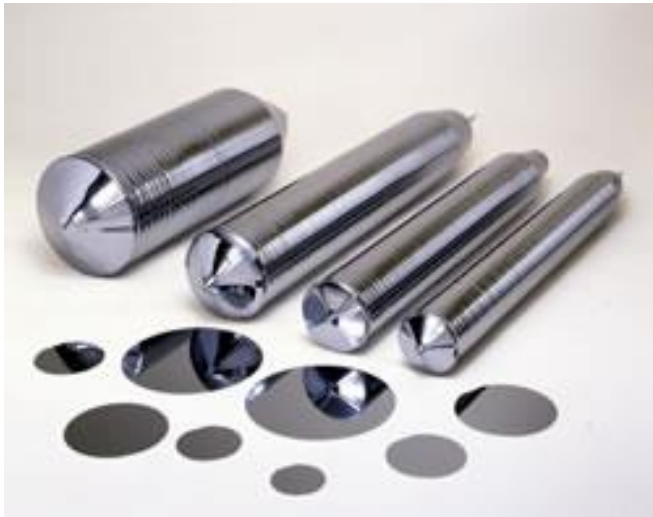
BAD

However, FZ silicon is more expensive than CZ silicon, and the crystal diameter is limited to eight inches (state of the art).



- The silicon crystal is **sliced** by using a diamond-tipped saw into thin wafers.
- Sorted by thickness.
- Damaged wafers removed during **lapping**.
- **Etch** wafers in chemical to remove any remaining crystal damage.
- **Polishing** smoothes uneven surface left by sawing process.

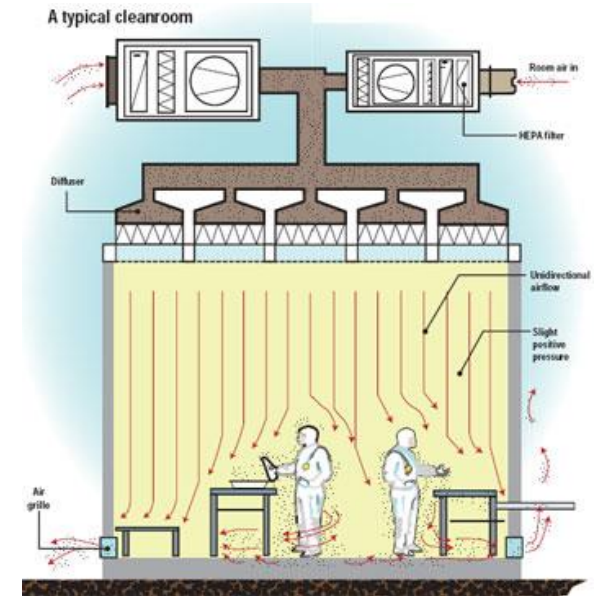




A finished wafer



Clean room



To ensure that the wafers remain free from dust and other airborne contaminants, the finished wafers are only handled within a **clean room**.

With stainless steel, perforated floors and ceiling tiles to promote air circulation, and sloped surfaces to avoid dust accumulation.



Yellow lighting is used to prevent light-sensitive material from being exposed to UV light, until desired as a part of chip fabrication.

Prior to entering the clean room, personnel must cover their clothing with a white **“bunny suit”**.

A polymeric nonwoven fabric that is nonlinting, antistatic.





Air Shower

Entry to the clean room, the worker must also walk over a sticky pad and pass through an **air shower** to remove dust particles from shoes and clothing.

- The ratings of clean rooms range from **Class 1 to Class 10,000** – an indication of the number of particles per cubic foot.
- As a familiar reference, in uncontrolled environments such as a typical home or office, the particle count is **5 million per cubic foot!**

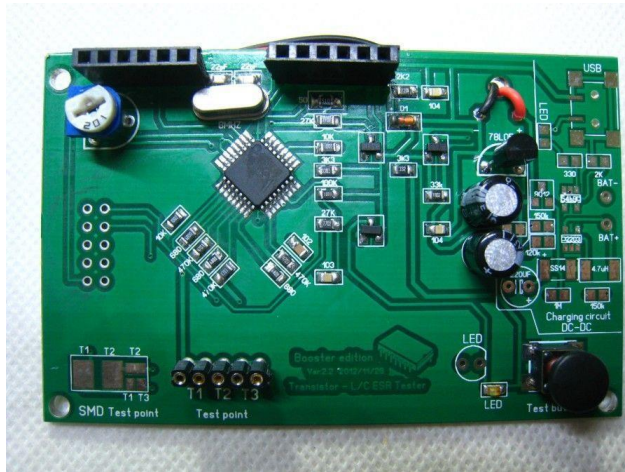




4.2.2 Integrated Circuits

An integrated circuit (IC), or chip, represents the brains behind all electronic devices.

An IC is a compilation of billions of complex subunits:



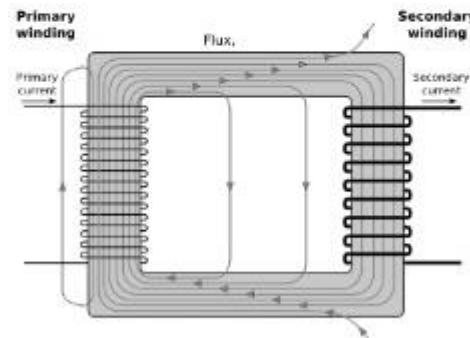
Resistor



Capacitor



Inductor



Transformer



Diode



Transistor



Though ICs have only been used since the early 1970s, consider some of the ways our lives have been changed:

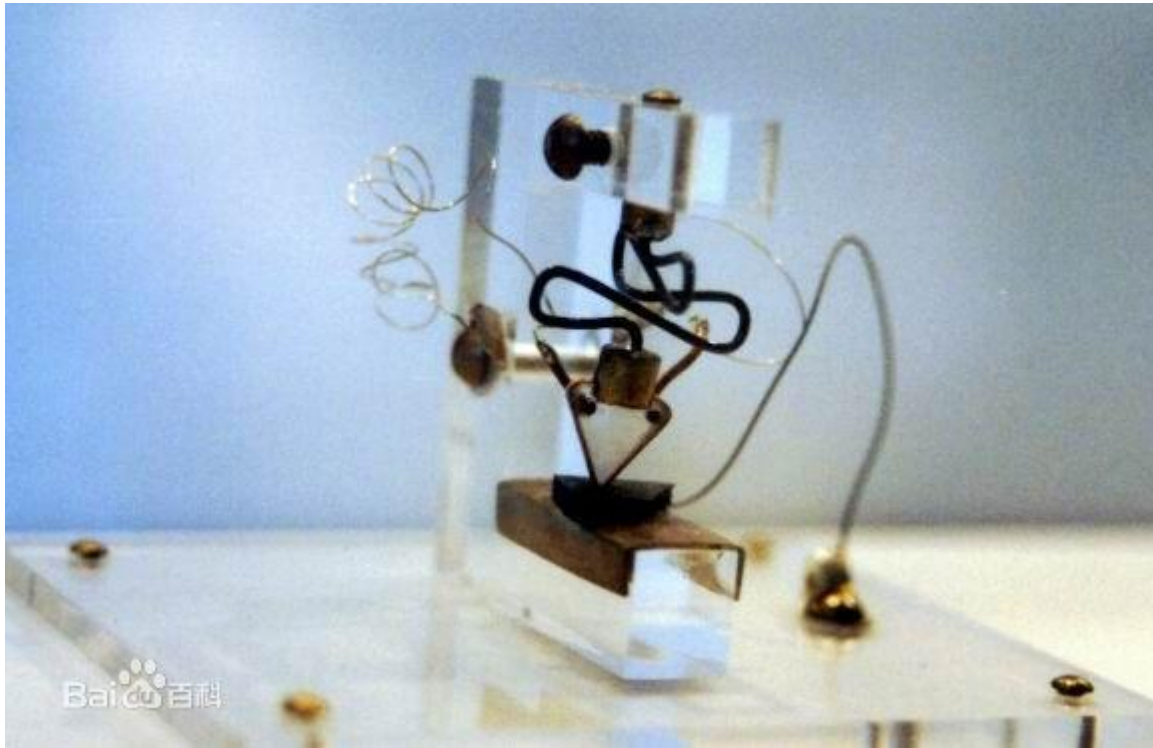
- Lightweight laptop computers are easily taken with us while traveling;
- Cars monitor their emissions and adjust engine conditions for maximum gasoline efficiency;
- Doctors are able to operate on patients from remote locations;
- Answers to virtually all questions we may pose are available within minutes via the Internet;
- Televisions may be mounted on walls, with a clarity that rivals viewing through a window;
- Phones may be used from virtually anywhere on Earth to keep in touch or check email;
- GPS systems tell us how to best arrive at our desired destination. . .

. . . What a world we live in

– what will the next 50 years bring? . . .



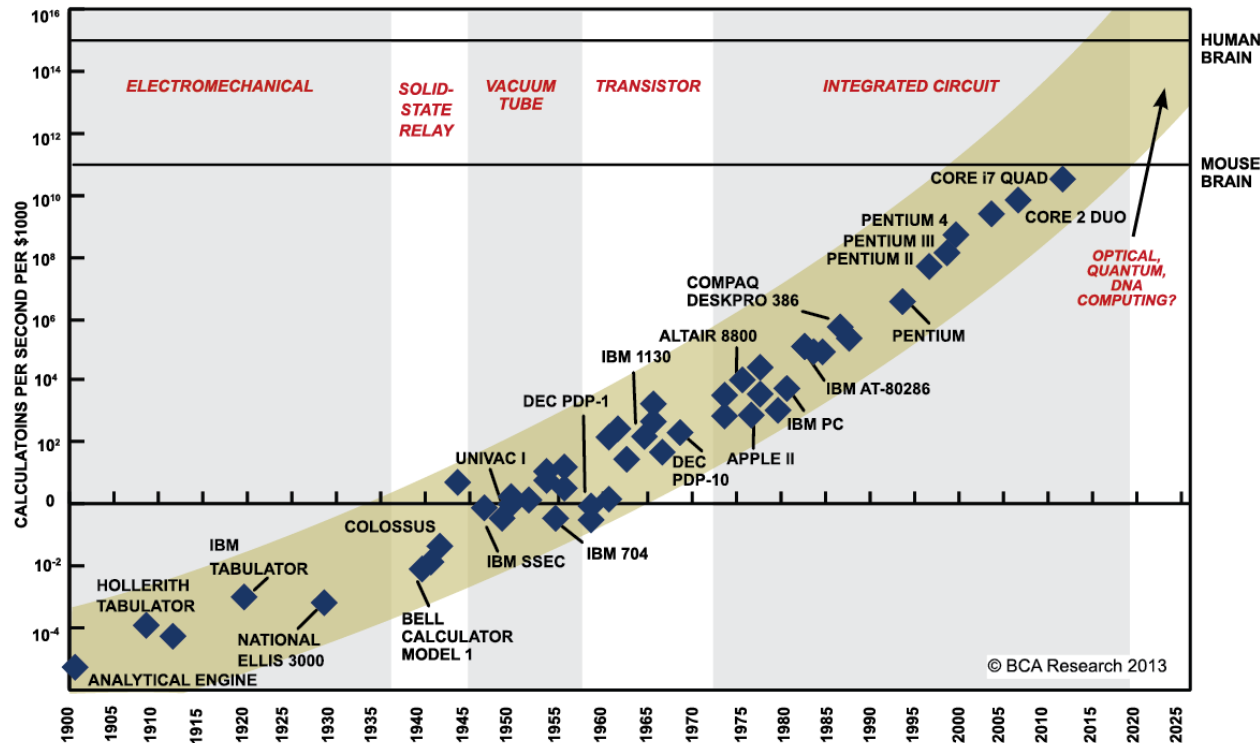
Field-effect transistor: structure and properties



First transistor
Bell Labs, 1948



Moore's Law



SOURCE: RAY KURZWEIL, "THE SINGULARITY IS NEAR: WHEN HUMANS TRANSCEND BIOLOGY", P.67, THE VIKING PRESS, 2006. DATAPPOINTS BETWEEN 2000 AND 2012 REPRESENT BCA ESTIMATES.



In 1965, Gordon Moore (a co-founder of Intel) predicted that the number of transistors on a chip would double every 2 years. **Moore's Law** has been upheld since this early prediction.

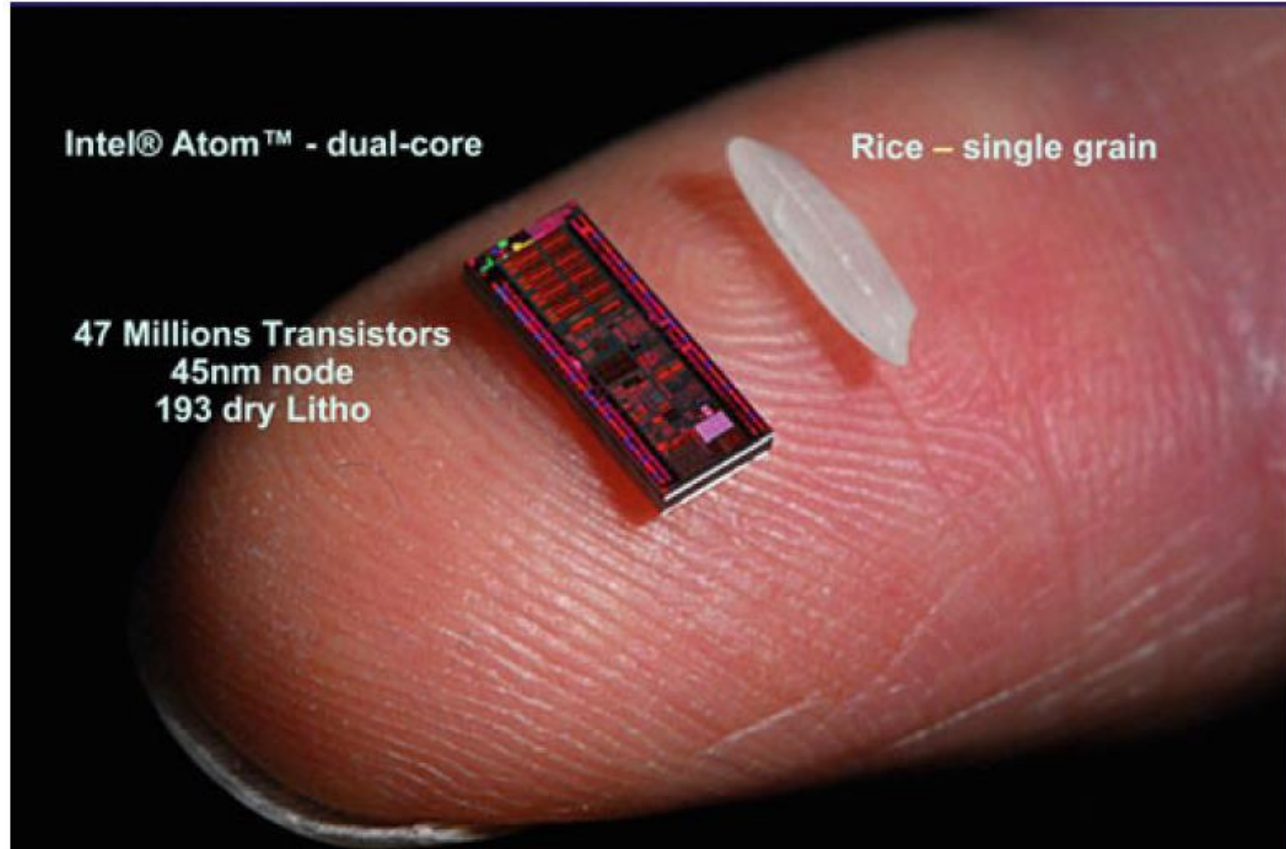


Figure 4.18. Dimensions of the Intel Atom™ dual-core processor, relative to a single grain of rice. Photograph reproduced with permission from Intel Corporation (<http://www.intel.com>). May be accessed online at: http://www.semi.org/cms/groups/public/documents/web_content/ctr_030805.pdf.



- The first applications for transistors were radios, which utilized a small number (*i.e.*, **<10**) of transistors.
- For current Intel chips, with a package size of 10–20 mm², now contain over a **billion** of individual nanoscopic transistors!
- Such a technological advancement is only possible through an exponential decrease in the price of single transistors.
- For instance, the first commercially available transistor was the Raytheon CK703, priced at \$18 in 1951. Taking inflation into account, the price of a single transistor would be *ca.* **\$120** today, resulting in computers that would be worth over **\$100 billion**!
- Fortunately, developments in chip fabrication processes have made it possible to reduce the production costs for a single transistor to less than **\$0.000001**.



small-scale integration (SSI):

The number of transistors within an IC: $N < 10$

medium-scale integration (MSI):

The number of transistors within an IC: $10 < N < 100$

large-scale integration (LSI):

The number of transistors within an IC: $100 < N < 1000$

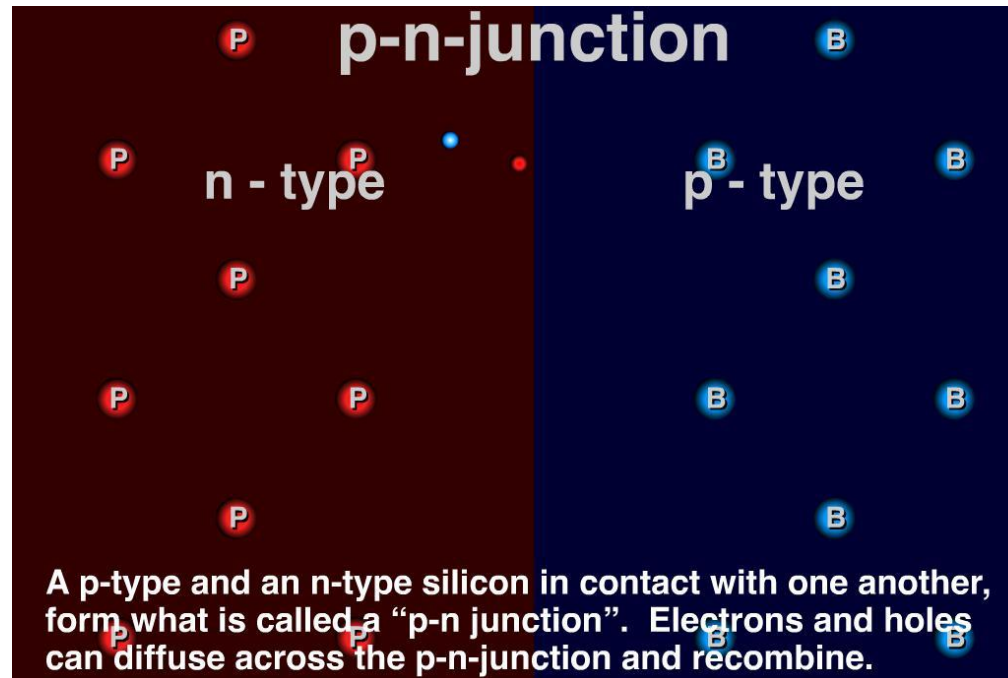
Very large-scale integration (VLSI):

The number of transistors within an IC: $N > 1000$

ultralarge-scale integration (ULSI):

The number of transistors within an IC: $N > 1,000,000$

p-n junction

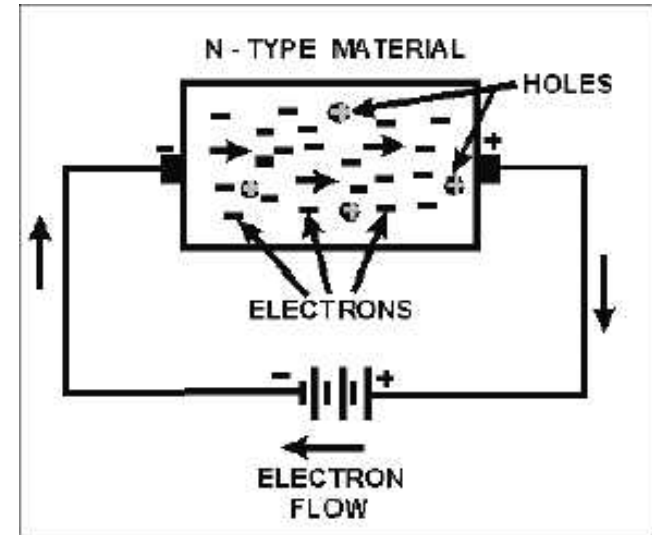


A PN Junction Diode is one of the simplest semiconductor devices around, and which has the characteristic of passing current in only one direction only.

However, unlike a resistor, a diode does not behave linearly with respect to the applied voltage, but the diode has an exponential current-voltage (I-V) relationship.



- The net flow of electrons and holes generate currents.
- The flow of "holes" within a solid-state material is, in all respects, equivalent to a flow of positive charge carriers.
- The process by which these charged particles move is called **Carrier Transport**.



There are 2 carrier transport mechanism in semiconductor

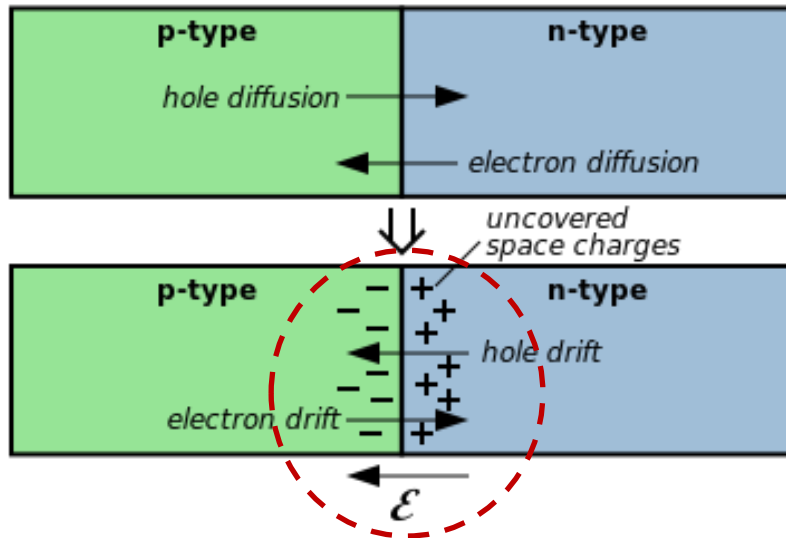
Diffusion – the flow of charge due to density gradients.

Drift – the movement of charge due to electric fields



p-n junction diode

Depleted region



In p-n junction



Majority carriers from each side diffuse to the opposite side (electron in n-side $\rightarrow$ p-side)



concentration equilibrium



opposite charge ions establish an electric field ("built-in potential" = $\mathcal{E}$)

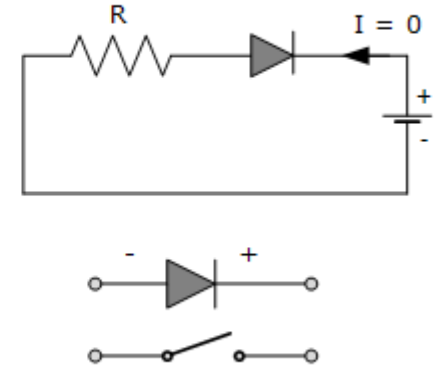
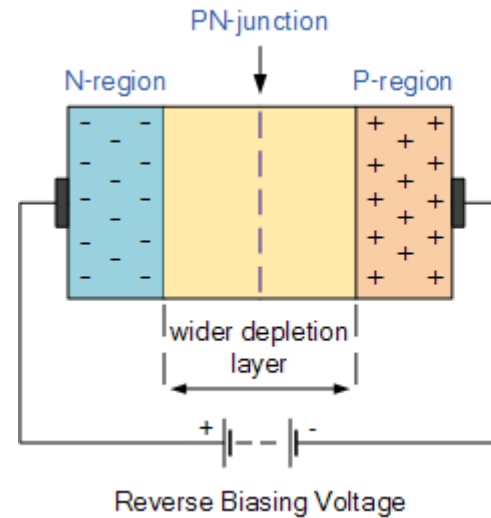
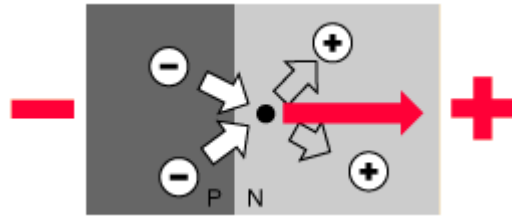
Prevents further diffusion of the majority carriers



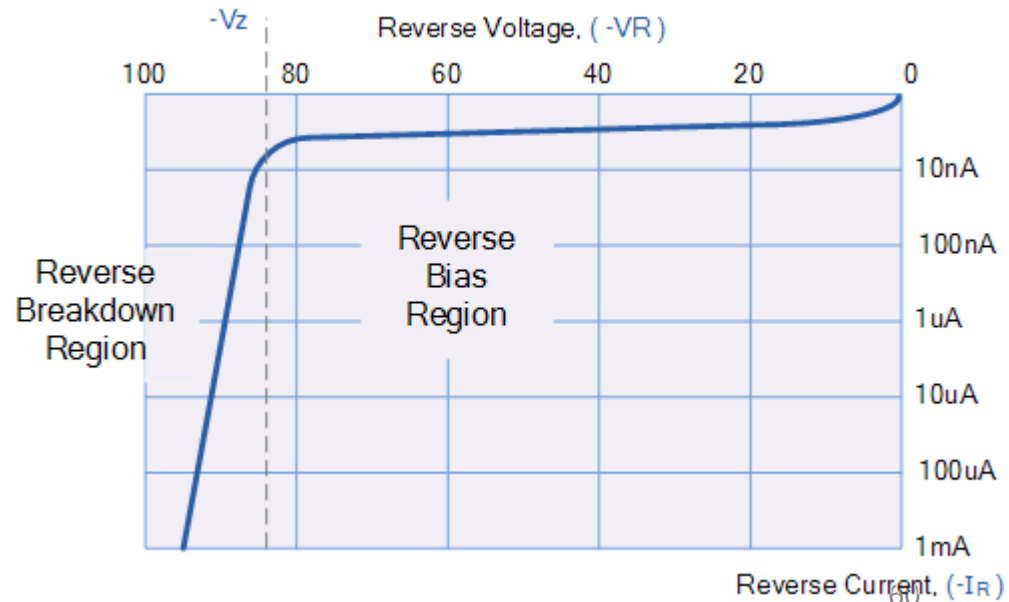
1. **Zero Bias** – No external voltage potential is applied to the PN junction diode.
2. **Reverse Bias** – The voltage potential is connected negative, $(-v_e)$ to the P-type material and positive, $(+v_e)$ to the N-type material across the diode which has the effect of Increasing the PN junction diode's width.
3. **Forward Bias** – The voltage potential is connected positive, $(+ve)$ to the P-type material and negative, $(-ve)$ to the N-type material across the diode which has the effect of Decreasing the PN junction diodes width.



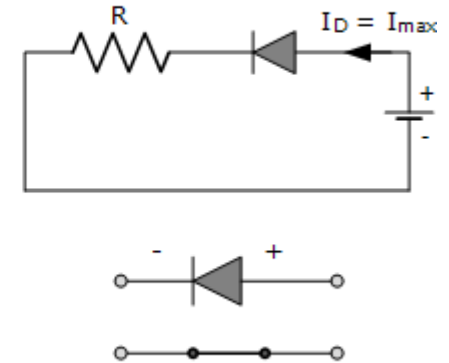
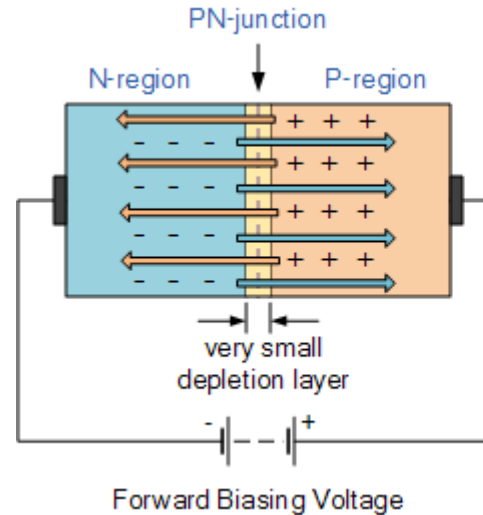
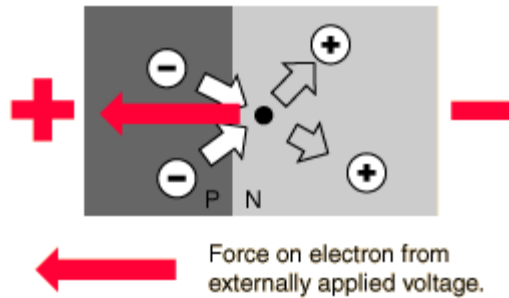
Reverse Bias



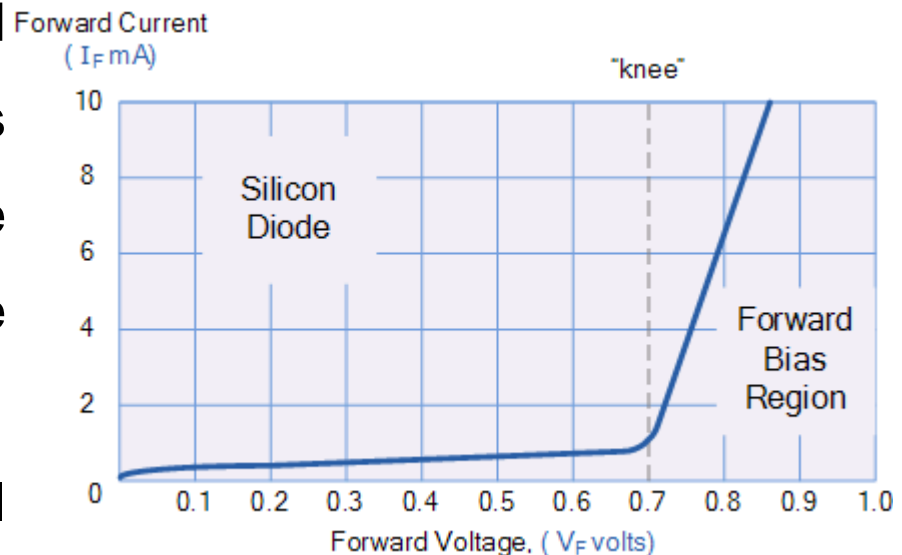
- An applied voltage with the indicated polarity further impedes the flow of electrons across the junction.
- A reverse voltage drives the electrons away from the junction, preventing conduction.

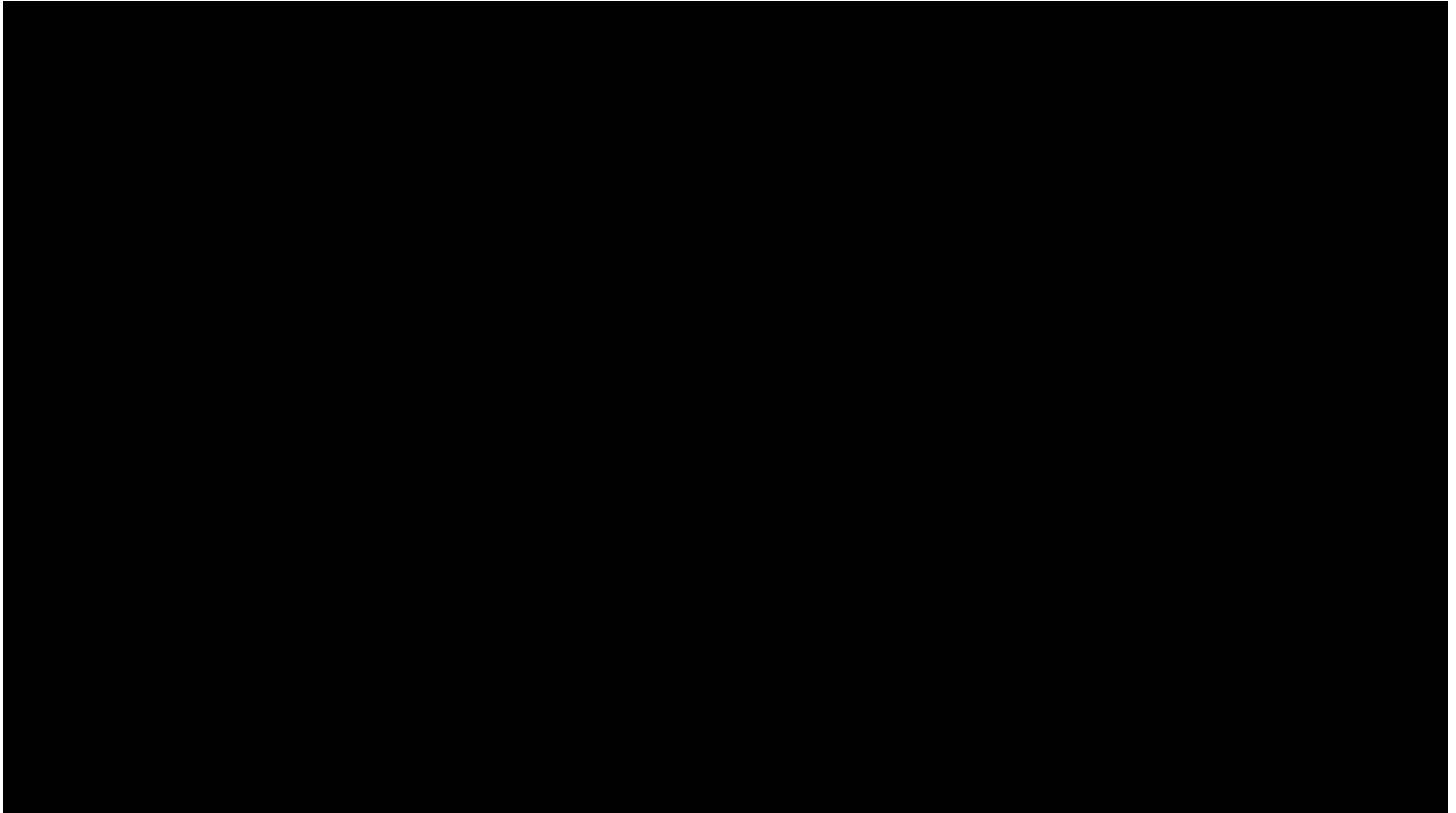


Forward Bias



- An applied voltage in the forward direction as indicated assists electrons in overcoming the coulomb barrier of the space charge in depletion region.
- Electrons will flow with very small resistance in the forward direction.

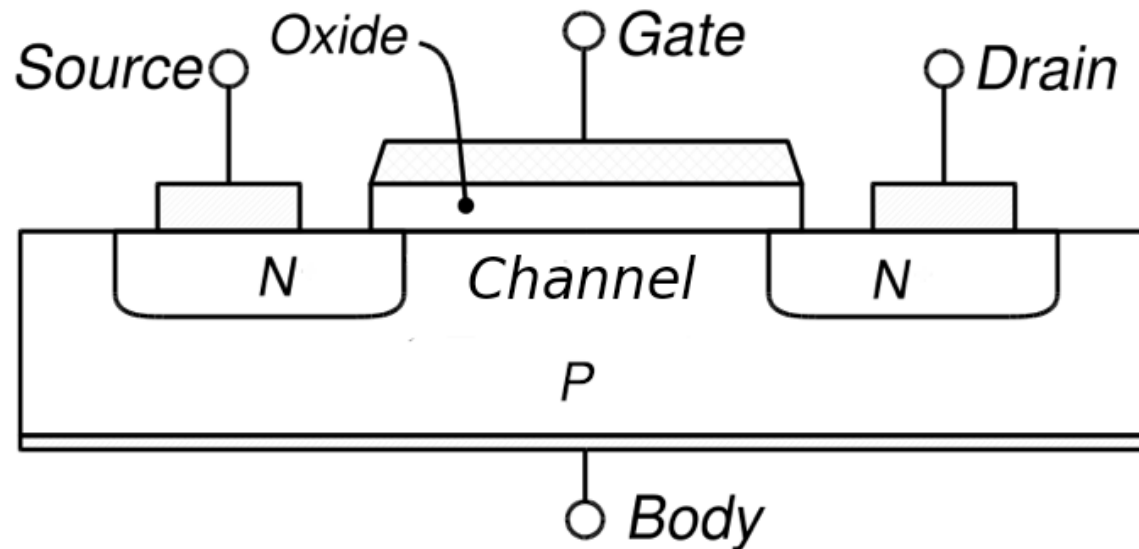


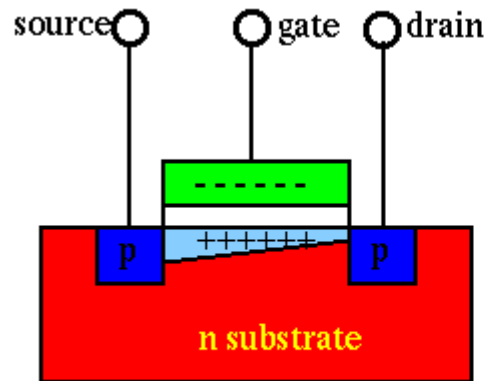




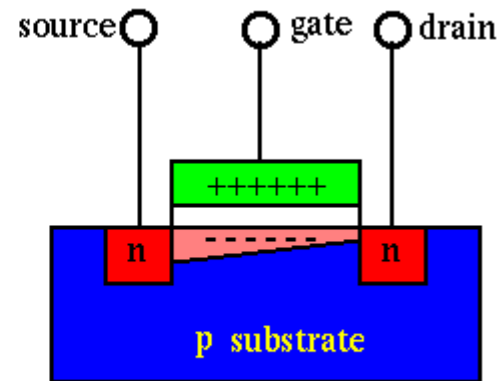
p-n transistor

Almost all transistors manufactured today are **MOSFET** transistors: **Metal Oxide Semiconductor Field Effect Transistors**.





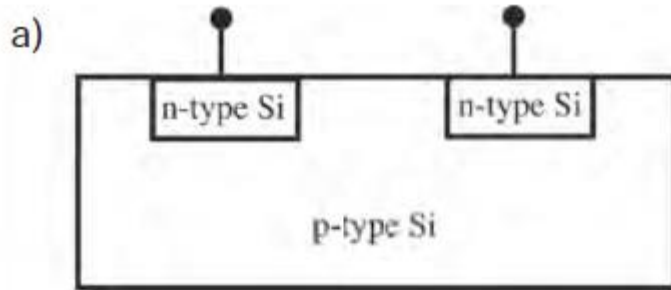
pMOS



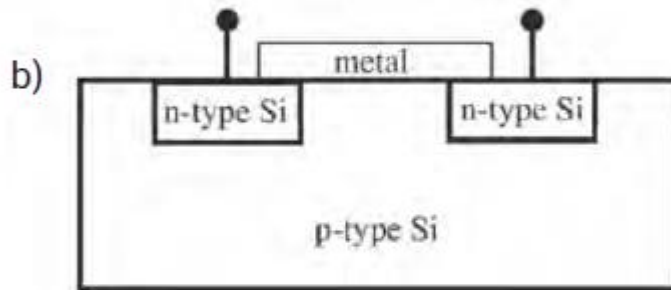
nMOS

NMOSFET
PMOSFET

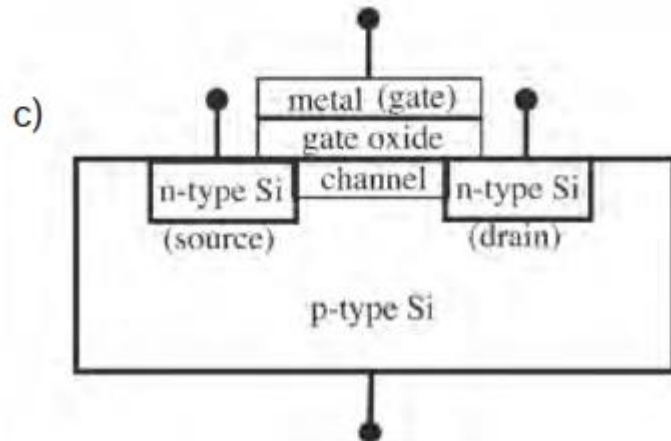
n-channel with p-type substrate
p-channel with n-type substrate



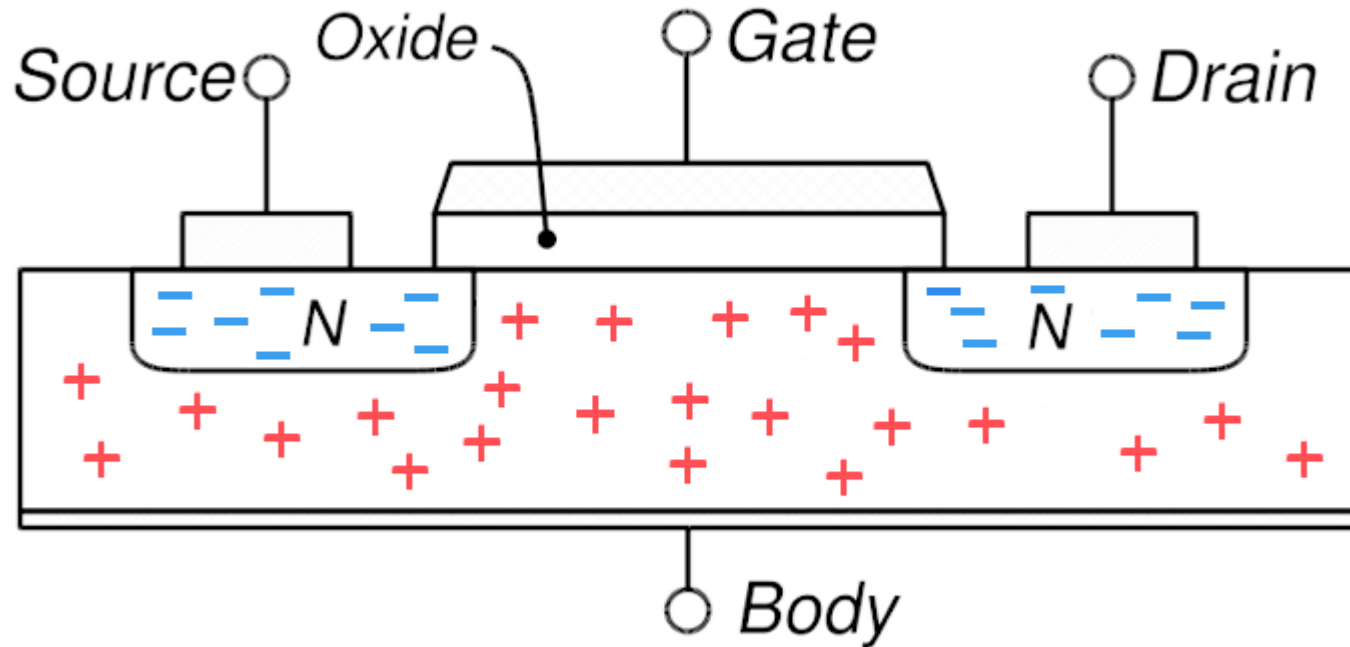
If a current is introduced to a simple configuration containing islands of n- or p-type Si within a matrix of an oppositely doped Si substrate, no electron flow will occur.



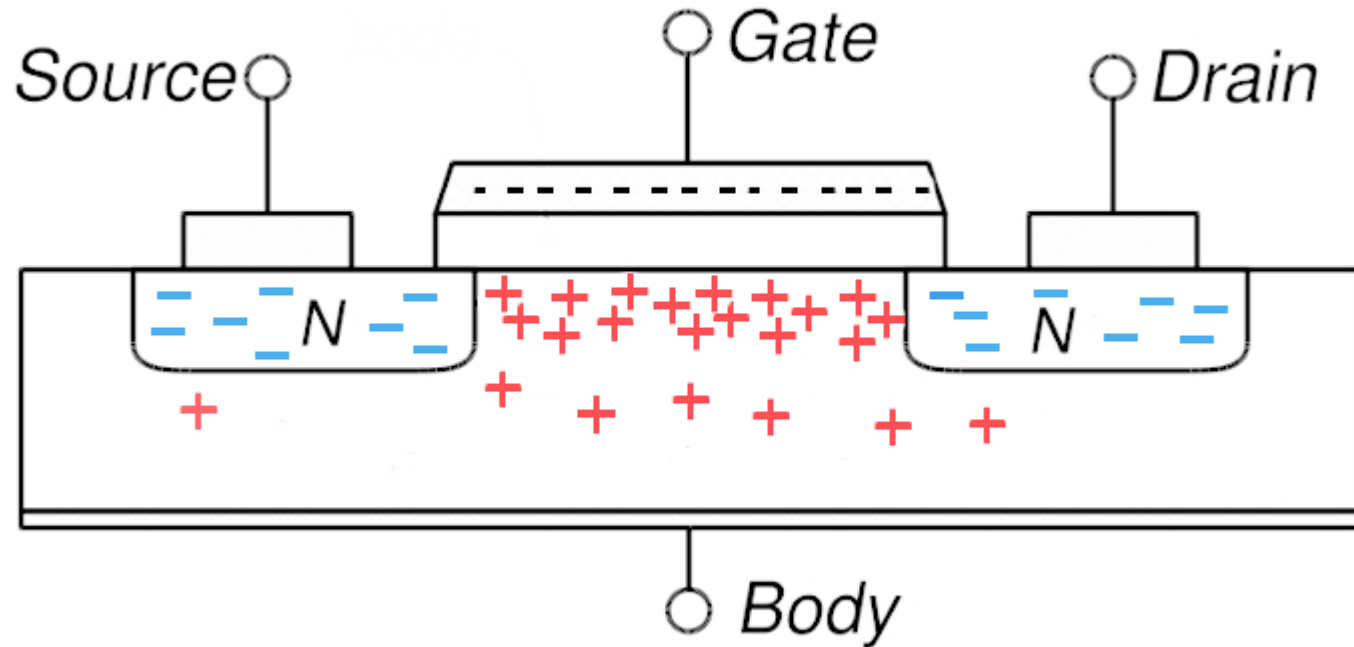
Electron flow will occur if a metal is allowed to contact the doped-Si regions; however, there will be no control over the electrical current



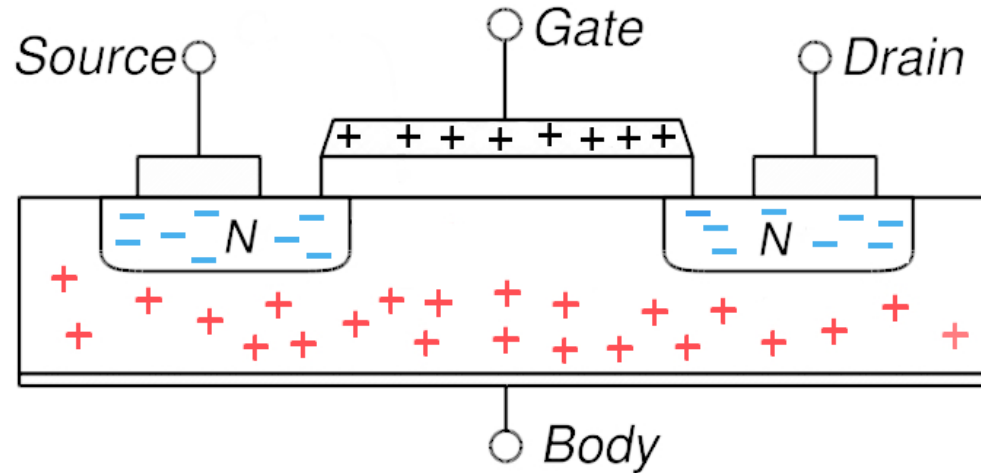
This architecture allows one to control the electron flow through a gate by varying the magnitude of the applied external voltage on the metal



No gate voltage and the natural distribution of holes throughout the body

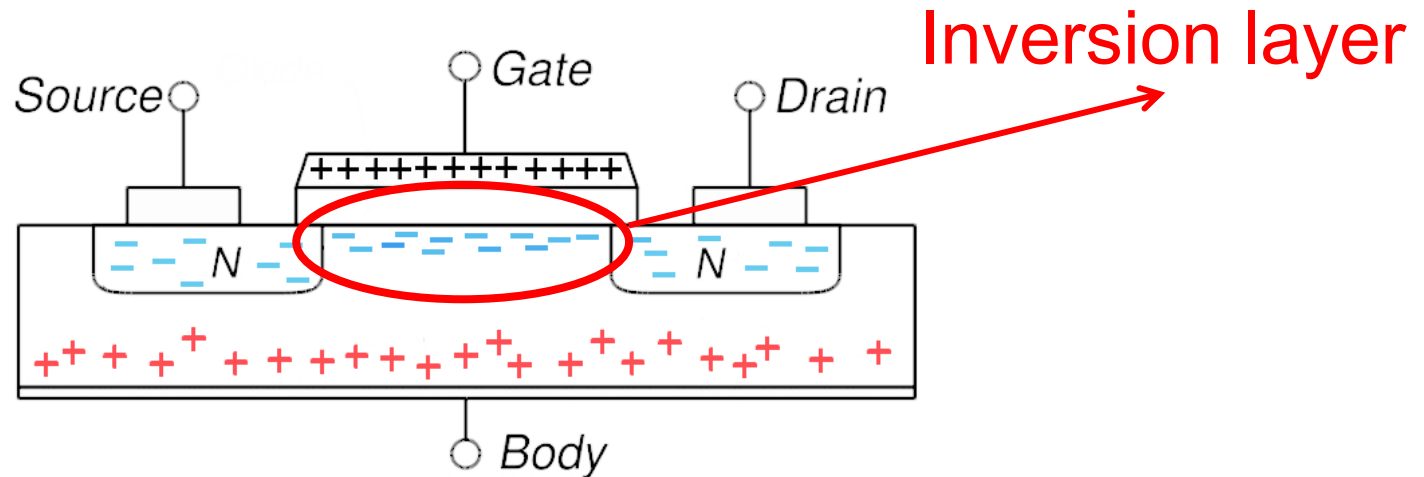


Negative gate voltage attracts holes and they accumulate in the channel region.



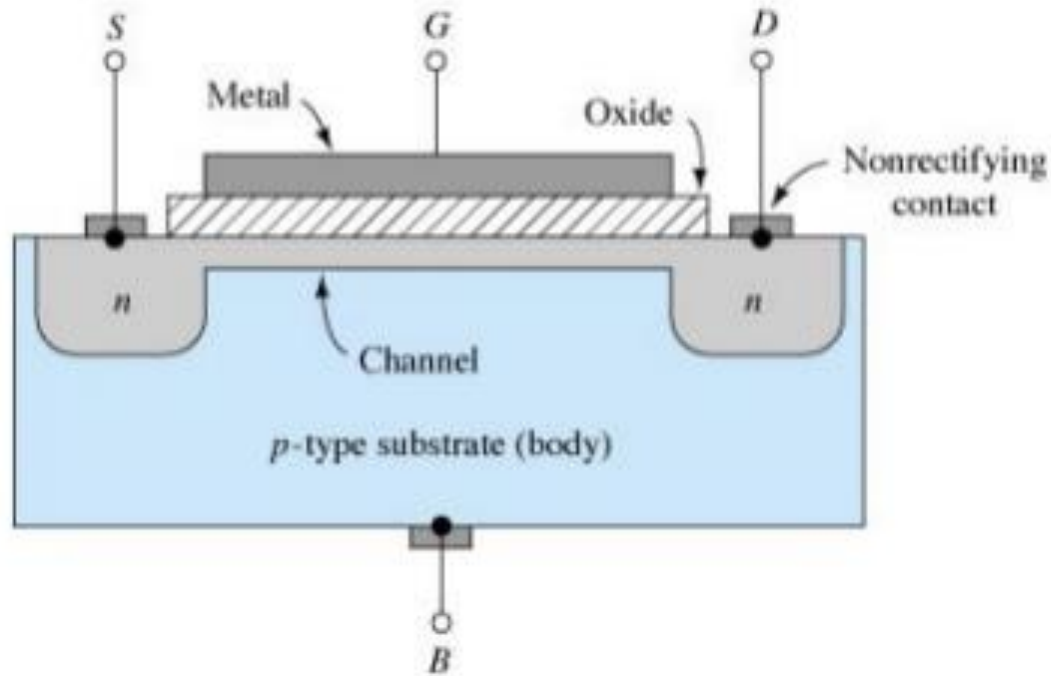
Positive gate voltage repels holes and the channel region is depleted of holes.

- The gate charges are balanced not by electrons, but by negative acceptor ions (*e.g.*, B^-) in the depletion layer.
- An increase in the gate bias causes the depletion layer to widen, yielding more acceptor ions in order to balance the gate charge.

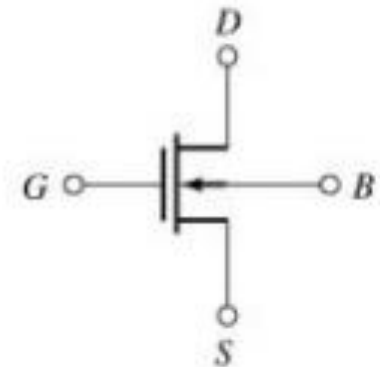


Very positive gate voltage repels all holes and actually attracts electrons (from source and/or drain) into the channel; the p-type region in the channel has been inverted in doping.

- At large applied voltages the surface of the Si is inverted from p-type to n-type due to the generation of free electrons that form a channel.



(a) Physical structure

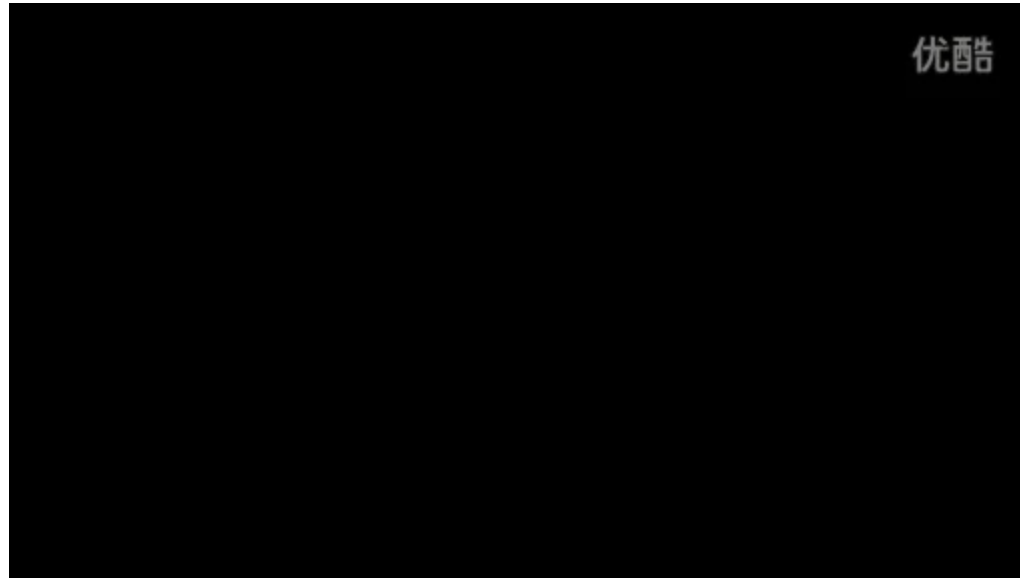


(b) Circuit symbol

Figure n -Channel depletion MOSFET.



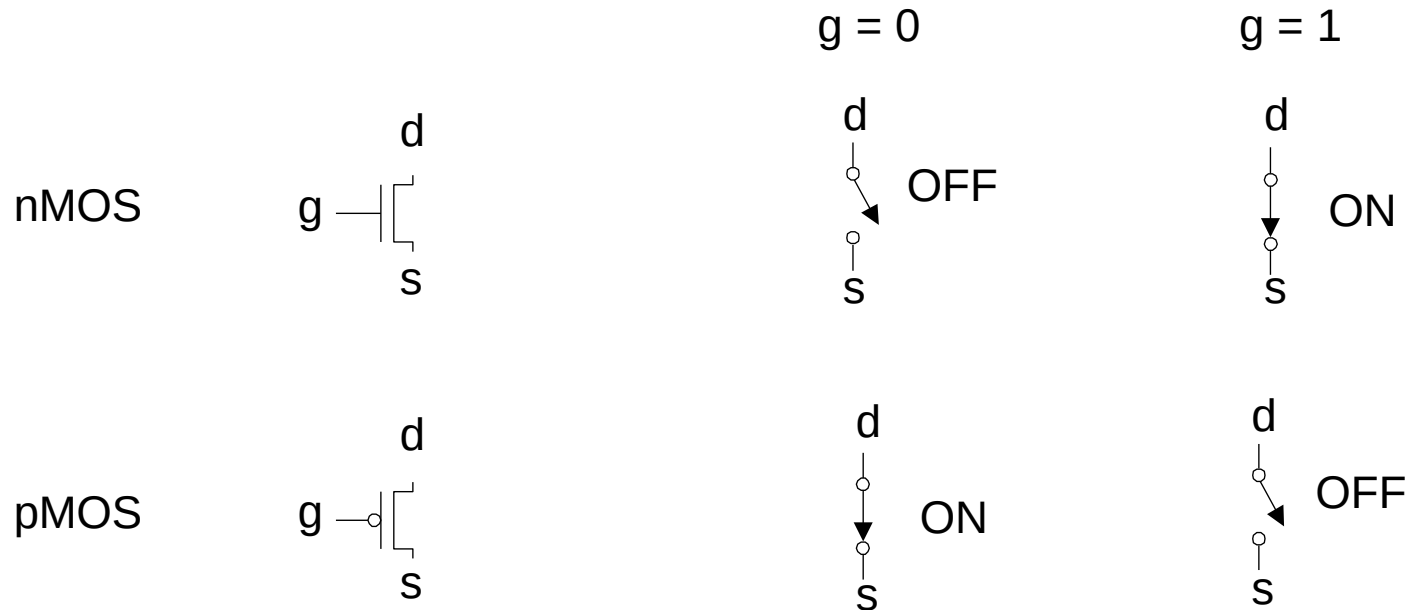
Put it most simply, a transistor is a switch. You can turn it on or off with a voltage. More precisely, you can make it conducting or highly insulating depending on the voltage you put on it (on the gate).





Integrated circuit fabrication

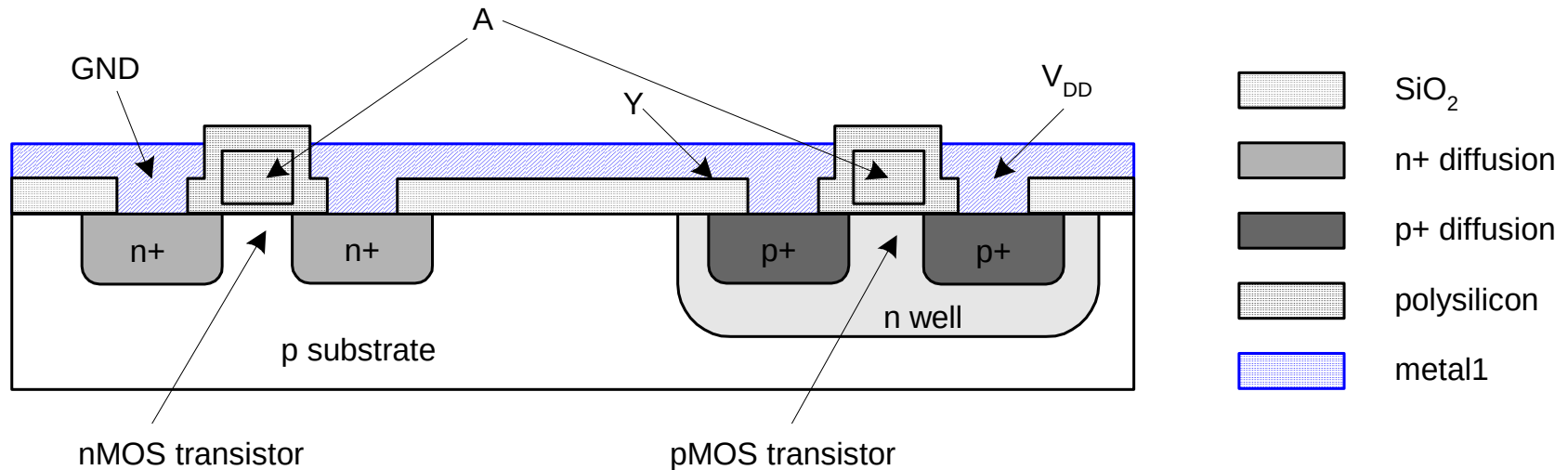
- We can view MOS transistors as electrically controlled switches
- Voltage at gate controls path from source to drain





The major class of digital ICs is the Complementary Metal–Oxide–Semiconductor (CMOS).

CMOS technology involves the complimentary operation of interconnected nMOSFET and pMOSFET pairs.





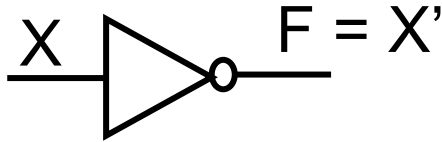
CMOS is basically a class of integrated circuits, and is used in a range of applications with digital logic circuits, such as microprocessors, microcontrollers, static RAM, etc.

Relative to nMOS or pMOS integrated circuitry, CMOS is the design-of-choice for the majority of ICs due to distinct advantages in overall design simplicity and reduction in power.

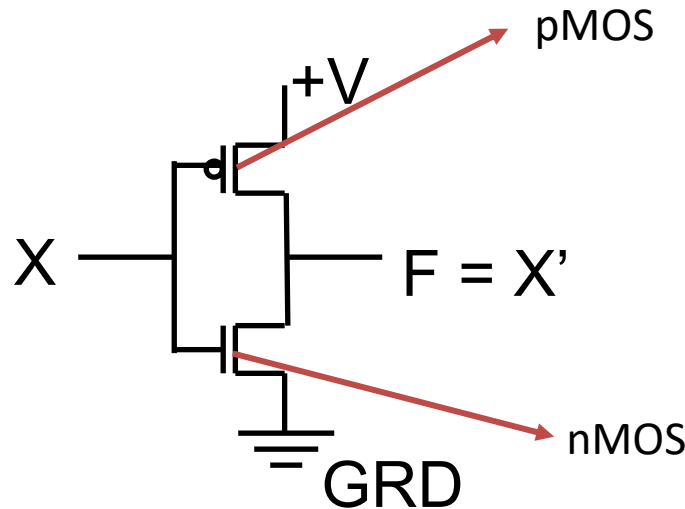
- small in size
- high operating speeds
- efficient usage of energy
- a wider voltage variation 3v to 15v dc
- very low static power supply drain
- a high degree of noise immunity



CMOS Inverter



Logic symbol



Transistor-level schematic

X	F
0	1
1	0

Operation:

- $X=1$ nMOS switch conducts (pMOS is open) and draws from GRD $F=0$
- $X=0$ pMOS switch conducts (nMOS is open) and draws from +V $F=1$



Threshold Voltage (V_t)

The gate voltage necessary to induce inversion in the channel region of the substrate.

The V_t is related to the difference in the work function between the gate and channel regions.

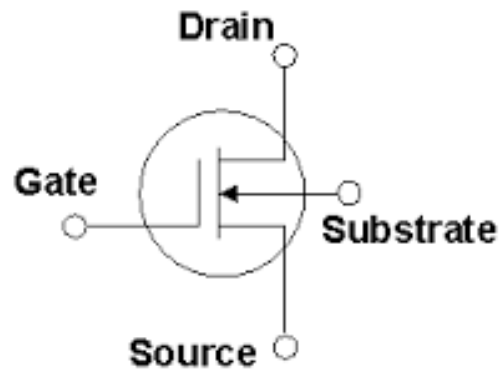
Work Function (ϕ)

The energy required to remove an electron from the surface of a material.



The smaller the better

- More densely packed on ICs
- Cheaper
- Higher computational speeds
- Less power



n-Channel MOSFET



$$I_{DS} = \frac{W}{2Lt_{ox}} \mu C (V_{GS} - V_T)^2$$

I_{DS} : the drain current

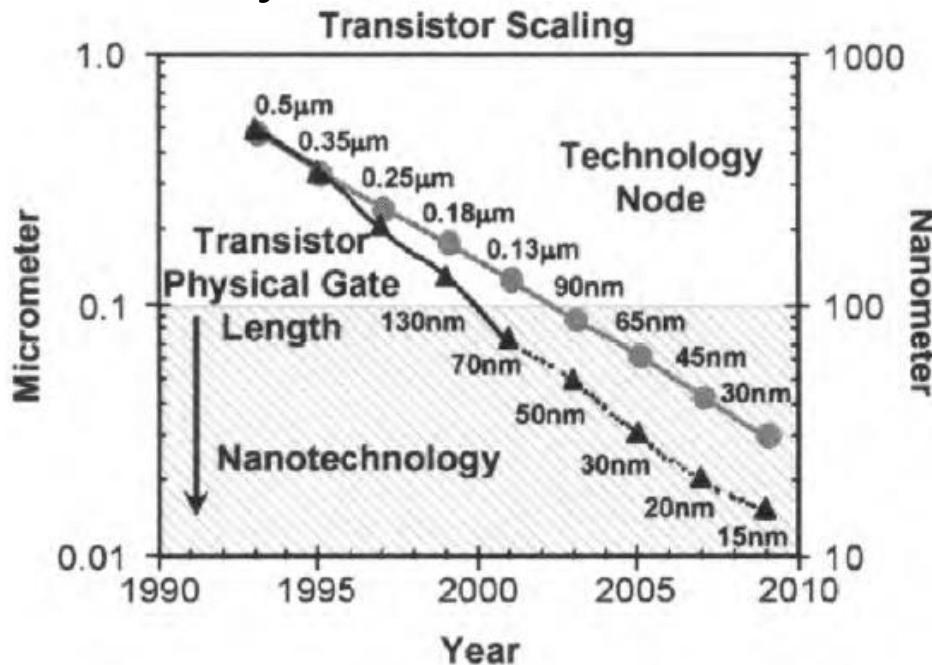
$\uparrow\uparrow$ { W : the channel width
 μ : the channel carrier mobility
 C : the capacitance density of the gate oxide
 V_{GS} : the gate voltage

$\uparrow\downarrow$ { L : the channel length
 t_{ox} : the gate oxide thickness

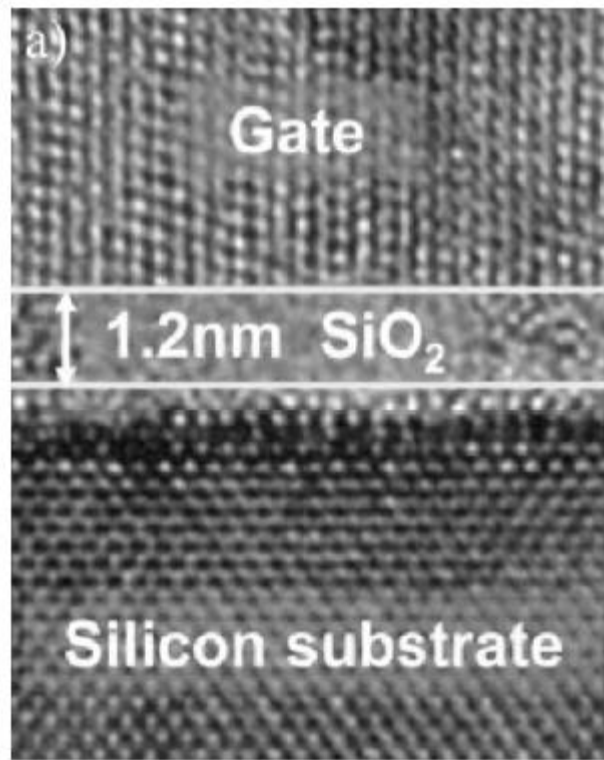


The gate oxide

- The insulator between the gate and channel;
- Ideally, the oxide layer should be as thin as possible.

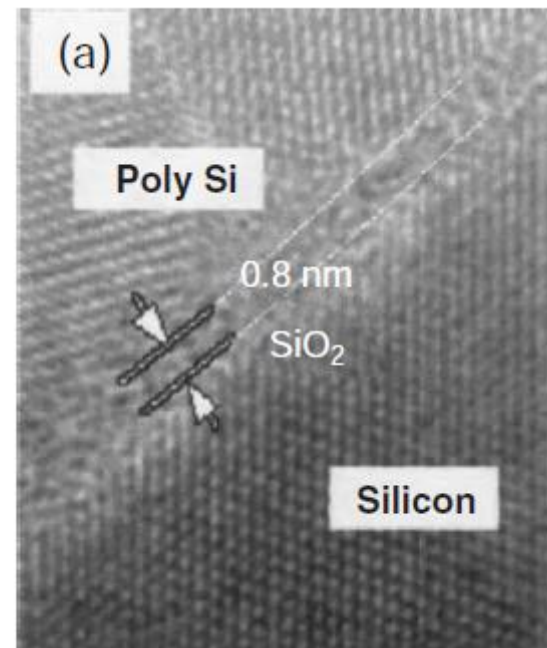


Scaling of transistor size. The top line indicates the technology node, whereas the bottom line indicates the physical size of the gate. It should be noted that the “technology node” no longer refers to physical dimensions of the transistor, but is rather an industry term related to a new fabrication process every 2 years, in accord with Moore’s Law.



Smaller and smaller

→ A five atomic layer

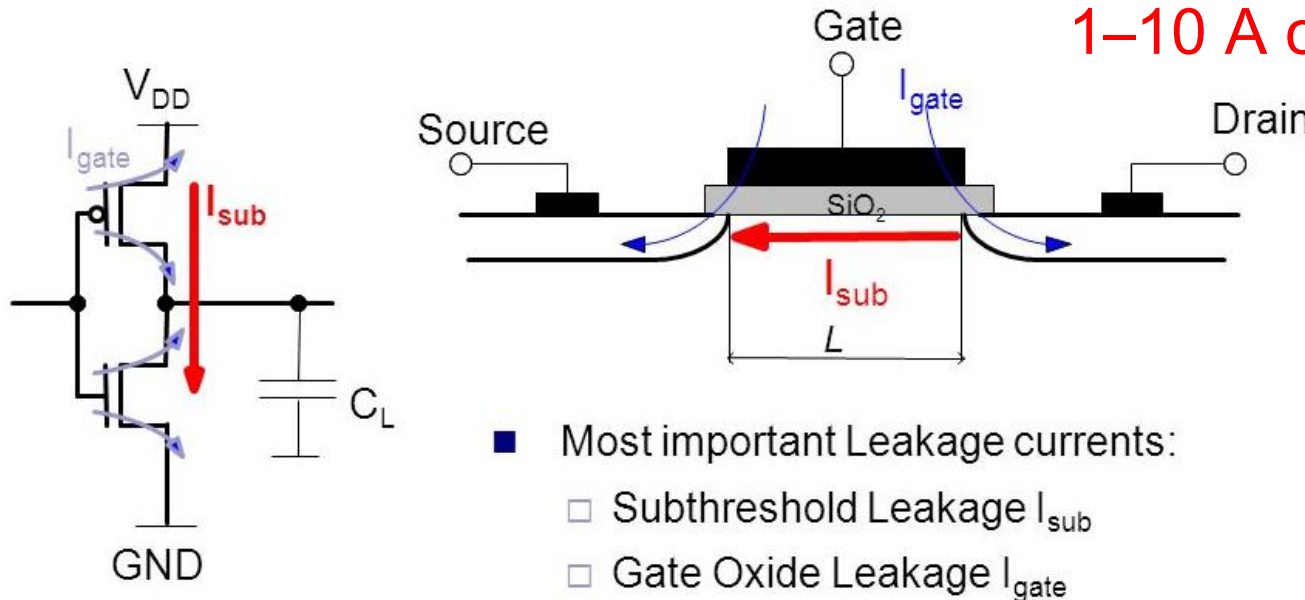


The smaller the better?

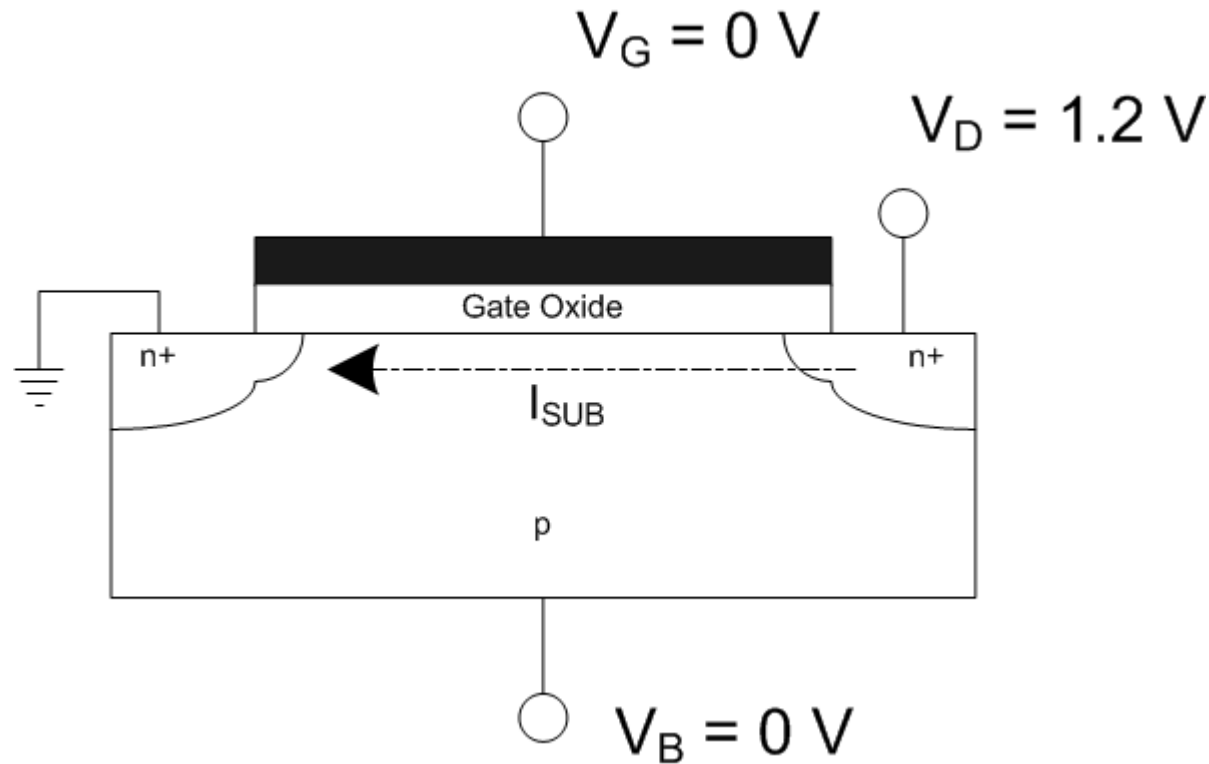
A thin insulating layer is prone to electron tunneling between the gate and channel, leading to increased power consumption and heat buildup.

Leakage current density

Current transistors:
 $1\text{--}10\text{ A cm}^{-2}$

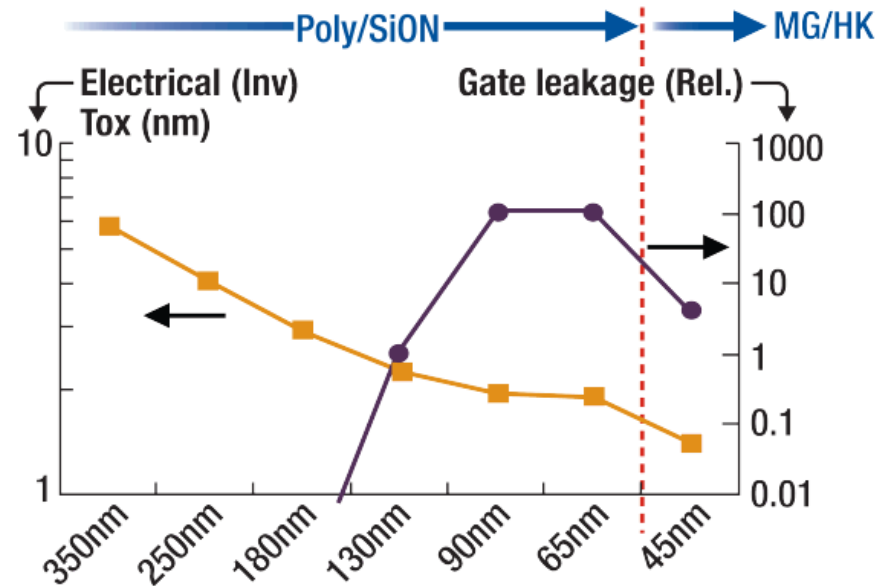
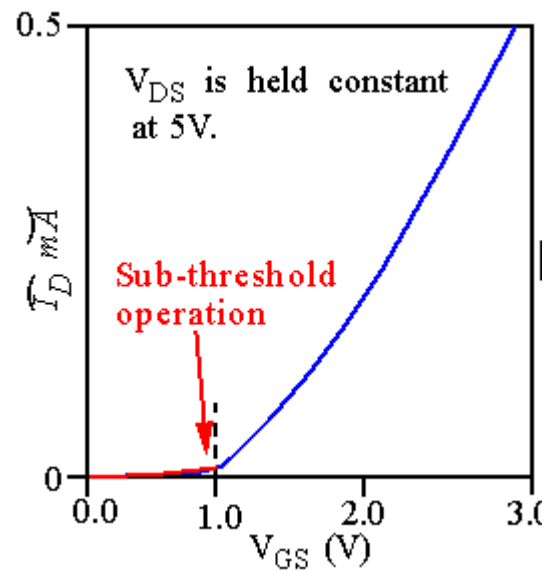
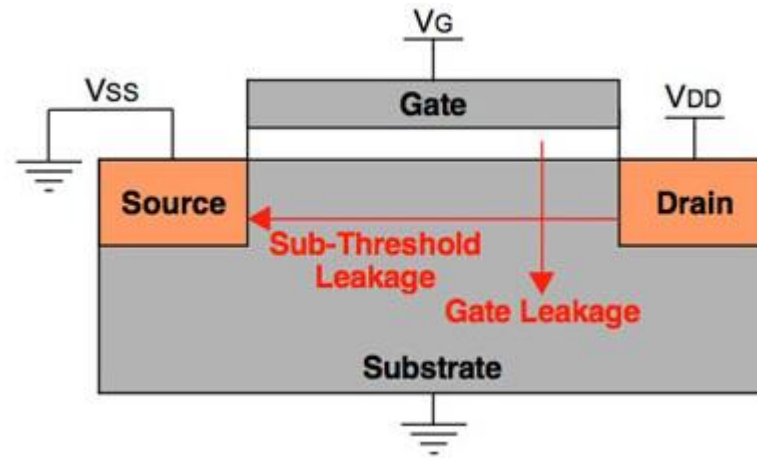


Empirical and theoretical studies have shown that the practical limit for dielectric effectiveness of the gate SiO_2 layer is *ca.* 0.7–1 nm.



Subthreshold leakage

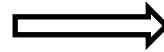
The current between the source and drain of a MOSFET when the transistor is in subthreshold region, or weak-inversion region, that is, for gate-to-source voltages below the threshold voltage.





To solve the Gate Leakage problems

SiO₂ gate oxide



Insulating film
High- κ dielectrics

$$\kappa = 3.82$$

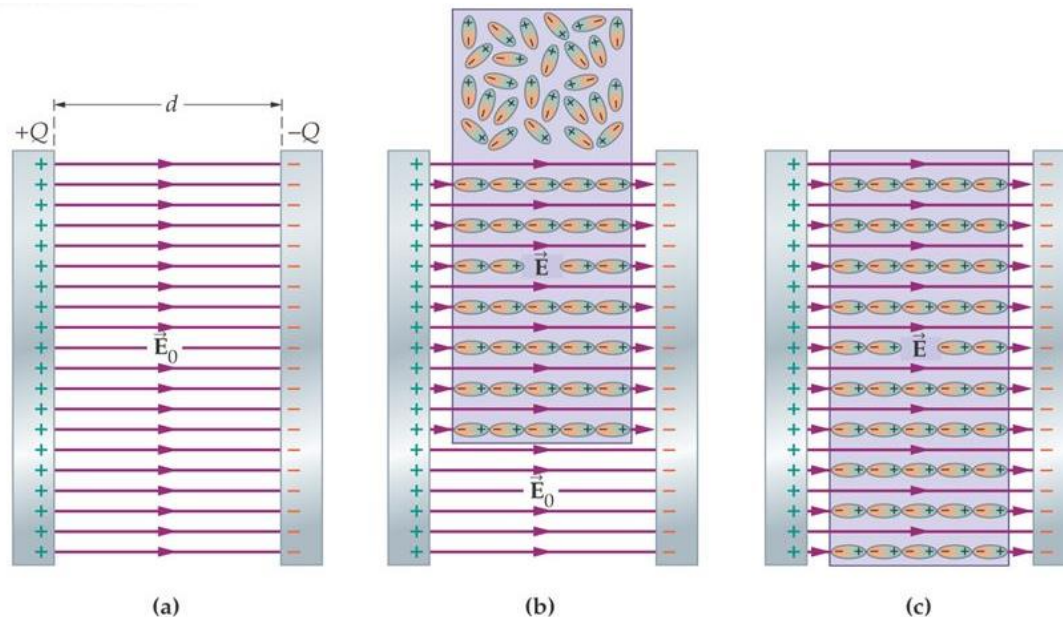
$$\kappa > 3.82$$

$$C_{vol} \propto \frac{\kappa}{t^2}$$

$$\kappa \uparrow \rightarrow t \uparrow$$

C_{vol} : the capacitance per unit volume;
 κ : the dielectric constant;
 t : the thickness of the dielectric medium.

- Materials that exhibit a high dielectric constant are strongly polarizable.
- This result in the buildup of electrical charges on the surface of the capacitor plates.



- Inorganic materials with ionic or partial ionic interactions will possess a relatively large dielectric constant (high- κ).
- Organic molecules with nonpolar bonds are typically low- κ materials.



Three types of high- κ dielectrics:

SiO_2 κ : ~ 3.82

- $4 < \kappa < 10$ (*i.e.*, relatively weak polarizability)

Al_2O_3 (7.5–10), Si_3N_4 (7.9–8.1), sodalime glass (7.2), $\text{Al}_6\text{Si}_2\text{O}_{13}$ (mullite, 6.3), MgAl_2O_4 (spinel, 8.6), MgSiO_4 (forsterite, 6.2), MgO (9.65), ZnO (8.15), BN (4.15), BeO (6.8), CaF_2 (6.8), BaCO_3 (8.5), SrCO_3 (8.85), $(\text{KAl}[\text{Si}_3\text{O}_{10}(\text{OH})_2])$ (mica muscovite, 6.5–8.7), and organic polymers (polyurethane: 7.1, polyvinyl fluoride: 7.4, polyvinylidene fluoride: 6.4).

- $10 < \kappa < 100$

MnO_2 (pyrolusite, 12.8), ThO_2 (18.9), Ta_2O_5 (27.6), UO_2 (24), Nb_2O_5 (67), HfO_2 (25), ZrO_2 (12.5–24.7), $\text{Zr}(\text{Hf})\text{O}_2$ (25), Y_2O_3 (15), La_2O_3 (25), Gd_2O_3 (23), Pr_2O_3 (25), and BaZrO_3 (43).

- $\kappa > 100$ (*i.e.*, strongly polarizable)

TiO_2 (rutile, 85–170), PbZrO_3 (200), $\text{Pb}(\text{Zr}_{0.52}\text{Ti}_{0.48})\text{O}_3$ (“PZT,” 1,800), SrTiO_3 (2,080), $\text{Ba}(\text{Sr}_{0.52}\text{Ti}_{0.48})\text{O}_3$ (3,000), $\text{Pb}(\text{Ni}_{0.33}\text{Nb}_{0.67})\text{O}_3$ (5,500), and BaTiO_3 (6,000).



The high- κ dielectric layers:

- Large dielectric constants
- Low conductivity

$$E_g(\text{SiO}_2) = 8.9 \text{ eV}$$

$$E_g(\text{Ta}_2\text{O}_5) = 4.5 \text{ eV}$$

$$E_g(\text{TiO}_2) = 3.5 \text{ eV}$$

Small bandgaps will result
in high leakage currents

$$E_g(\text{Al}_2\text{O}_3) = 8.7 \text{ eV}$$

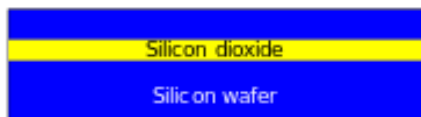
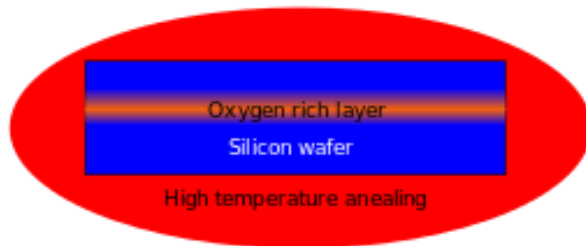
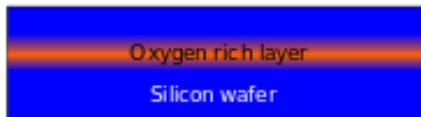
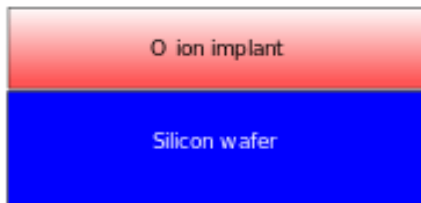
$$E_g(\text{HfO}_2) = 5.7 \text{ eV}$$

$$E_g(\text{ZrO}_2) = 7.8 \text{ eV}$$

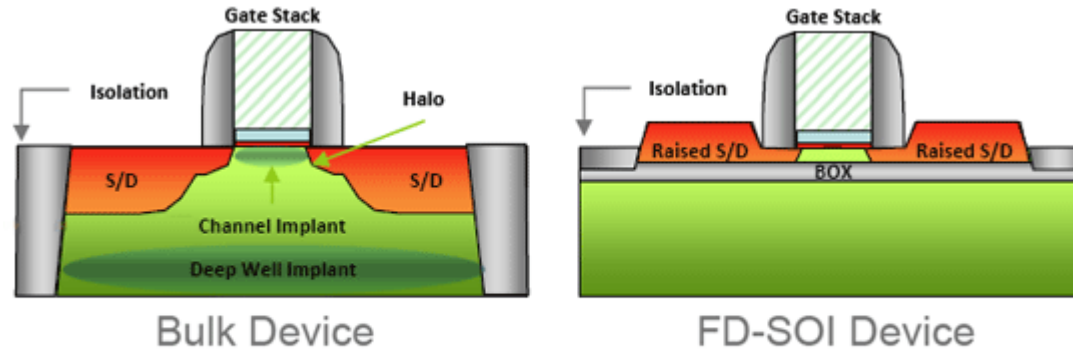
Promising candidates

- The thermodynamic and kinetic stabilities with the Si channel (especially under high-temperature processing conditions).
- Little/no electrically active defect sites.

Silicon on insulator (SOI) technology ---Removing the substrate



- For high-frequency circuits (about 5 GHz and above), capacitive coupling to the Si substrate limits the switching frequency.
- Also, leakage into the substrate from the small devices can cause extra power dissipation.
- These problems are being avoided by making circuits on insulating substrates (either sapphire or silicon dioxide) that have a thin, approximately 100 nm layer of crystalline silicon, in which the MOSFETs are fabricated.



The fully depleted SOI transistor at 20 nm is significantly simpler than even a simplified version of the bulk CMOS transistor.

- One of the major source of parasitic capacitance is from the source and drain to substrate junctions.
- SOI can reduced the capacitance at the source and drain junctions significantly — by eliminating the depletion regions extending into the substrate.



- The source and drain cavities with greater thicknesses:
 - ✓ The electrons will have more room to maneuver, resulting in less electron–electron repulsions and lower overall resistance.
 - ✓ This translates to a lower operating voltage for the transistor, which results in low power consumption with less heat generation during its operation.



- The insulation (high- κ) between source and drain cavities:
 - ✓ Allowing the threshold voltage of the SOI CMOS device to be extremely low, with minimal offstate (*i.e.*, subthreshold) leakage.
 - ✓ It is reported that this simple substrate alteration results in chips that operate more than 30% faster, and use *ca.* 20% less power than standard CMOS analogs.
 - ✓ The use of less power per chip is important for extending the battery life of portable electronics, as well as dealing with the heat generated by ICs. If one extrapolates the enormous heat generated from high-density bulk Si chips into the near-future, it may be necessary to use liquid coolants within the IC package— clearly not a desirable (or feasible) option.⁹¹



Let's build ICs

More than 300 carefully designed processing steps are used to form a dense packing of nanoscale objects onto a thin wafer of silicon.

□ Silicon wafer

- Cutting

The silicon wafers used for IC fabrication are obtained through precise slicing of high-purity ingots using a diamond saw.

The wafer edges are ground to a round finish polished until they have a defect-free mirrored finish.



- RCA cleaning

The RCA (Radio Corporation of America) clean is a standard set of wafer cleaning steps which need to be performed before high-temperature processing steps (oxidation, diffusion, CVD) of silicon wafers in semiconductor manufacturing. Including the following steps

1. Removal of the organic contaminants (organic clean + particle clean);
2. Removal of thin oxide layer (oxide strip, optional);
3. Removal of ionic contamination (ionic clean).



First step (SC-1): organic clean + particle clean

SC stands for Standard Clean

- The wafer is degreased using acetone, isopropyl alcohol, and ultrapure water (UPW) rinses.
- The native SiO_2 layer is removed from the surface by exposing the wafer to a 50:1 UPW:HF solution for 30 s, followed by a prolonged UPW rinse.
- The wafer is exposed to a 10:2:1 DIW: H_2O_2 : NH_4OH solution at 75°C for 10 min, followed by a final UPW rinse.
- Nitrile gloves and Teflon tweezers are used to handle the wafer during treatments.

The base-peroxide mixture removes organic residues. Particles are also very effectively removed, even insoluble particles, since SC-1 modifies the surface and particle zeta potentials and causes them to repel. This treatment results in the formation of a thin silicon dioxide layer (about $10 \text{ \AA}$) on the silicon surface, along with a certain degree of metallic contamination (notably Iron) that shall be removed in subsequent steps.



Second step (optional): oxide strip

The optional second step (for bare silicon wafers) is a short immersion in a 1:100 or 1:50 solution of HF + H₂O at 25 °C for about 15 s, in order to remove the thin oxide layer and some fraction of ionic contaminants.

If this step is performed without ultra high purity materials, it can lead to recontamination since the bare silicon surface is very reactive.



Third step (SC-2): ionic clean

- The wafer is exposed to a 10:2:1 UPW:H₂O₂:H₂SO₄ solution at 75°C, followed by a UPW rinse. This effectively removes metals/ions from the surface.
- The wafer is rinsed with DIW for *ca.* 20 min., followed by drying under N₂ flow.
- When finished, the polished wafer surface should be reflective with no observable residues.

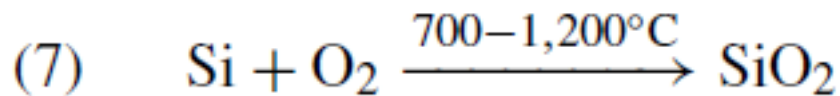
This treatment effectively removes the remaining traces of metallic (ionic) contaminants, some of which were introduced in the SC-1 cleaning step. It also leaves a thin passivating layer on the wafer surface, which protects the surface from subsequent contamination (bare exposed silicon is contaminated immediately).



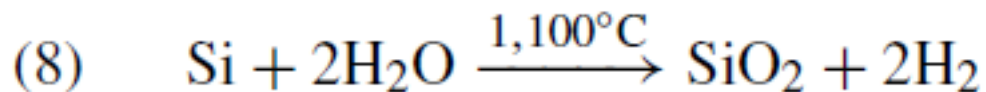
The fabrication of a CMOS IC involves a repeating sequence of film deposition, patterning, and dopant implantation procedures.

□ Film deposition

- Deposition of SiO_2 onto the wafer surface
 - this procedure involves the controlled deposition of an amorphous SiO_2 coating ($<1 \mu\text{m}$) of known thickness onto the wafer surface.



Dry Oxidation



Wet Oxidation

The resultant thickness of the silica film is determined by temperature and pressure/flow-rate of O_2 .

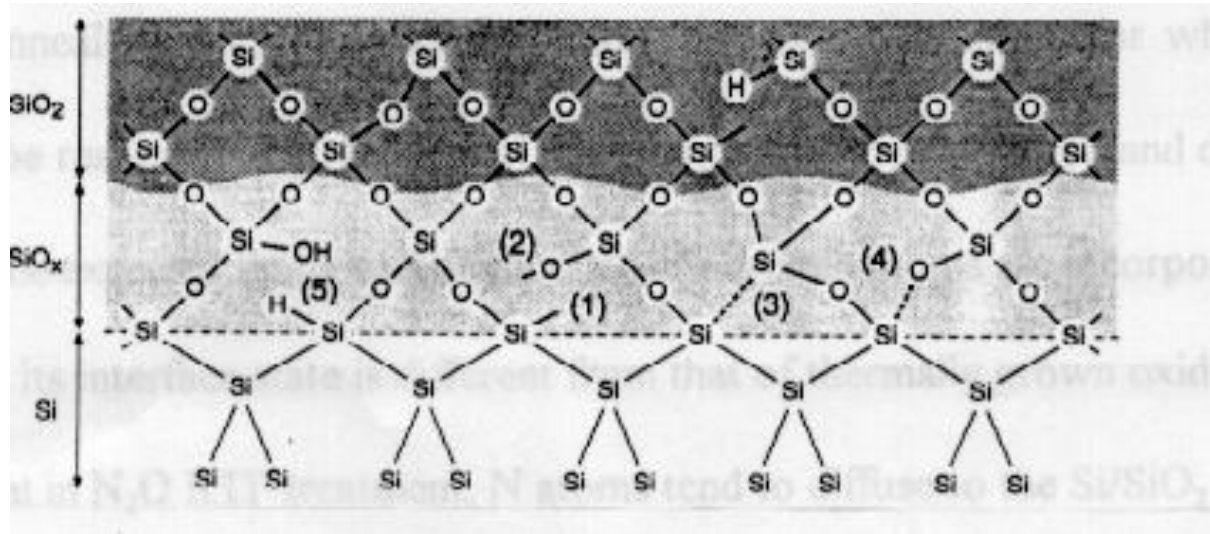


- It should be noted that SiO_2 deposition does not occur through simple addition to the wafer surface. Rather, as the surface becomes oxidized, oxygen atoms are also able to diffuse from the surface to react with underlying Si atoms.
- The incoming O_2 or H_2O molecules adsorb dissociatively to the Si surface, resulting in the O/OH groups being either attached to dangling Si bonds, or inserted between Si–Si bonds of the same layer.



- As the temperature is increased, there is a greater degree of oxygen insertion into Si–Si bonds of underlying layers, which adds to the thickness of the SiO_2 film.
- A chlorine containing co-reactant gas such as HCl or C_2HCl_3 is often present during Si oxidation to stabilize the Si/SiO_2 interface. It has been shown that small amounts ($<5\%$) of chlorine in the gas phase helps to tie up sodium or potassium ions that may be present as contaminants, as well as strengthening the growing oxide film by preventing oxidation-induced stacking faults.

- Following oxidation, it has been shown that 0.001% of the silicon atoms in the region immediately surrounding the Si/SiO₂ interface remain unsaturated.



In the transition region, the chemical composition is not stoichiometric (SiO_x: $x < 2$).
(1), silicon dangling bond; (2), nonbridging oxygen; (3), weak Si-Si bond; (4), strained Si-O bond; (5), hydrogen-containing species.

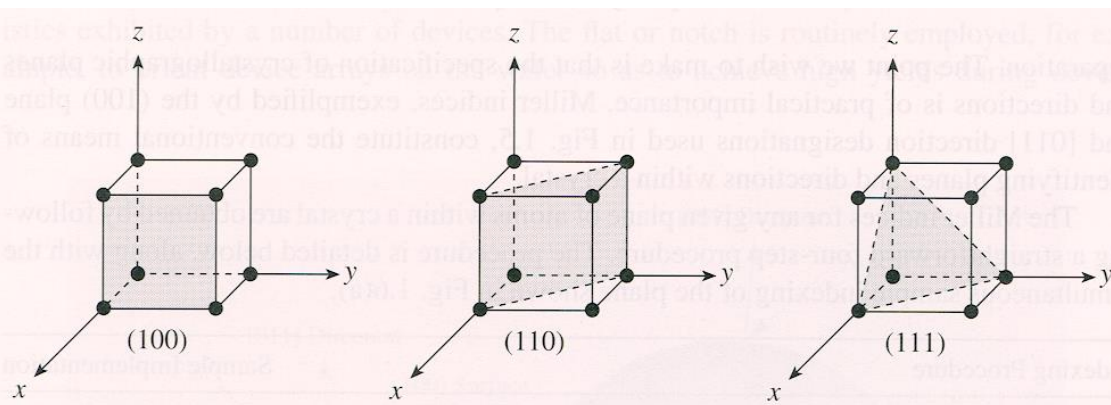
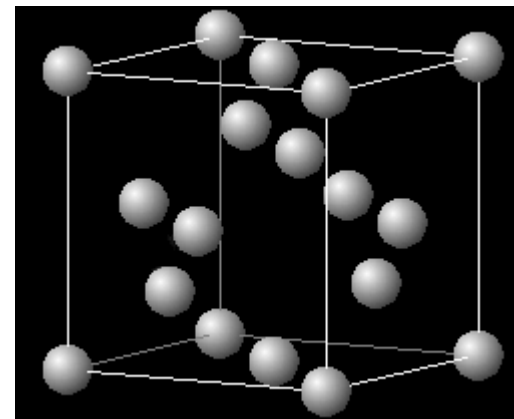


- The presence of these defects results in a layer of positive charge that is located within ± 2 nm from the interface, which may affect the V_T of the MOSFET, and performance of the IC as a whole.
- These effects are minimized by:
 1. using Si(100) substrates;
 2. passivated through high-temperature annealing in the presence of hydrogen;

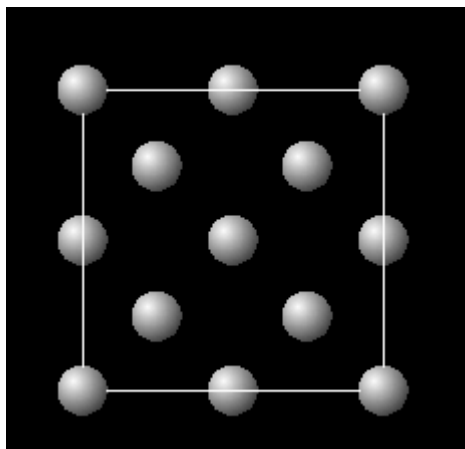


Crystallographic Planes in Si

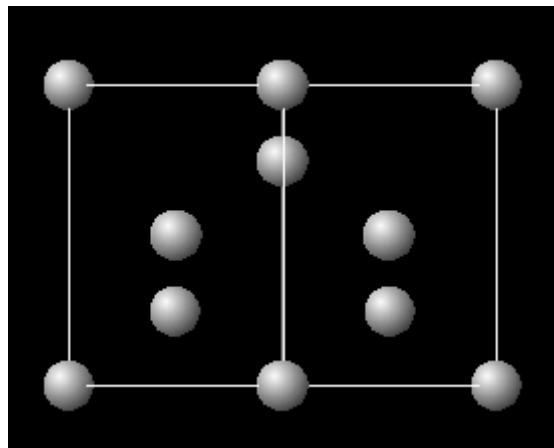
Unit cell:



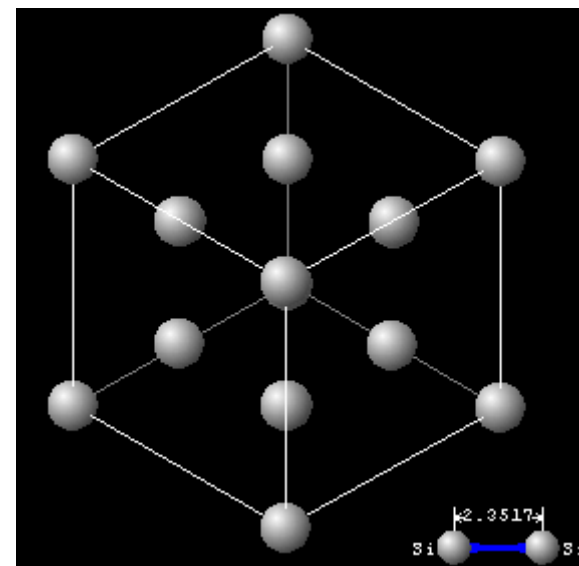
View in $\langle 100 \rangle$ direction



View in $\langle 110 \rangle$ direction



View in $\langle 111 \rangle$ direction





	Si (100)	Si (110)	Si (111)
Atomic densities	Si (100) <	Si (110) <	Si (111)
Si-Si bond densities	$6.77 \times 10^{14} \text{ cm}^{-2}$	$9.59 \times 10^{14} \text{ cm}^{-2}$	$11.76 \times 10^{14} \text{ cm}^{-2}$
Si etching rate (40% alcoholic KOH solution)	$6,000 \text{ Å min}^{-1}$	$13,000 \text{ Å min}^{-1}$	90 Å min^{-1}

The Si(110) orientation etches fastest due to its more corrugated structure, relative to the other atomically flat surfaces.



Majority of ICs utilize a Si(100) substrate **WHY?**

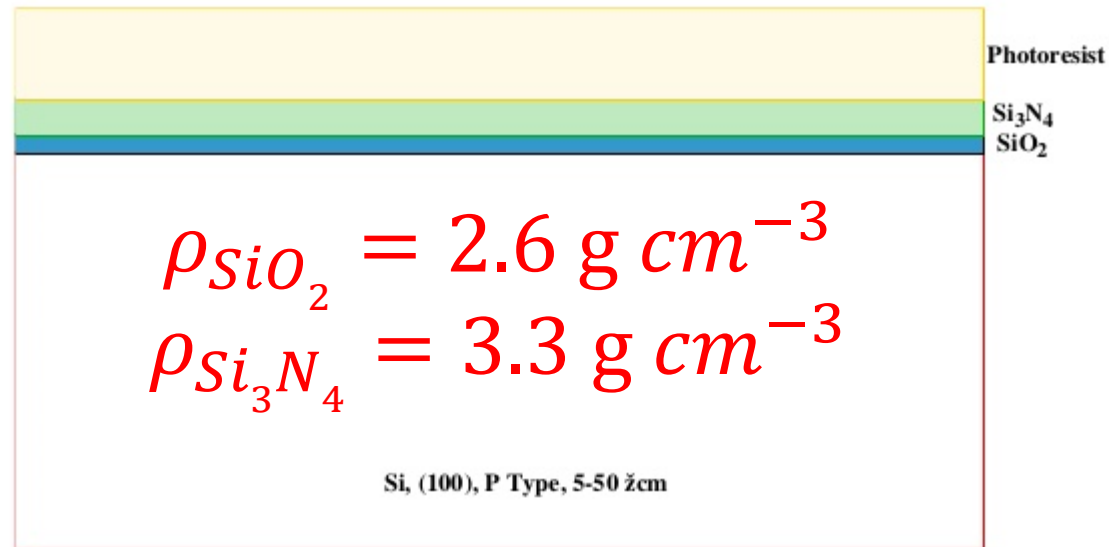
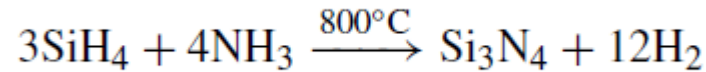
That is, since the Si(100) surface is the least dense, there will be a relatively low number of coordinatively unsaturated (*i.e.*, dangling) Si bonds at the surface.

As you might imagine, the presence of dangling bonds on a surface greatly affects its reactivity.

One way to remove these defects is through passivation. For example, a large number of dangling bonds are tied up by H atoms during HF etching to remove native SiO₂ (the RCA clean procedure above).



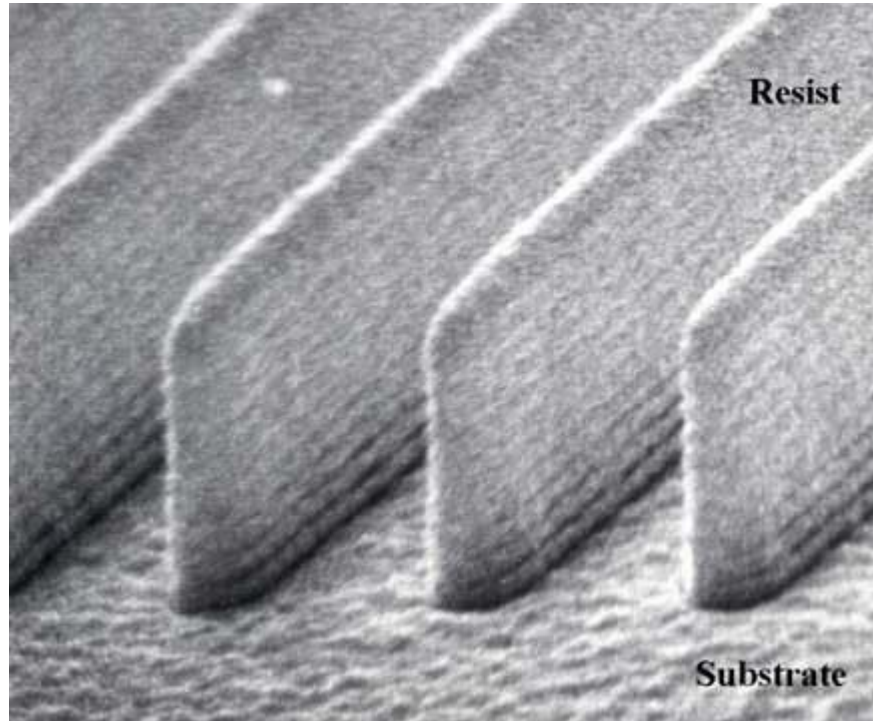
3. The deposition of silicon nitride, Si_3N_4 over the SiO_2 layer.



- The thicknesses of the SiO_2 and Si_3N_4 films are typically on the order of 15 and 75 nm, respectively.
- Si_3N_4 is used as an effective passivating layer to prevent oxidation of underlying SiO_2 and Si regions.



□ Patterning steps based on photolithography



Photolithography, also termed optical lithography or UV lithography, is a process used in microfabrication to pattern parts of a thin film or the bulk of a substrate. It uses light to transfer a geometric pattern from a photomask to a light-sensitive chemical "photoresist", or simply "resist," on the substrate.



- Pretreatments

- Dehydration bake

- To remove adsorbed water; 15 minutes in 80-90°C convection oven.

- Wafer priming

- Adhesion promoters are used to assist resist coating

- Primers form bonds with surface and produce a polar (electrostatic) surface

- most are based upon siloxane linkages (Si-O-Si)

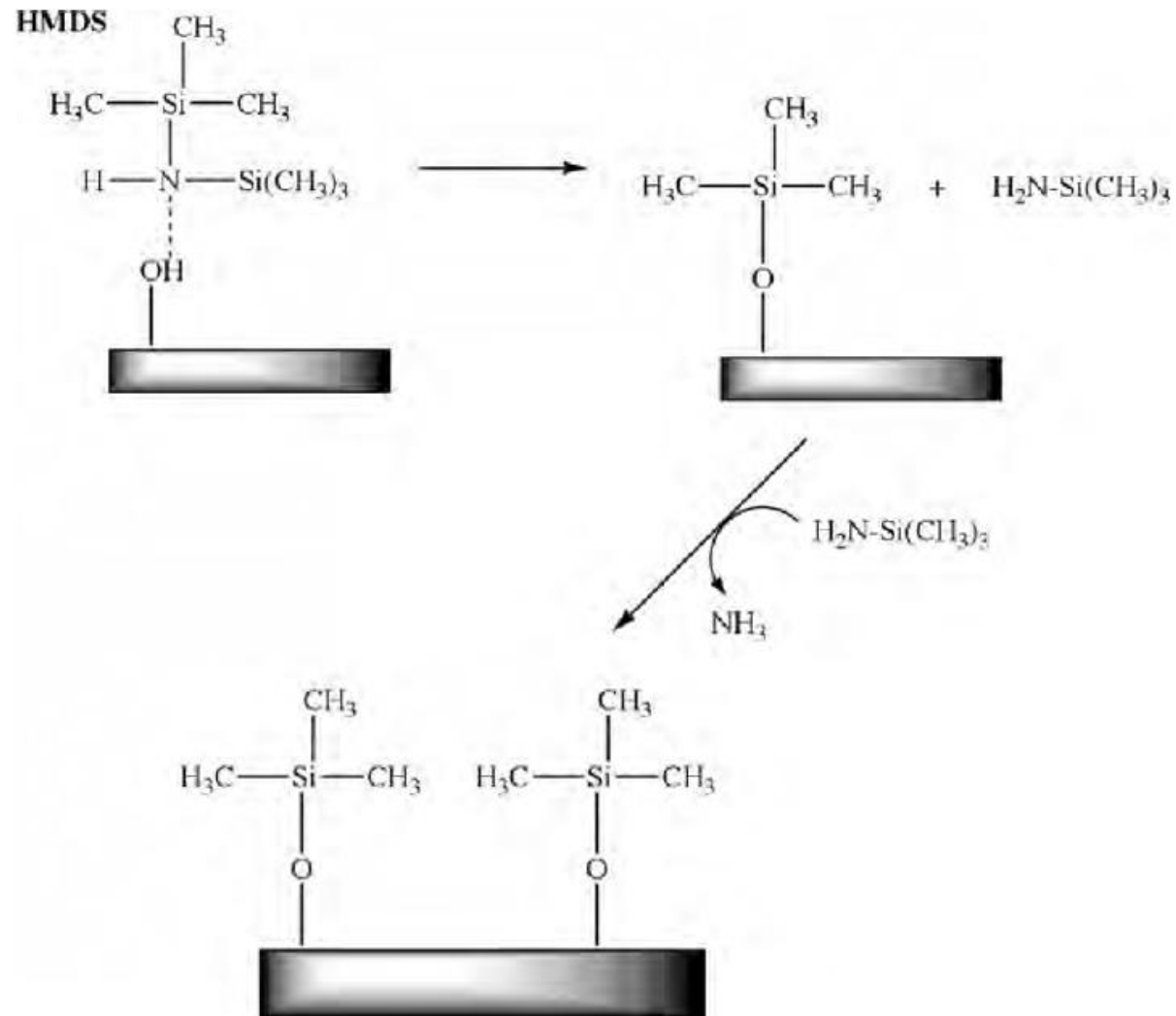
- 1,1,1,3,3,3-hexamethyldisilazane(HMDS),

- $(\text{CH}_3)_3\text{SiNHSi}(\text{CH}_3)_3$

- trichlorophenylsilane (TCPS), $\text{C}_6\text{H}_5\text{SiCl}_3$

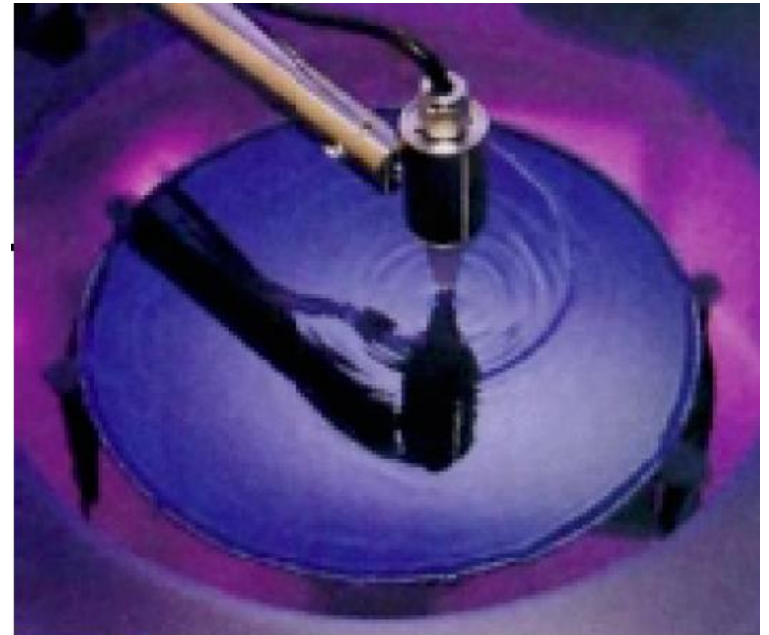
- bis(trimethylsilyl)acetamide (BSA),

- $(\text{CH}_3)_3\text{SiNCH}_3\text{COSi}(\text{CH}_3)_3$



- Coating of photoresist

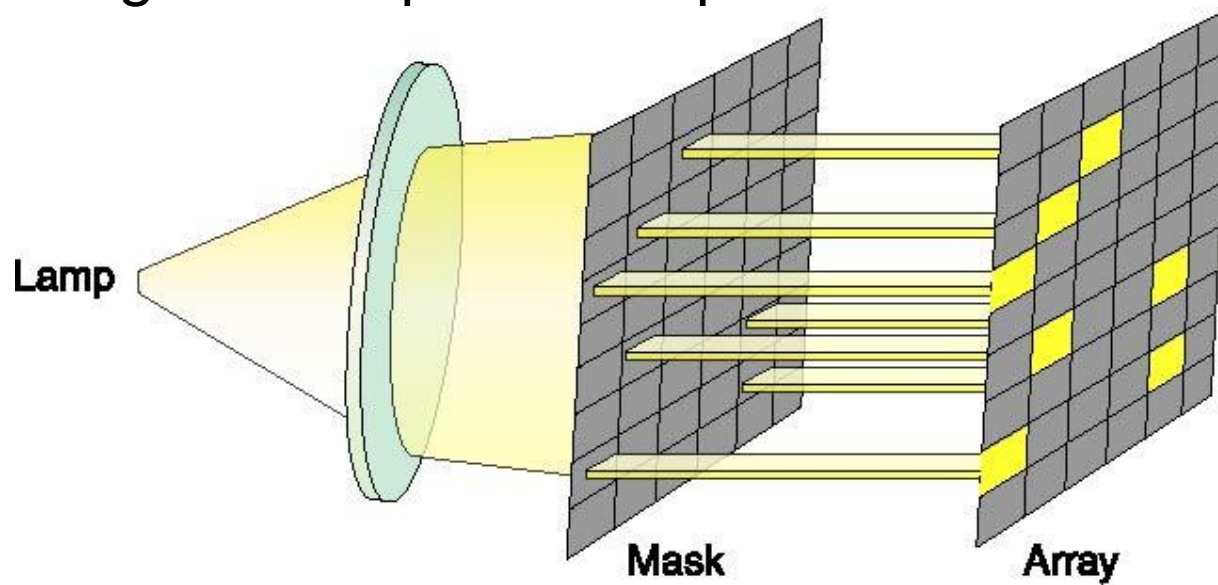
- ✓ The wafer is uniformly coated with a thick light-sensitive liquid called photoresist. The coating is applied while the wafer is spinning (1500-8000 rpm)
- ✓ The photoresist thickness is in the range: $0.5\text{-}2\mu\text{m}$. Thickness uniformity of $\sim 5\text{nm}$ is required.



- ✓ After spin coating, resist contains up to 15% organic solvent. This is removed by soft-baking at $75\text{-}100^{\circ}\text{C}$ for approximately 10 mins. This step also releases stress.

- Exposure to UV

- ✓ Parts of the wafer are selected for exposure by carefully aligning a mask between an ultraviolet light source and the wafer.
- ✓ In the transparent areas of the mask, light passes through and exposes the photoresist.



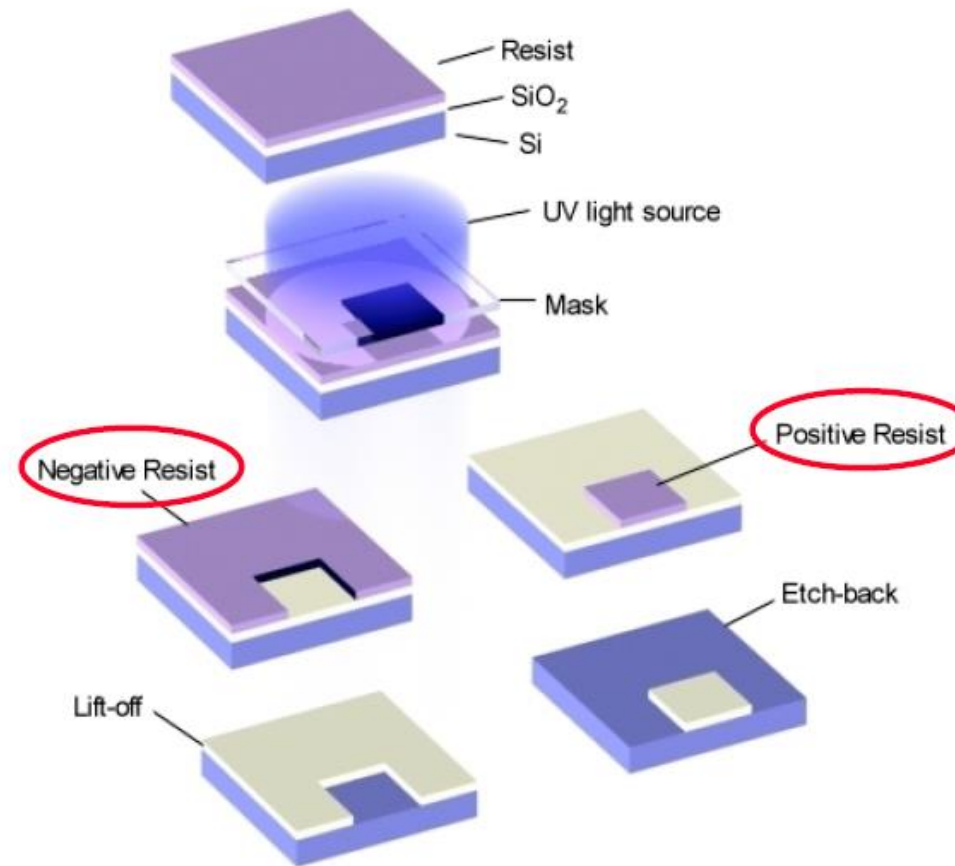
There are two types of photoresist:

- **negative tone**

UV light causes the negative resist to become polymerized, and more difficult to dissolve. With photosensitive groups such as epoxy, vinyl, or aryl halides on the photoresist backbone.

- **positive tone**

UV light changes the chemical structure of the resist so that it becomes more soluble in the developer.

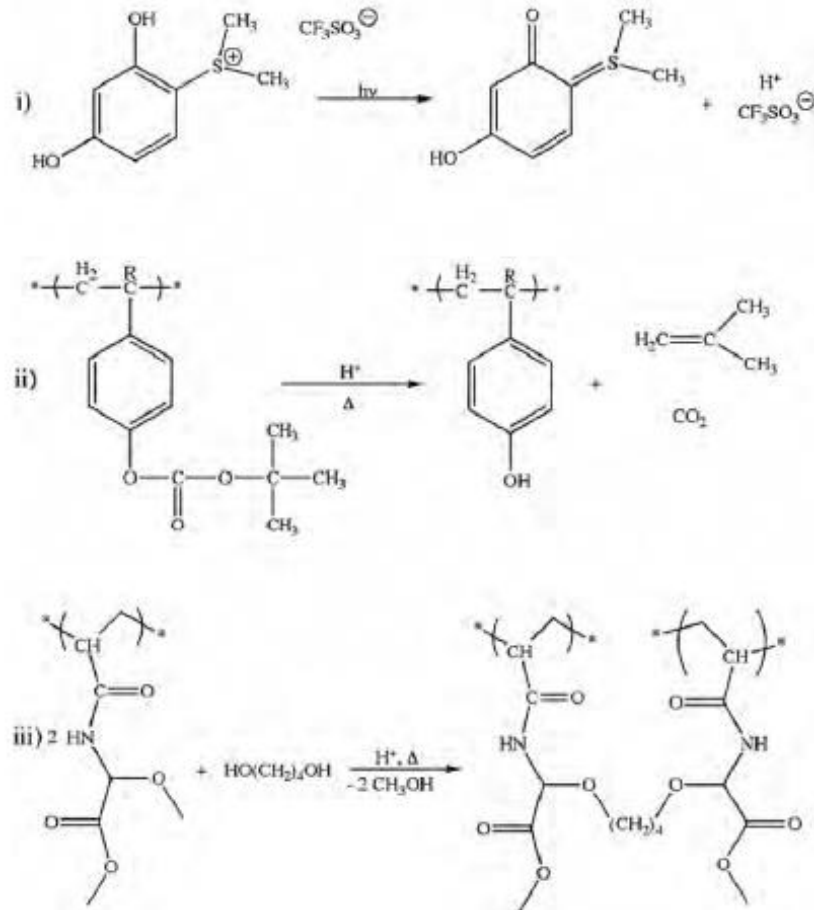




- CAM

In order to improve the line resolution of photoresists, a technique known as chemical amplification (CAM) is often used for photolithographic applications.

The enhancement of quantum efficiency and sensitivity results from a photo-generated acid that will either catalyze a crosslinking (negative photoresists), or de-protection (positive photoresists) mechanism.



CAM reactions are illustrated in (i)–(iii):

(i) represents photo-induced acid generation;

(ii) is an acid-catalyzed de-protection mechanism (positive tone resist);

(iii) is an acid-catalyzed crosslinking mechanism (negative tone resist).



- Etching

Etching is used to remove exposed underlying layer(s) from the surface of a wafer during manufacturing.

“Wet” etching

- Dipping the wafer in an acidic solution.
- Resulting in isotropic etching and a line resolution of $>2\text{ }\mu\text{m}$.



“Dry” etching

- Using a plasma
- Resulting in anisotropic conditions and line resolutions $<0.25\text{ }\mu\text{m}$.



“Wet” etching

-- HNO_3 , HF , and CH_3COOH (or KOH solutions).

“Dry” etching

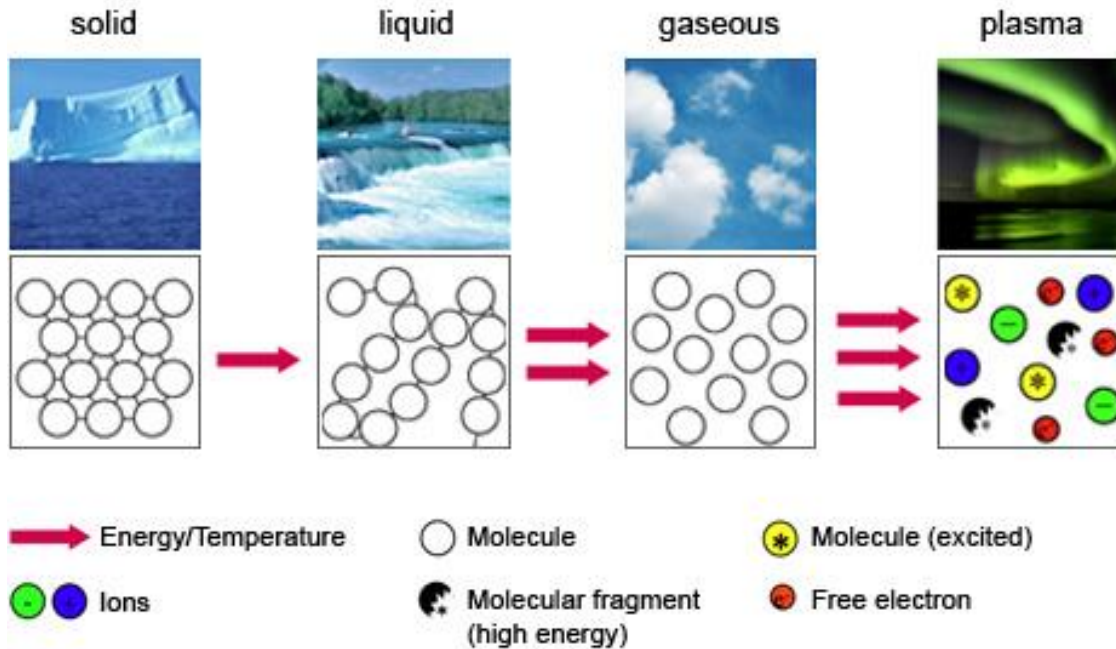
- For SiO_2
 - CF_4/H_2 and $\text{C}_2\text{F}_6/\text{CF}_3$
- For n-type Si
 - CF_4/O_2 , SF_6/O_2 , $\text{C}_2\text{F}_6/\text{O}_2$, $\text{Cl}_2/\text{C}_2\text{F}_6$;
- For undoped Si
 - Cl_2/Argon .
- For Si_3N_4
 - CF_4/H_2 or NF_3/Cl_2
- For the remaining photoresist
 - $\text{O}_2/\text{H}_2\text{O}$



Plasma

Plasma is one of the four fundamental states of matter, the others being solid, liquid, and gas. A plasma has properties unlike those of the other states.

A plasma can be created by heating a gas or subjecting it to a strong electromagnetic field applied with a laser or microwave generator. This decreases or increases the number of electrons, creating positive or negative charged particles called ions.



light (corona) around the sun

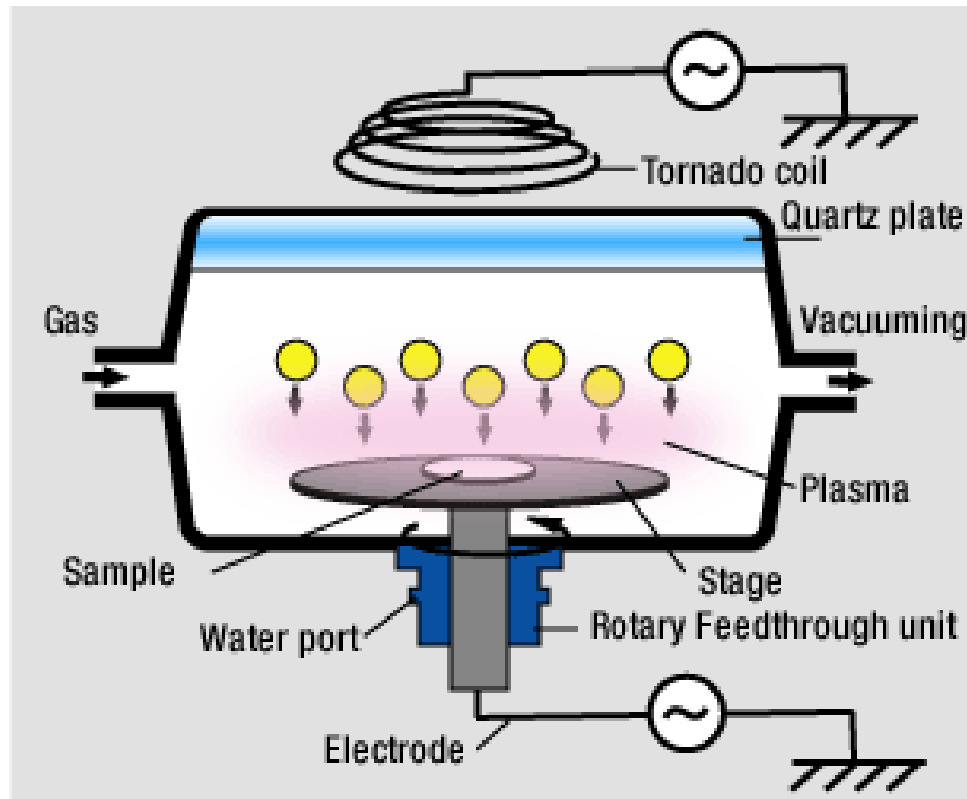


thunder



polar light in the Arctic and Antarctic

Plasma etching



Etching resolution

$$R = \frac{0.61\lambda}{NA}$$

R: the resolution limit (line spacing capable of being solved);

λ : the wavelength of exposure

NA: the numerical aperture of the exposure tool (the light-gathering power of a lens).数值孔径



Figure 1

In current semiconductor fabrication, photolithography with 193 nm UV irradiation is able to pattern features that are 37 nm wide. However, the rapid miniaturization of ICs demands improvements in photolithography in order to sharpen the line resolution at size regimes far below 37 nm.



Decreasing the wavelength of exposure

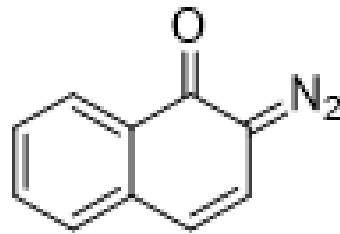
Table 4.1. Resolution Limits for Various Lithographic Techniques

Lithographic technique	Exposure wavelength (nm)	Resolution
Photolithography ^a mid-UV (MUV)	350–450	0.35–3 μm
Photolithography deep-UV (DUV)	248	$\geq 0.25 \mu\text{m}$
Photolithography deep-UV (DUV)	193	$< 70 \text{ nm}^{\text{b}}$
Photolithography extreme-UV (EUV)	13.5	$< 65 \text{ nm}$
Electron-beam lithography ^c	<i>ca.</i> 1	$< 50 \text{ nm}$
X-ray lithography ^d	0.4–20	$< 50 \text{ nm}$

- The best opportunity to image small features is the use of extreme UV light (EUV), which is actually in the range of soft X-rays (*ca.* $\lambda = 13.5 \text{ nm}$). A decrease in exposure wavelength translates to new types of photoresists that will be stable upon contact with higher energy radiation.
- The DNQ photoresist system is effective in the MUV range;
- Photoresists that contain CAM moieties are used exclusively for DUV and shorter wavelengths.

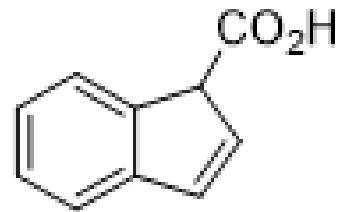


DNQ photoresist system



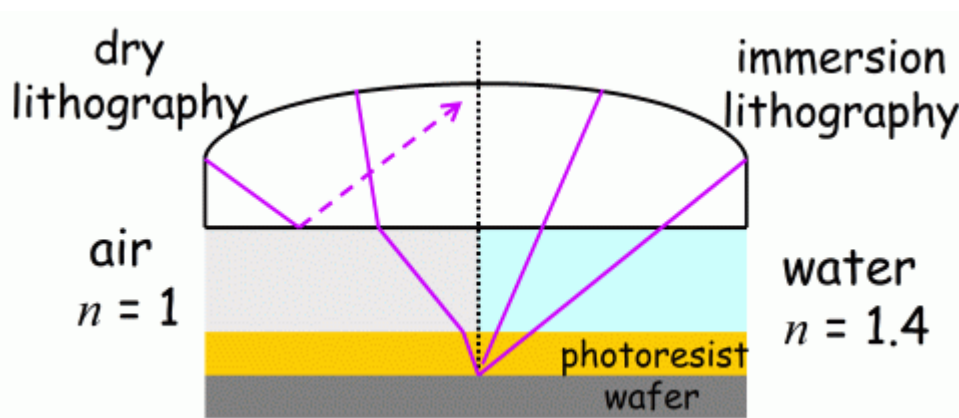
Dizaonaphthoquinone

重氮萘醌



Indene carboxylic acid

Immersion Lithography



$$\lambda = \lambda' / n$$

$$R = \frac{0.61\lambda}{NA} = \frac{0.61\lambda'}{NA \times n}$$

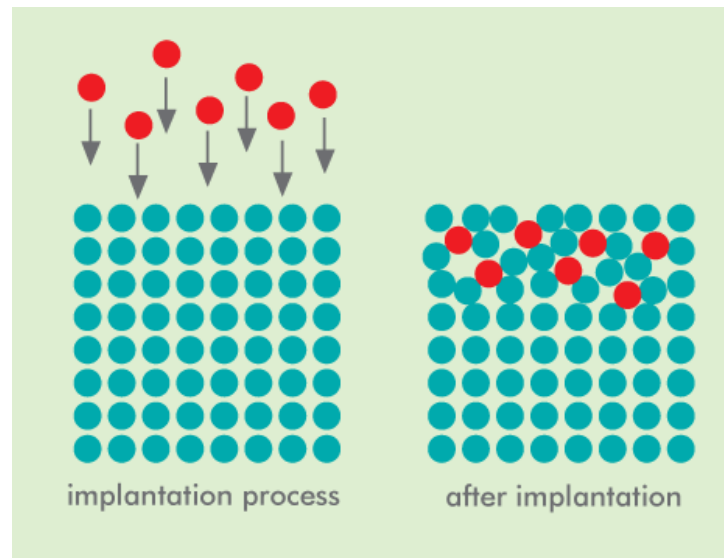
This technique consists of surrounding the lithographic lens with a fluid of greater refractive index (*e.g.*, water: $n = 1.47$ at 193 nm; air: $n = 1.0$). Since the refractive index of the medium is directly proportional to the aperture (NA), an improvement is made to the imaging resolution.



□ Ion implantation

The placement of n- and p-type dopants onto the top of the silicon surface.

A technique known as ion implantation involves the acceleration of B (p-type) or Sb/As/P (n-type) ions to very high kinetic energies (ca. 50–200 keV), which then collide with the Si lattice.





- The chosen energy must be sufficient to penetrate through the SiO_2 layer and a desired depth of Si substrate, but not the exposed photoresist layer that defines the dopant region.
- For the same accelerating voltage, ions of smaller mass will penetrate to deeper/broader regions of the solid relative to heavier ions, due to differences in their kinetic energies and concomitant collisions with Si atoms.



- An energy on the order of 15 eV is enough to dislodge a Si atom from its crystalline lattice, forming a Frenkel defect.
- However, the energies of the impinging dopant ions are an order of magnitude larger, which will cause significant damage to the Si crystal structure.
 - For instance, bombardment of a Si surface with a single arsenic ion with an energy of 30 keV will displace 1,000 Si atoms – all on a timescale of *ca.* 10^{-13} s! Multiply this by thousands of bombarding ions during the implantation process, and one has a picture of a highly perturbed Si crystal.
- Thermal annealing is able to cause the dislodged Si atoms to find vacant sites, removing the Frenckel defects and thus repair the damage.
 - Prolonged high-temperature annealing at temperatures of 1,000–1,100°C is subsequently used to further restore the periodicity of the Si crystal lattice.



Gate electrode deposition

the formation of the gate electrodes onto the nMOS and pMOS devices. This consists of the deposition of a *ca.* 500 nm thick polysilicon film onto the wafer surface, and subsequent photolithography/etching steps.

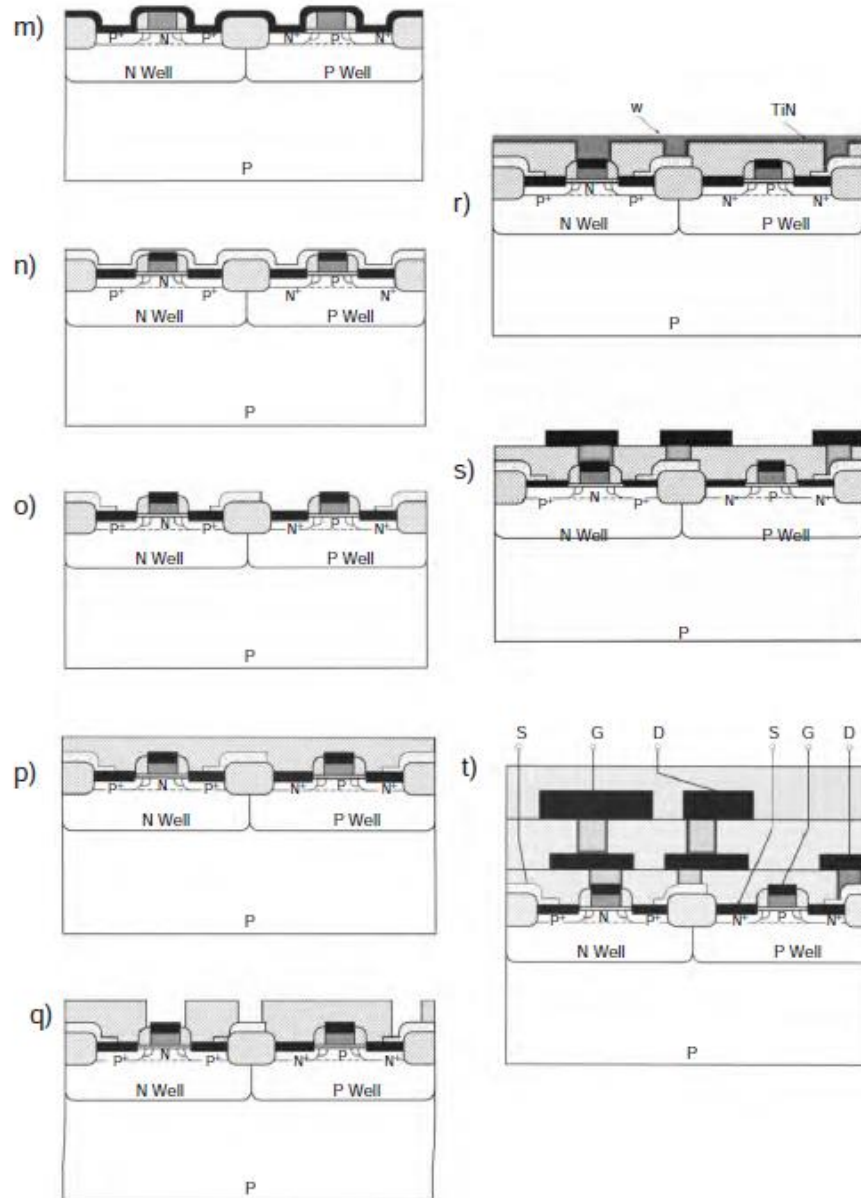


metalation

The selective deposition of metals that form the interconnection between the active devices on the IC.

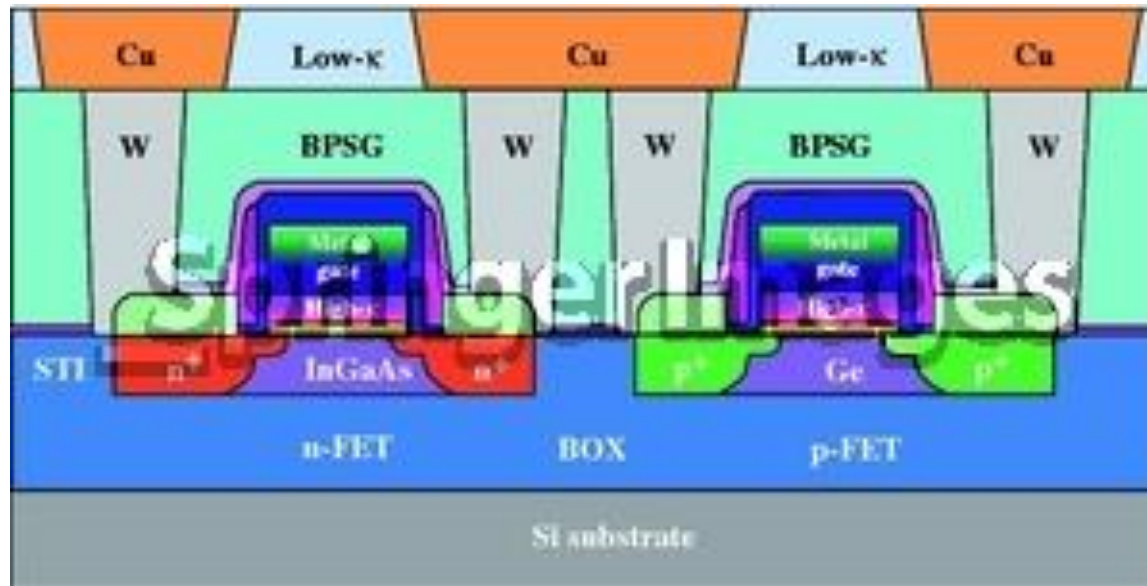
For example, the deposition of titanium onto the IC surface, followed by annealing at a temperature of 600°C under a nitrogen environment.

- Although Ti will readily react with the ambient N_2 to form TiN, titanium will also readily react with Si atoms to form $TiSi_2$ in the regions where a Ti/Si interface exists (at the top of the polysilicon gate, and source/drain cavities).
- The dark regions in step n indicate the regions of $TiSi_2$, with TiN formed in all other areas.
- Metal silicides are used to reduce the resistivity of the polysilicon gate; in addition to $TiSi_2$, other analogues such as WSi_2 , and $CoSi_2$ are frequently utilized.



A thick layer (*ca.* 1 μm) of insulating oxide is then deposited onto the evolving IC, which serves as the interconnect dielectric medium.

Low- κ dielectrics





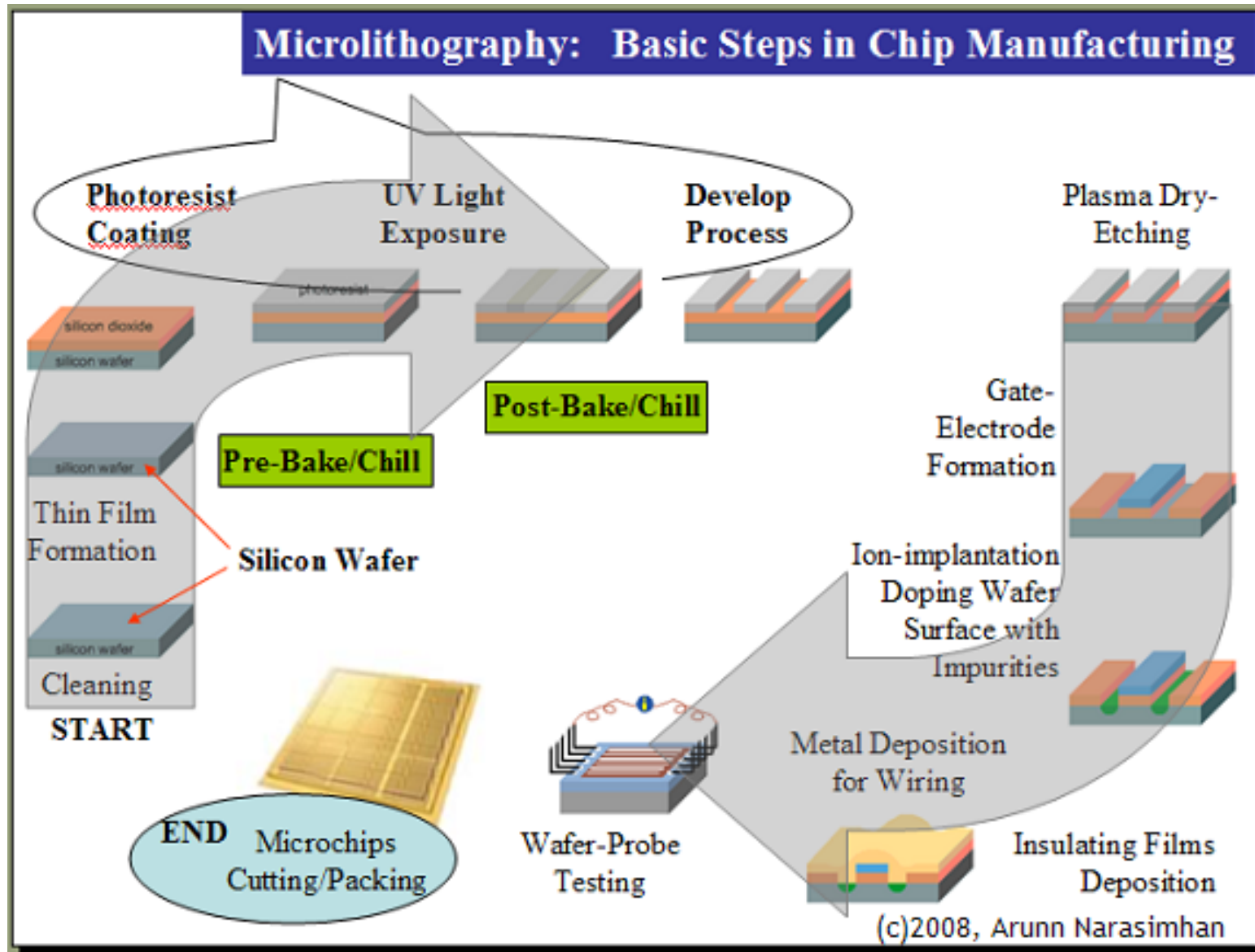
- In ICs, insulating dielectrics separate the conductive portions (*e.g.*, wire interconnects and individual transistors) from one another. In order to fabricate chips with higher speeds, the transistors must be placed closer and closer together, thus resulting in a thinner insulating layer. This leads to charge buildup and crosstalk, which adversely affects the maximum operating speed and performance of the chip.
- The use of low- κ dielectrics is important to reduce the parasitic capacitance, hence enabling faster switching speeds and lower heat dissipation.
 - Doping SiO_2 with fluorine reduces the κ from 3.9 to 3.5;
 - Other approaches that yield values of $\kappa < 1.5$ involve the use of nanoporous dielectrics based on poly(amidoamine-organosilicon) (PAMAMOS) films.



The final steps involve deposition of the interconnect metal.

Copper is now the metal-of-choice due to its more desirable electrical resistivity.

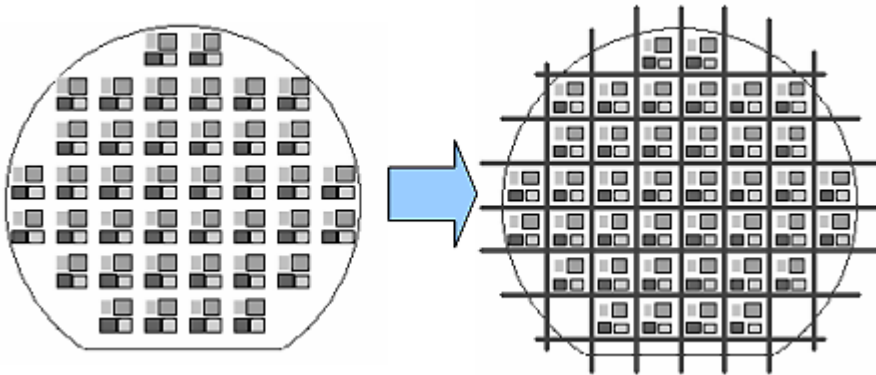
Due to its low resistivity and high density, titanium nitride is an efficient barrier level that prevents surface oxidation of Cu, as well as the inter diffusion of Cu into adjacent layers.



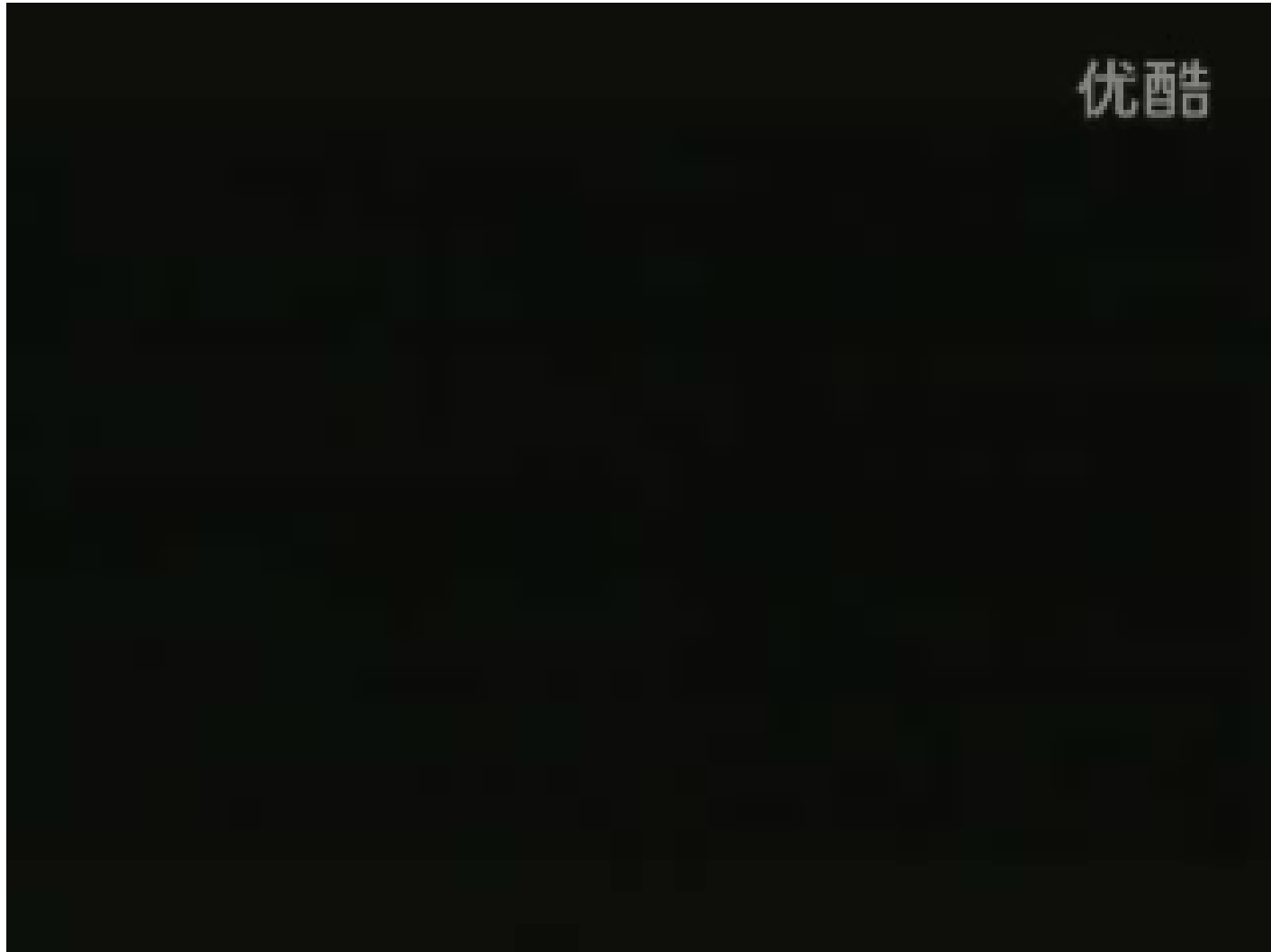


These multistep processes were not performed from start to finish, one IC at a time.

Typically, integrated circuits are produced in large batches on a single wafer. The wafer is cut (“diced”) into many pieces, each containing one copy of the circuit. Each of these pieces is called a die.

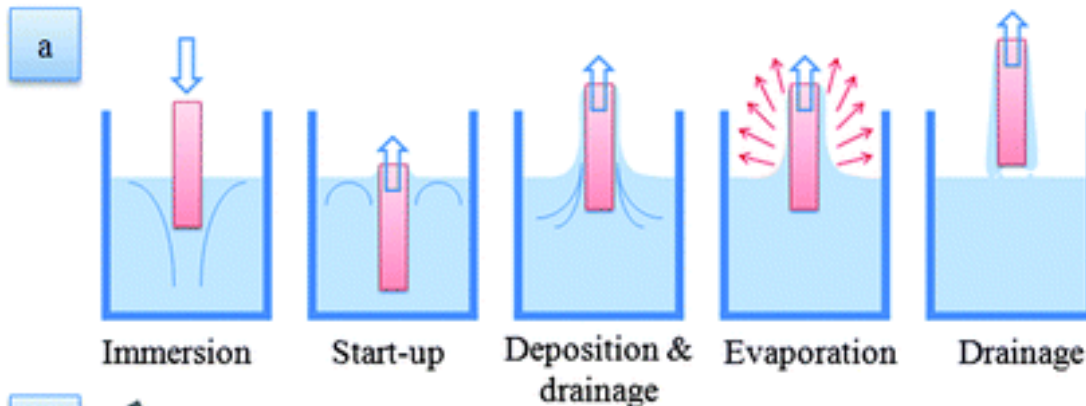


Significant price drop!

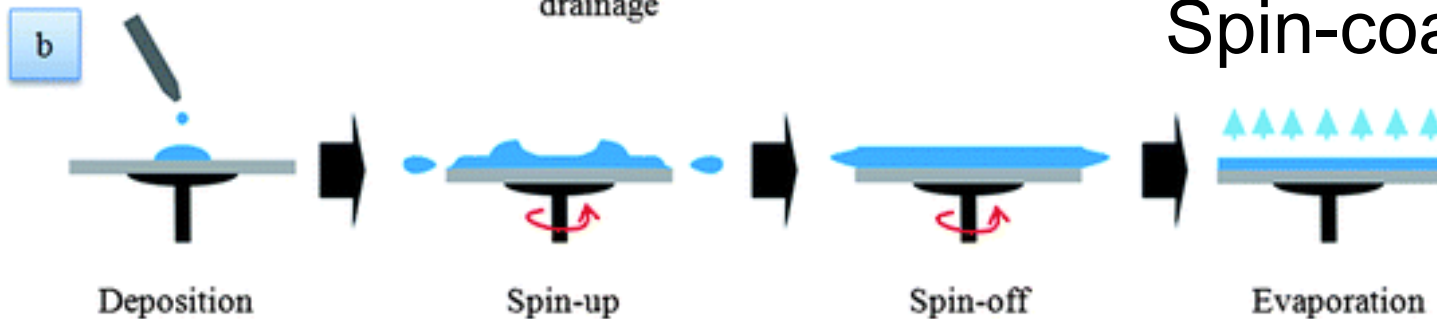




Thin-film deposition methodologies



Dip-coating



Spin-coating

It is possible to control the film thickness during spin-coating through varying the solvent, spin rate, drop height, *etc.*



Human hair
 $50 \mu m$

How to control the film thickness $< 1 \mu m$?

Bottom up method:

Vapor deposition techniques ($\text{\AA} \text{ min}^{-1}$)

- Physical vapor deposition (PVD)
 - evaporation or sputtering
- Chemical vapor deposition (CVD)
 - involves a chemical reaction



- CVD
 - Reactive gases interact with substrate
 - Used to deposit Si and dielectrics
 - Good film quality
 - Good step coverage
- PVD
 - Used to deposit metals
 - High purity
 - Line of sight

PVD

- **Evaporation**

PVD may occur through the evaporation of atoms/molecules from a precursor solid in vacuo.

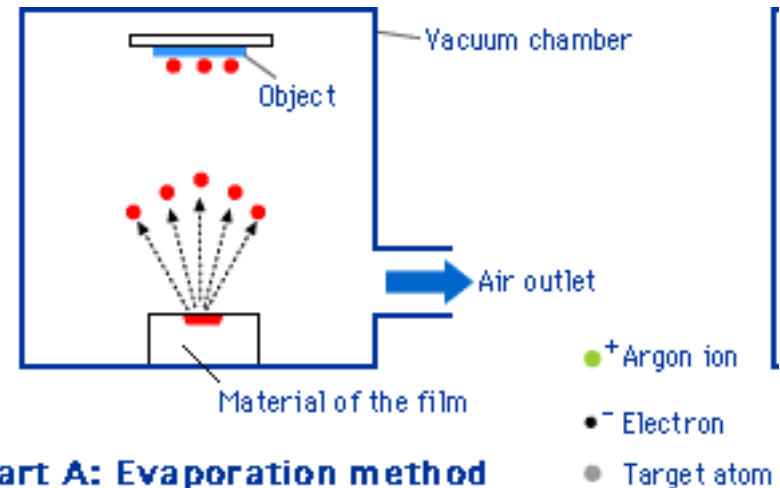


Chart A: Evaporation method

- **Sputtering**

PVD may also occur through use of a high-energy Ar plasma source that causes the vaporization of atoms from a solid target.

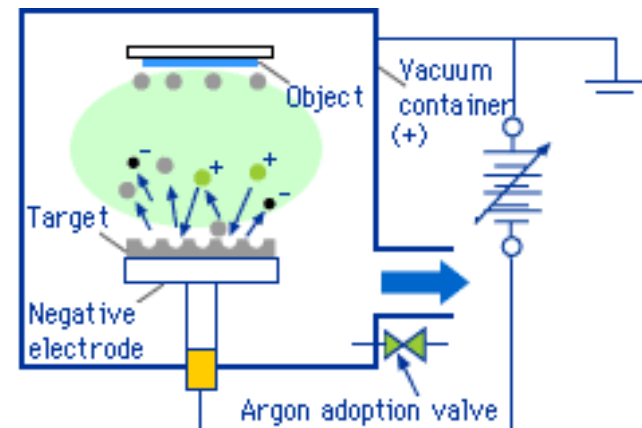


Chart B: Sputtering method



- Since all atoms of the solid in both techniques will enter the gas phase, the resultant film is only as pure as the solid precursor that is used.
- As a result, high-purity precursors must be used for PVD.
 - For example, a piece of gold foil of purity >99.99999% is typically used as a target for Au sputtering.



Two methods for the evaporation of precursors:

- Resistance heating

This method employs a simple alumina crucible that is heated by a W filament.

Temperatures as high as 1,800°C may be reached inside the chamber, which is enough for some metals or metal salts to vaporize.

Deposition rates for this method are 1–20 Å s⁻¹.



Two methods for the evaporation of precursors:

- **Electron beam collision**

The use of an electron beam to assist in the precursor evaporation results in temperatures on the order of $3,000^{\circ}\text{C}$, being more suited for the deposition of refractory metals/alloys and metal oxides such as alumina, titania, and zirconia.

Since the temperature of the chamber interior is much higher than the walls, the gas-phase ions/ atoms/ molecules condense on the sidewalls as well as the substrate; this may lead to film contamination as the nonselective coating flakes off the chamber walls.



The properties of films deposited by PVD

- Hard with a high atomic density due to slow and efficient nucleation/growth;
- The thickness of PVD coatings ranges from a few angstroms to $>30\text{ }\mu\text{m}$;
- Since the substrate is maintained at room temperature, there are no limitations related to the thermal stability of the substrate.
 - This is an important consideration for coating materials that will decompose at high temperatures such as plastics, paper, *etc.*



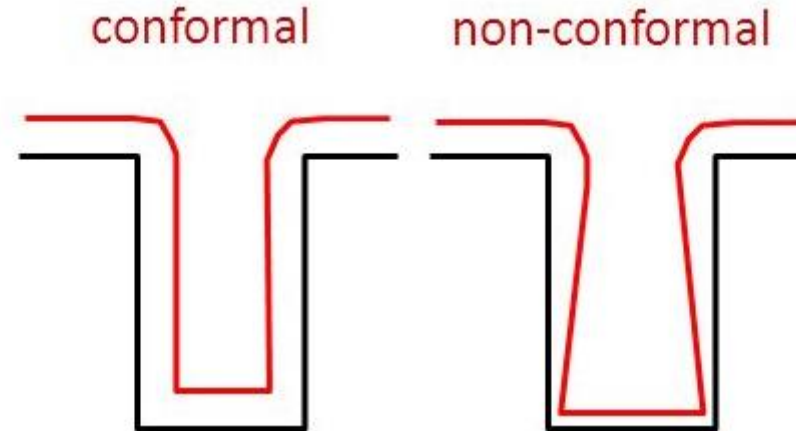
The limitations of PVD

- For high-purity metal or carbon films, an ultrahigh vacuum (UHV) environment (typically $< 1 \times 10^{-6}$ Torr) must be used during PVD.

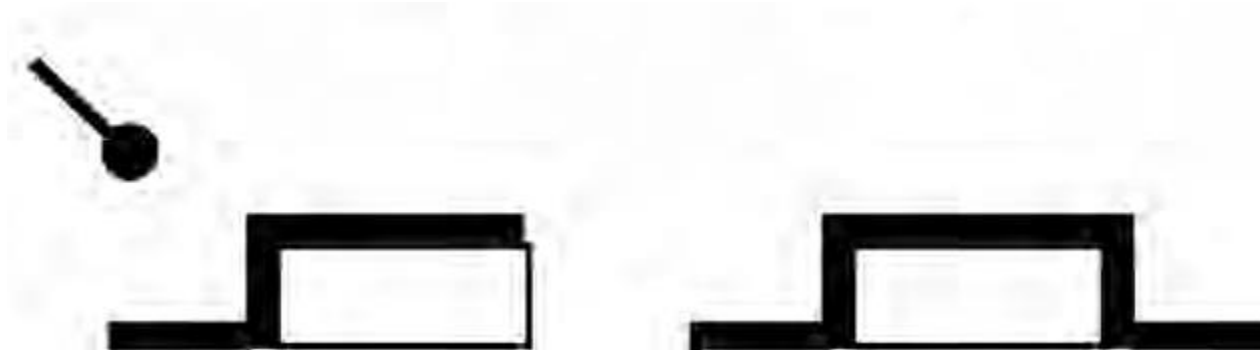
This is necessary in order to prevent the gas-phase reaction of metal/carbon atoms with atmospheric gases (*e.g.*, H_2O , O_2 , N_2) that would preferentially form metal oxides, hydroxides, or nitrides rather than the desired film.

- Since such a high-purity metal and high vacuum chamber must be used, PVD is relatively quite expensive.
- Limited selectivity and conformality.

While PVD works well to deposit material on surfaces in line-of-sight of the source, non-conformal deposition of complex or rough surfaces (*e.g.*, fibers) is a critical limitation. This issue becomes more important for integrated circuits with feature sizes less than 100 nm, often with high *aspect ratios* (height/width ratio of the surface feature).



line-of-sight





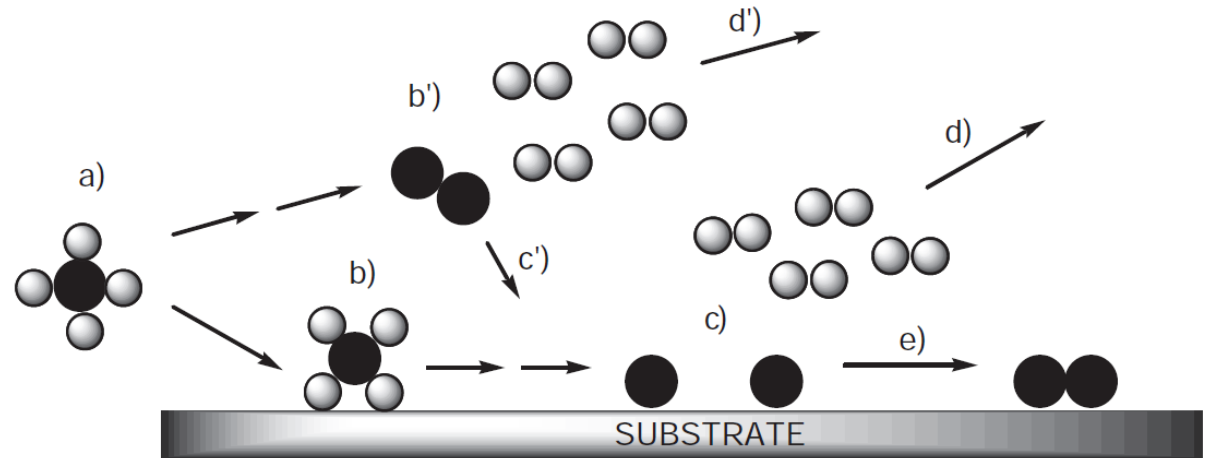
CVD

- CVD is a process in which gaseous precursors are reactively transformed into a thin film, coating or other solid-state material on the surface of a catalyst or substrate.
- It should be stressed that CVD is no longer limited to thin film growth; this method is now the preferred route to generating fiber-optic preforms, as well as an increasingly diverse nature of nano-structural architectures

Metal-organic CVD (MOCVD)

The use of an organometallic precursor, containing a central metal and ancillary ligands.

MOCVD



- (a) The gas-phase precursor molecules enter the deposition zone;
- (b) They are physisorbed to the substrate surface through weak van der Waals interactions;
- (c) The ancillary ligands are removed through thermolysis, leaving the desired residual species on the surface;
- (d) Ligands being removed from the deposition chamber *via* the carrier gas flow;
- (e) Strong covalent interactions are then formed between the surface and adsorbed species, resulting in chemisorption. The surface-bound species may migrate along the surface and/or react with other surface species, eventually nucleating on thermodynamically favored positions *en route* toward thin-film growth.



Advantages:

- The simplicity of setup;
- Being able to maintain a uniform temperature over a large number of substrates.

Limits:

- Secondary coating of the reactor walls is unavoidable;
- Frequent cleaning of these reactors is therefore necessary, as deposits may easily flake off these surfaces and contaminate the growing film.

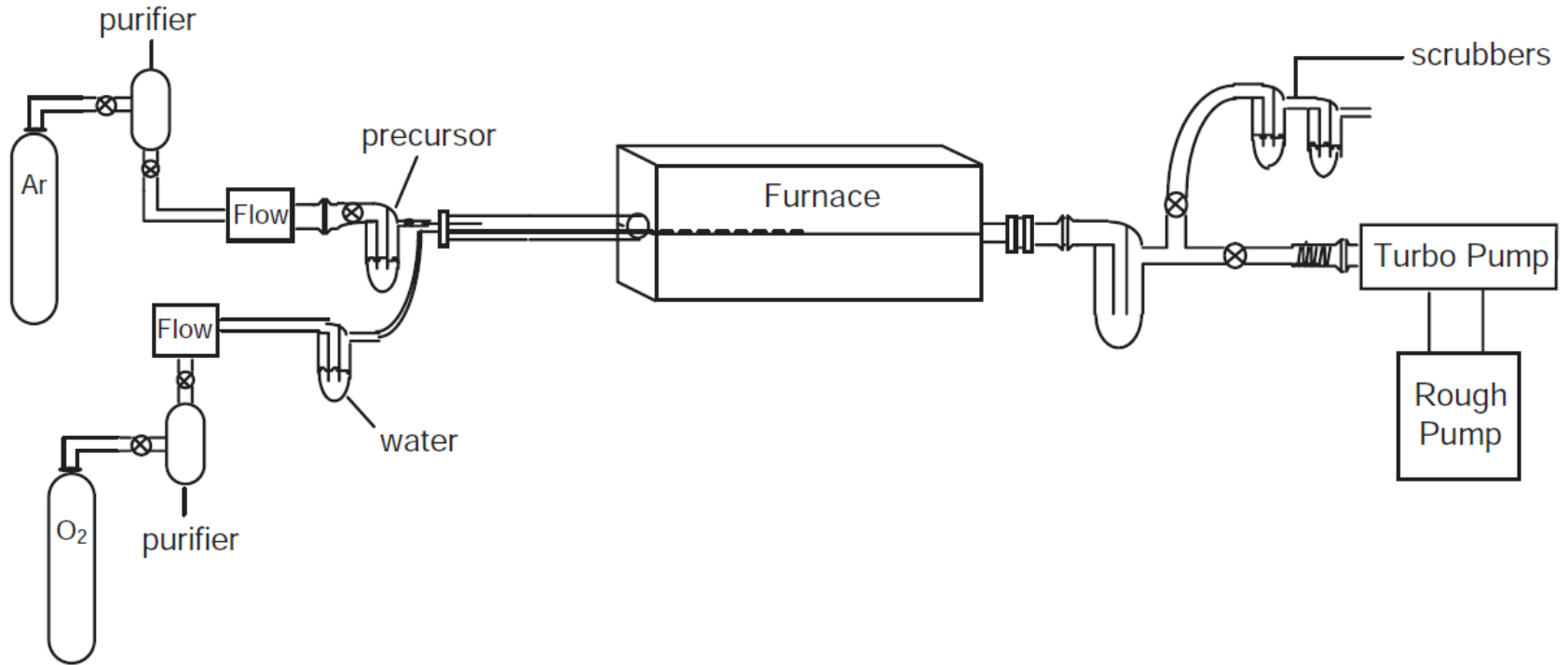


Reactors for CVD

- Hot-walled reactor

A tube furnace completely surrounds the deposition chamber containing the substrate.

The desired deposition temperature depends on the thermal stabilities of the substrate to be coated, and the reactor (i.e., glass ($T < 600^{\circ}\text{C}$) or quartz ($T < 1,100^{\circ}\text{C}$)).



Schematic of a horizontal hot-walled CVD reactor. Shown is a two-precursor system, where the water sensitive precursor contacts water vapor directly over the heated substrate.



- Cold-walled reactor
 - ✓ In cold-walled reactors, only the substrate is heated.
 - ✓ This focuses the surface and gas-phase reactions to a region immediately surrounding the substrate surface.
 - ✓ However, this reactor type generally results in narrower deposition zones and slower growth rates than hot-walled analogues.



If no prefix is affixed to CVD

Usually denotes simple thermolysis of a precursor within a standard cold- or hot-wall reactor.

If more energetic sources of energy are used to degrade the precursor:

- laser-assisted CVD (LACVD),
- plasma-enhanced CVD (PECVD),
- microwave plasma CVD (MPCVD)
- catalyzed CVD (Cat-CVD)
- hot-wire CVD (HWCVD)



CVD precursors

- low-melting powders with high volatilities and low decomposition temperatures are regarded as the best candidates for CVD, as they are often less susceptible to premature hydrolysis during transfer to the CVD system.
- Co-reactant gases such as NH_3 , H_2 , H_2S , H_2O , *etc.* are often added in order to assist the removal of organic moieties from the organometallic precursor molecule, preventing their incorporation in the growing film.



The volatility of the precursor is directly related to its molecular structure.

- large molecular weights and/or extensive intermolecular forces will significantly decrease its volatility.
- An effective method to enhance precursor volatility is to incorporate partially and fully fluorinated ligands.
- The enhancement may be rationalized either by an increased amount of intermolecular repulsion due to the additional lone pairs or that the reduced polarizability of fluorine (relative to hydrogen) causes fluorinated ligands to have less intermolecular attractive interactions.



Although it was once essential that volatile precursors be used, this is no longer a synthetic limitation.

Supercritical fluids (*e.g.*, CO₂) have been used to solvate certain precursors, facilitating their use for CVD.

Two variations of this technique may be used:

- Supercritical fluid transport (SFT)

Using the fluid as an aerosol-like delivery vehicle.

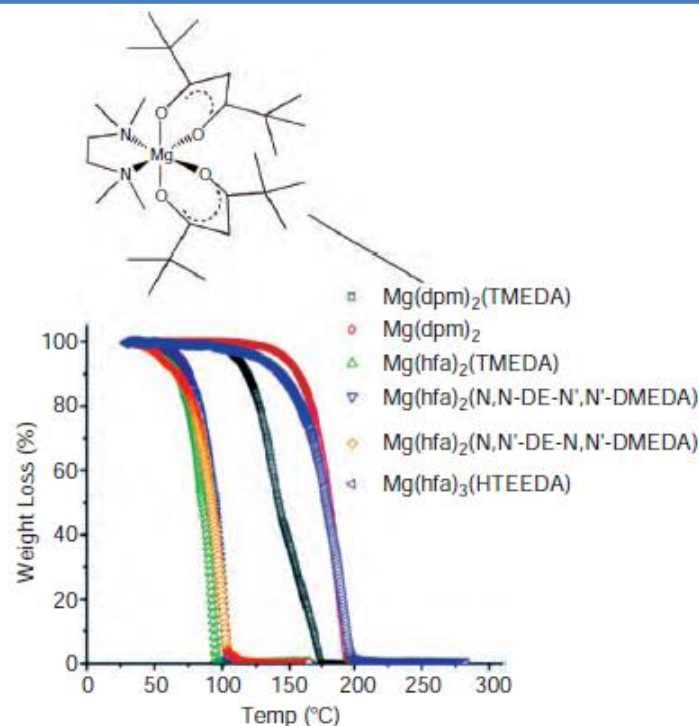
- Supercritical fluid deposition (SFD)

in situ Thin film growth within a high pressure reactor.



Prior to CVD studies, it is essential that the potential precursor undergo **thermogravimetric analysis (TGA)** to determine the most suitable temperature regime for thin film Growth.

A sharp TGA curve with no remaining residue indicates that the precursor vaporizes without significant ligand decomposition – most desirable for CVD applications.





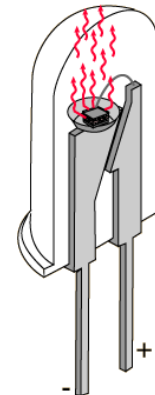
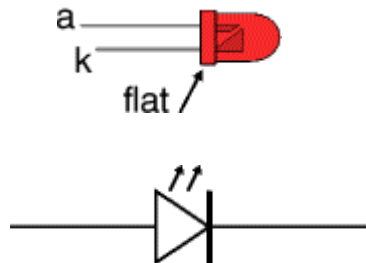
Semiconductor II-VI: ZnS, ZnSe
Semiconductor III-V: GaAs, GaN, GaP

Light-emitting diodes (LEDs)
Solid-state lasers¹⁵⁶



A light emitting diode (LED) is essentially a PN junction opto-semiconductor that emits a monochromatic (single color) light when operated in a forward biased direction.

LEDs convert electrical energy into light energy. They are frequently used as "pilot" lights in electronic appliances to indicate whether the circuit is closed or not.





LEDs
From display to lighting





Once an inexpensive white LED becomes commercially available, our society will forever change as we shift from our longstanding reliance on inefficient and short-lived incandescent bulbs.

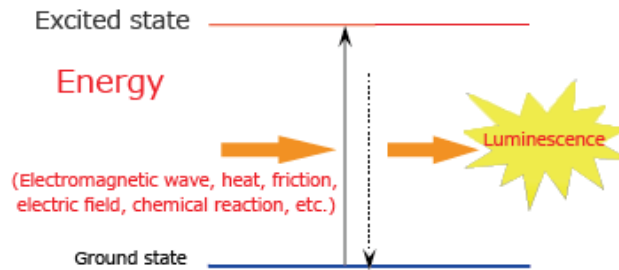


Fig. 1 Luminescence

Luminescence

Thermoluminescence

Electroluminescence

Chemoluminescence

Photoluminescence

Incandescence – the generation of light from a material as a result of its high temperature.

LEDs



Fluorescence
phosphorescence



Bandgap Engineering

Colour	Wavelength (nm)	Band Gap Energy, E (eV)
Infrared	$\lambda > 760$	$E < 1.63$
Red	$610 < \lambda < 760$	$1.63 < E < 2.03$
Orange	$590 < \lambda < 610$	$2.03 < E < 2.10$
Yellow	$570 < \lambda < 590$	$2.10 < E < 2.18$
Green	$500 < \lambda < 570$	$2.18 < E < 2.48$
Blue	$450 < \lambda < 500$	$2.48 < E < 2.76$
Violet	$400 < \lambda < 450$	$2.76 < E < 3.10$
Ultraviolet	$\lambda < 400$	$3.1 < E$

$$E = hc/\lambda$$

$$\lambda(\mu\text{m}) = \frac{1.240}{E_g(\text{eV})}$$

GaAs

$$E_g = 1.35 \text{ eV}$$

GaP

$$E_g = 2.35 \text{ eV}$$

$\text{GaAs}_x\text{P}_{1-x}$

$$E_g = 1.4\text{-}2.3 \text{ eV}$$

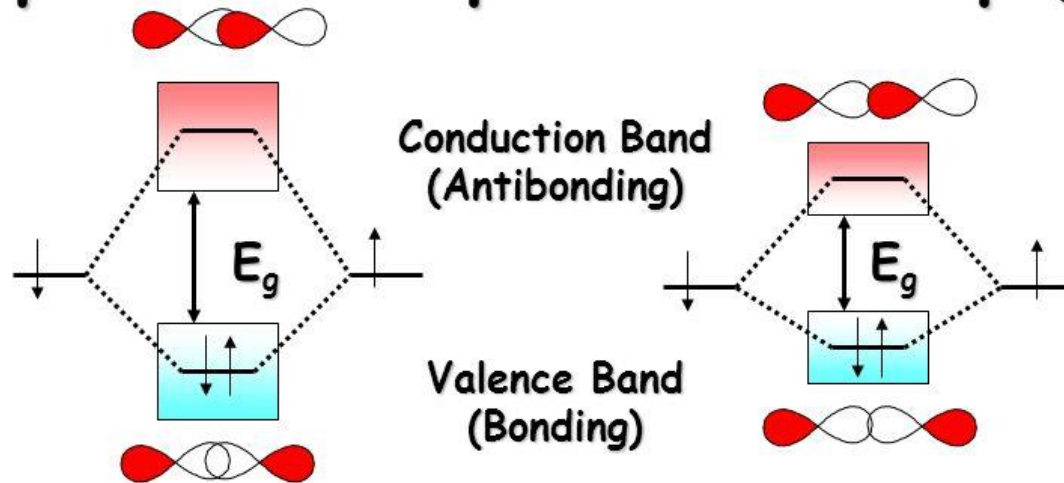


Bandgap $\propto$ the bond strength between atoms in the lattice

As the bond strength increases, there is more efficient overlap between molecular orbitals that gives rise to a larger bandgap between bonding and antibonding MOs (*i.e.*, valence and conduction bands of the infinite lattice, respectively).

	Bond strength (kcal mol ⁻¹)	Bandgap (eV)	Wavelength (nm)
GaN	98.9	3.50	~350
GaAs	50.1	1.35	~900

Spatial Overlap and Band Gap (E_g)

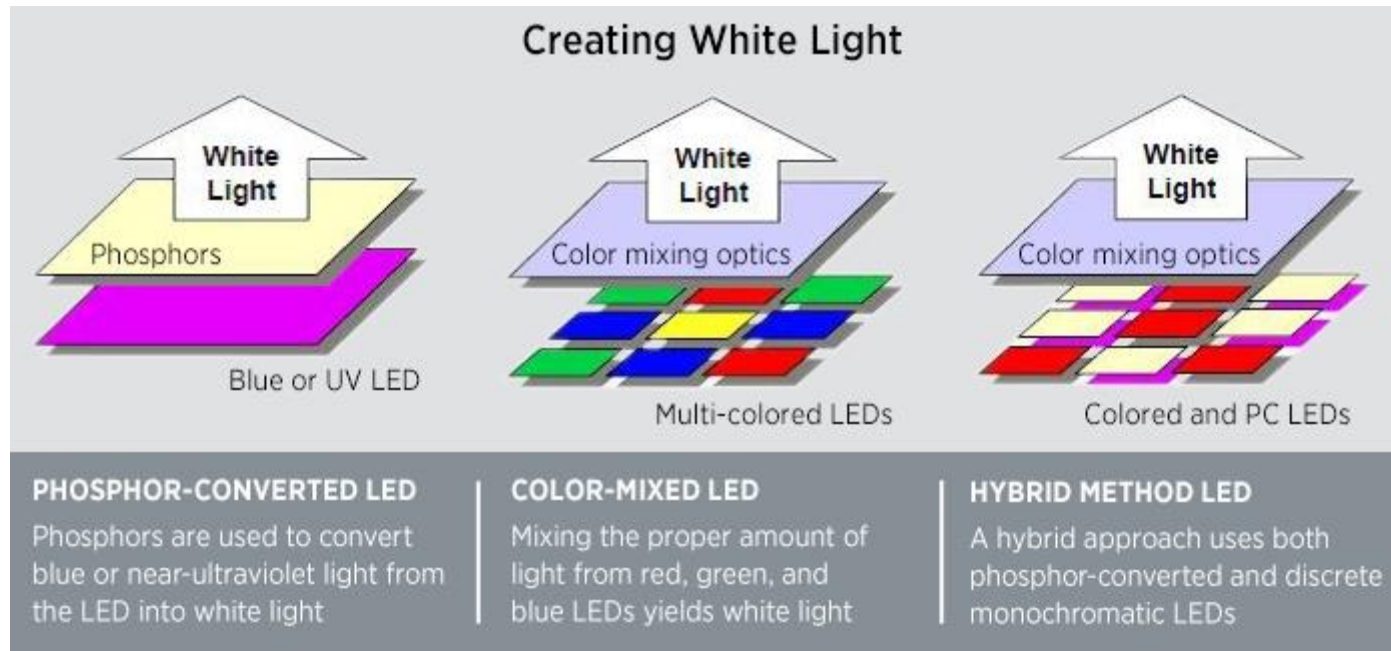


What are the effects of increasing the spatial overlap? Van-Vechten & Phillips assumed spatial overlap is proportional to $d^{-2.5}$ (where d is the bond distance).

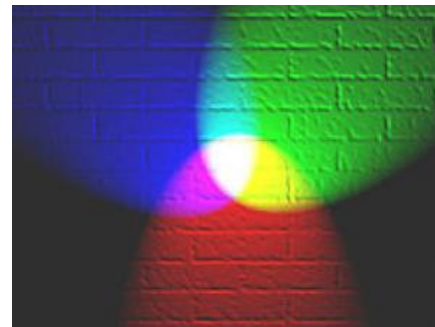
- **Primary Effect:** Increases the overall energy level of the conduction band (more highly antibonding) [$E_g \uparrow$]
- **Secondary Effect:** Increases the bandwidth [$E_g \downarrow$]



White emitting LEDs



RGB





White emitting LEDs--- Phosphor conversion

- White light-emitting diodes are usually blue InGaN LEDs with a coating of a suitable material.
- Cerium(III)-doped YAG (e.g., $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$) is often used; it absorbs the light from the blue LED and emits in a broad range from greenish to reddish, with most of output in yellow.
- This yellow emission combined with the remaining blue emission gives the “white” light, which can be adjusted to color temperature as warm (yellowish) or cold (blueish) white.



Organic light-emitting diodes (OLEDs)



- Flexible
- Light weight
- Printable

Function of an OLED

Kodak, 1987

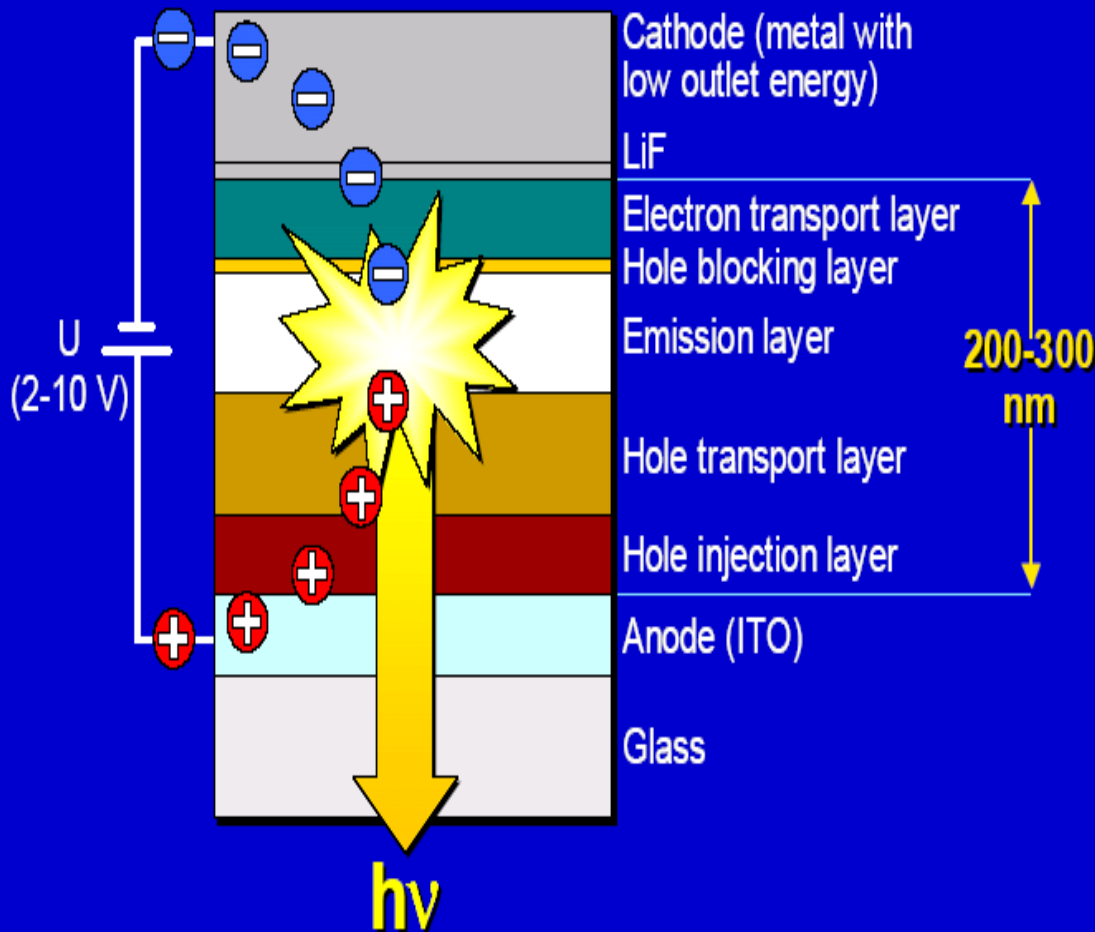
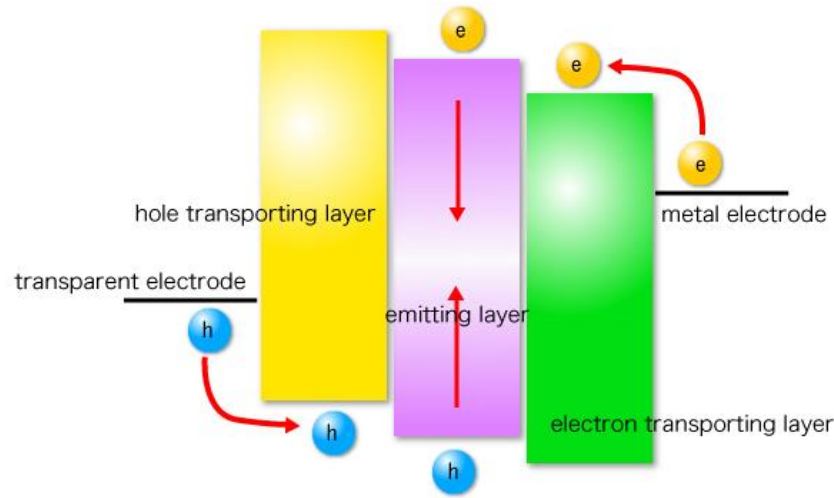


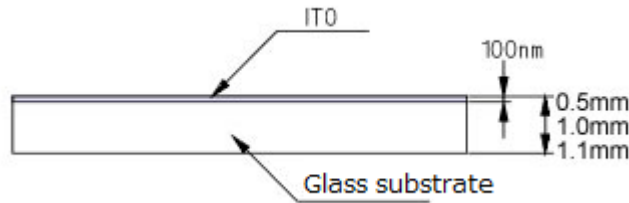
Fig. 3

OLED is a display device that sandwiches carbon based films between the two electrodes and when voltage is applied creates light.

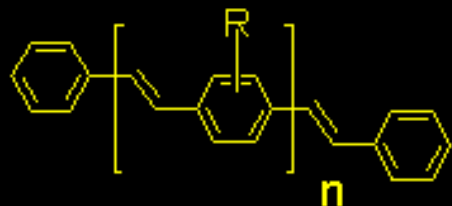
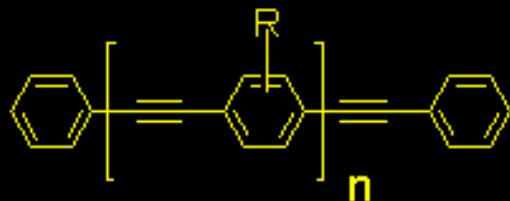
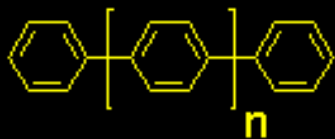
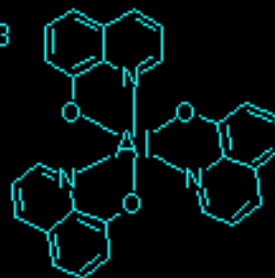


- The recombination of a hole and electron creates a quasiparticle known as an *exciton*, which releases a photon of energy.
- Organic molecules in the emissive layer facilitate exciton formation.
- A challenge in OLED design is to ensure that an equal number of holes and electrons meet in the emissive layer. This is not trivial, since holes migrate much slower than electrons in conductive organic molecules.

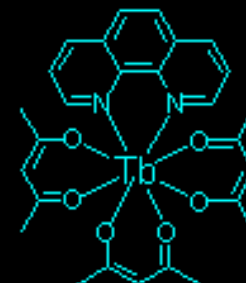
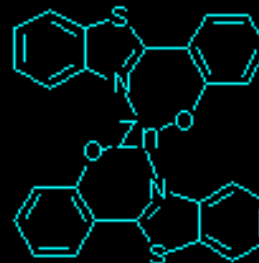
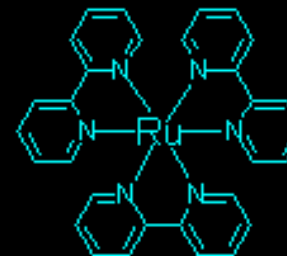
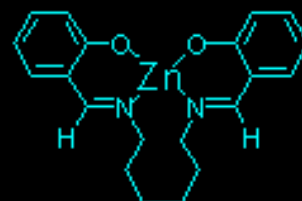
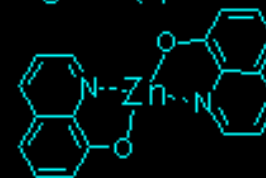
Anode: tin-doped In_2O_3 (ITO, typically 1:9 Sn: In)



- ITO thin film is transparent and highly conductive (*ca.* $1 \times 10^4 \text{ W}^{-1} \text{ cm}^{-1}$).
- The substitutional replacement of In^{3+} ions with Sn^{4+} ions results in n-doping of the lattice, injecting electrons into the conduction band.
- The doping of Sn into the In_2O_3 lattice may also result in SnO sites; this introduces holes in the lattice that reduces its conductivity.

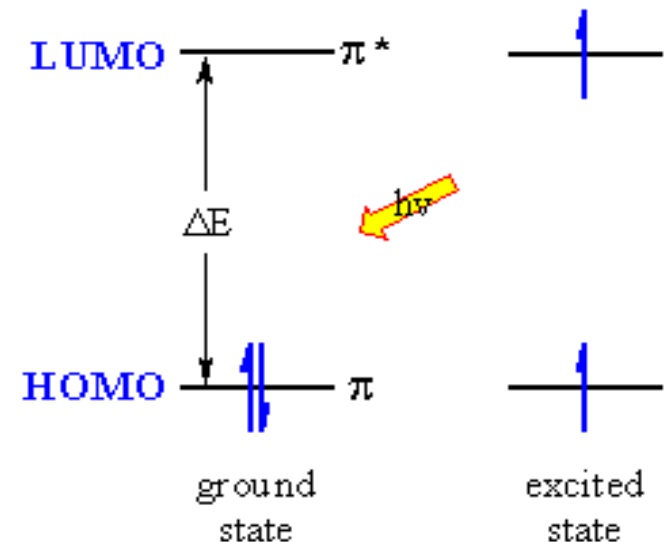
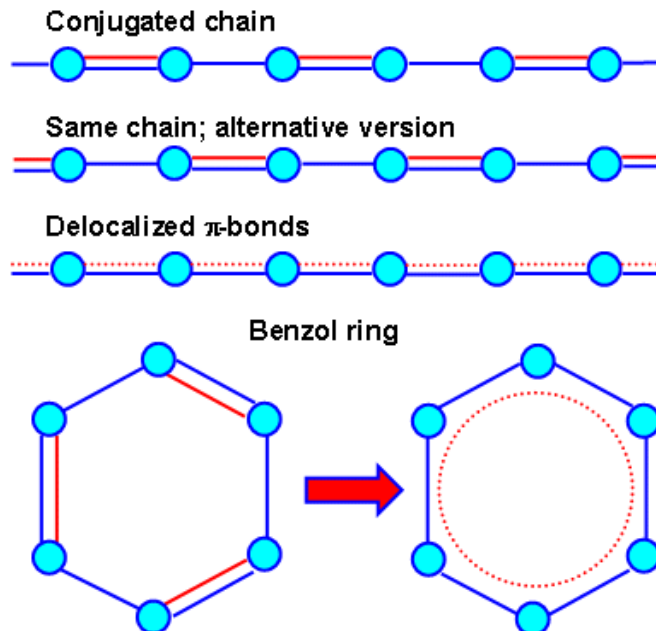
**Conjugated-polymers****A** poly(*p*-phenylenevinylene) = PPV**B** poly(*p*-phenyleneethynylene) = PPE**C** poly(arylene), e.g. poly(*p*-phenylene)**Small organometallic molecules**tris(quinolinolate)aluminum(III) MQ₂ : M = Be, Mg, Zn, etc.AlQ₃

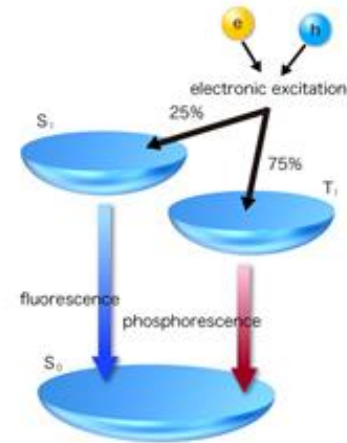
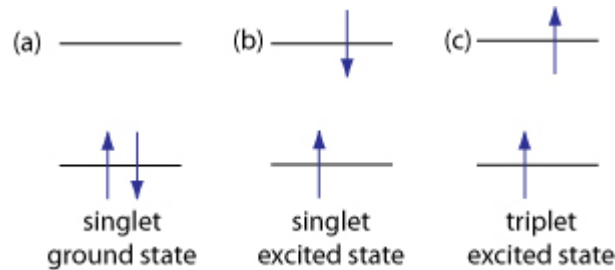
e.g. bis(quinolinolate)zinc(II)





- The organic molecules used in OLEDs/PLEDs are π -conjugated, meaning that p_z orbitals on adjacent $(-C=C-C=)_n$ units overlap, resulting in π valence and π^* conduction bands.
- The observed color will depend on the HOMO–LUMO gap of the molecule, which may be fine-tuned by altering the length of the conjugation, or nature of the molecular backbone (*e.g.*, substitution of electron withdrawing/donating groups, *etc.*).





- A limitation of traditional OLEDs/PLEDs is their relatively poor quantum efficiency.
- An exciton may exist in either a singlet (total spin, $S = 0$) or triplet state ($S = 1$), with the triplet outweighing the singlet by a 3:1 ratio.
- The molecules in the emissive layer are referred to as fluorophores, which yield fluorescent radiation only when a singlet exciton is formed. Hence, the theoretical maximum efficiency that is possible is only 25%.

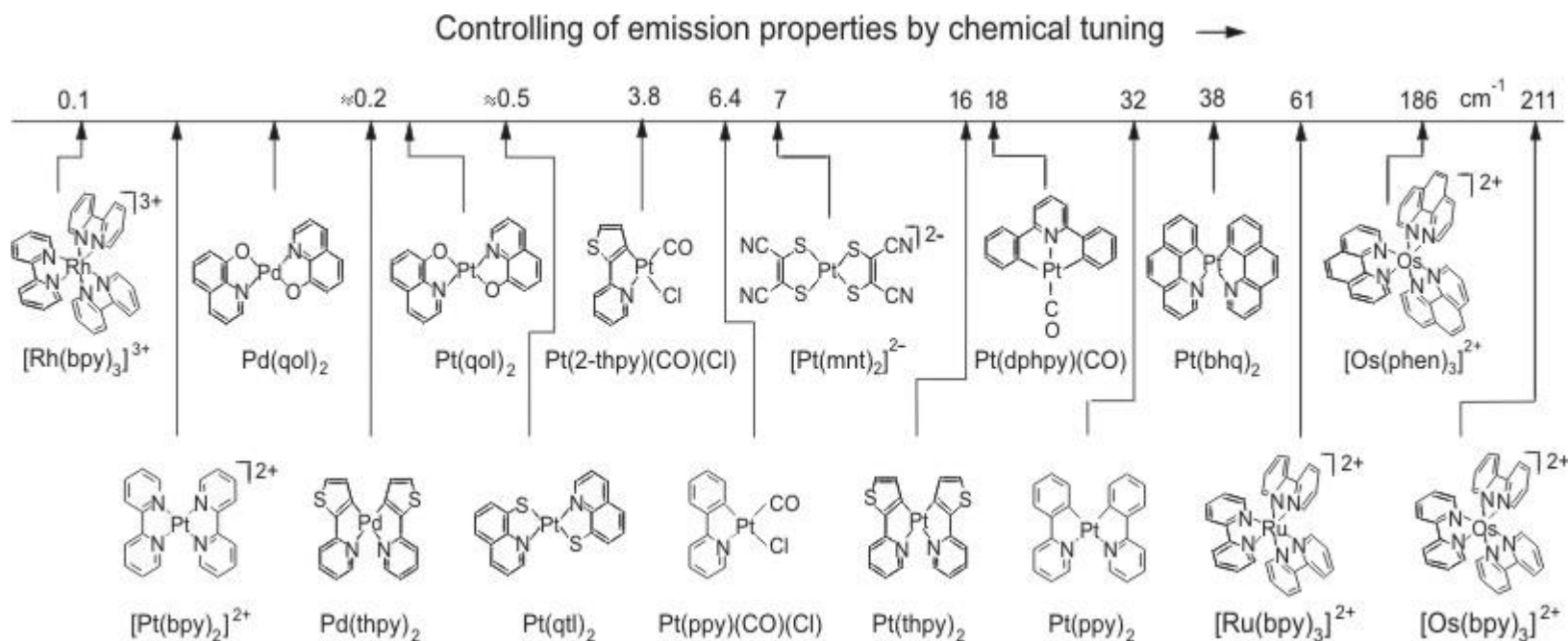


- In an effort to improve the efficiency, a number of recent studies have been devoted to using organometallic compounds (transition metals with organic ligands), which allow for rapid intersystem crossing (ISC) from the excited singlet to light-emitting triplet states.
- The process of collecting the excitons in the lowest excited triplet state is referred to as triplet harvesting, and results in a 100% quantum efficiency of the OLED.



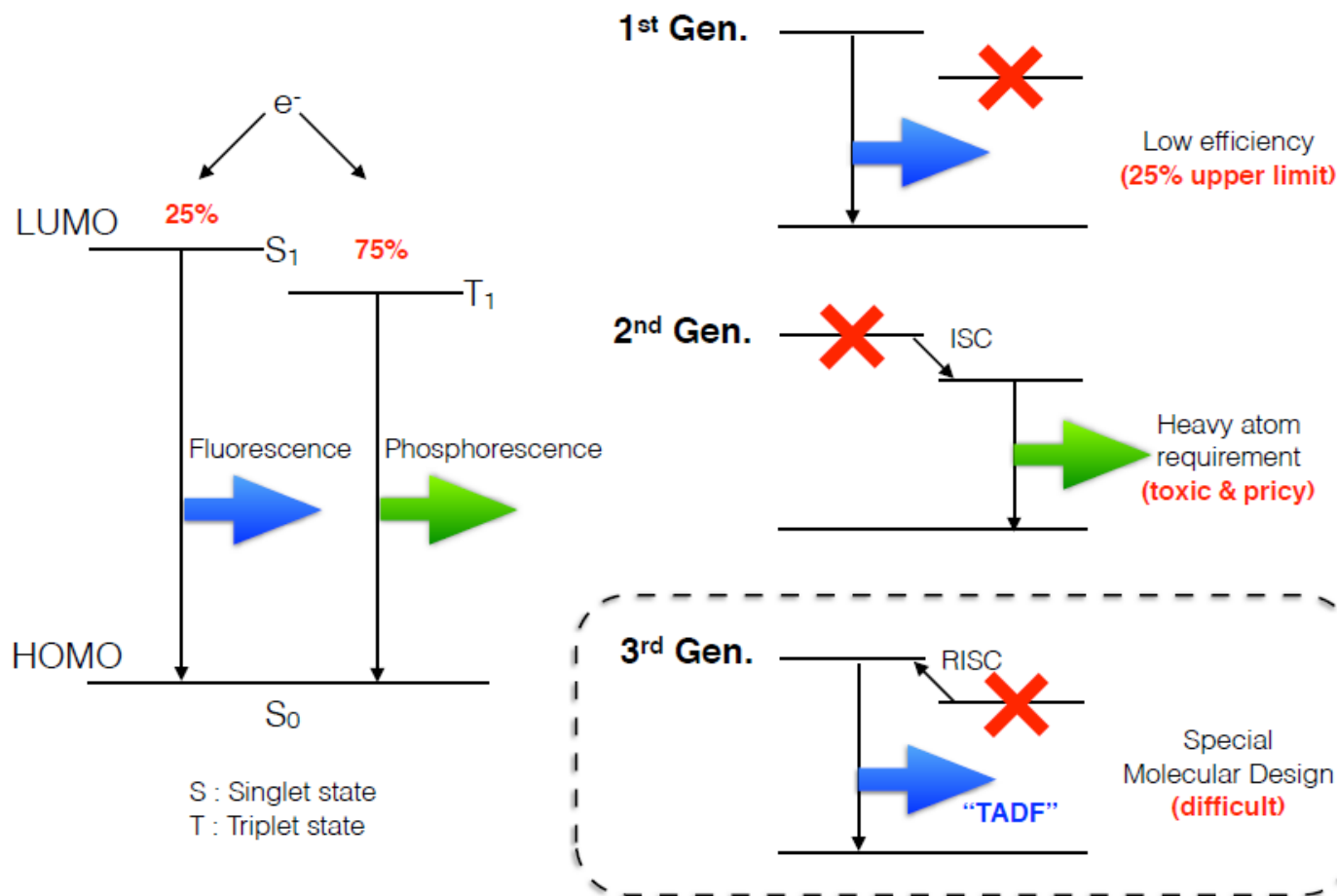


This effect is most pronounced for complexes of 4d and 5d transition metals with well-shielded valence electrons, which exhibit large spin–orbit coupling and concomitant mixing of singlet and triplet states.



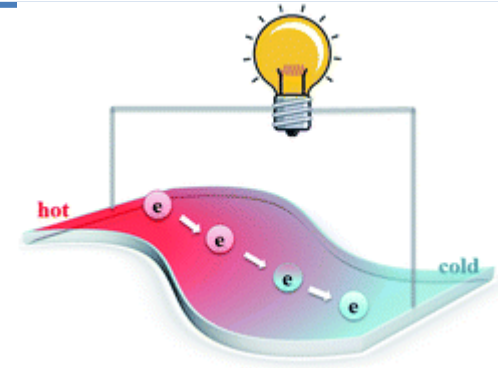


Three Generations of OLED



Thermally Activated Delayed Fluorescence

4.4 Thermoelectric materials



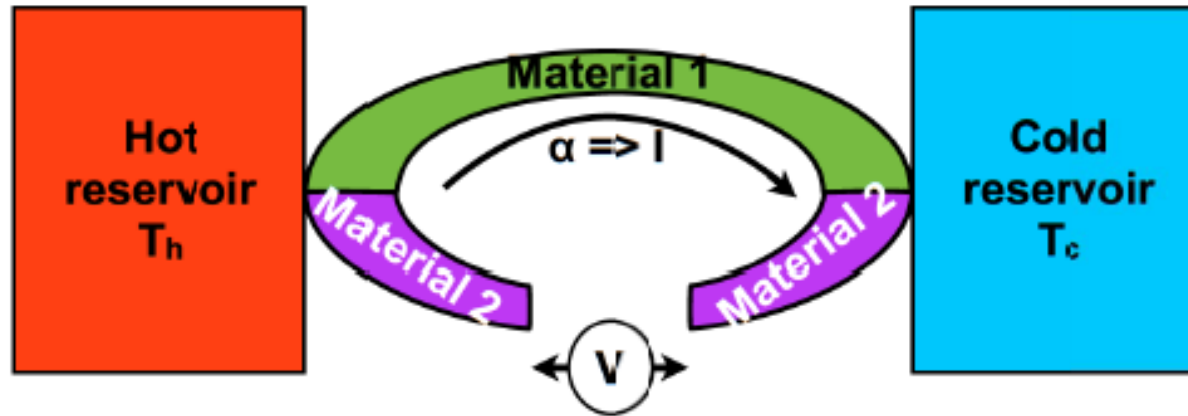
- The thermoelectric effect refers to phenomena by which either a temperature difference creates an electric potential or an electric potential creates a temperature difference.
- These phenomena are known more specifically as:
 - The Seebeck effect (converting temperature to current);
 - The Peltier effect (converting current to temperature);



Applications

- Air conditioning (distributed, environmentally friendly)
- Spot cooling of electronic chips, superconductors etc.
- Thermal suits for fire-fighting, soldier etc
- Waste heat recovery (automobiles, utilities etc)
- Geothermal power generation

Seebeck Effect



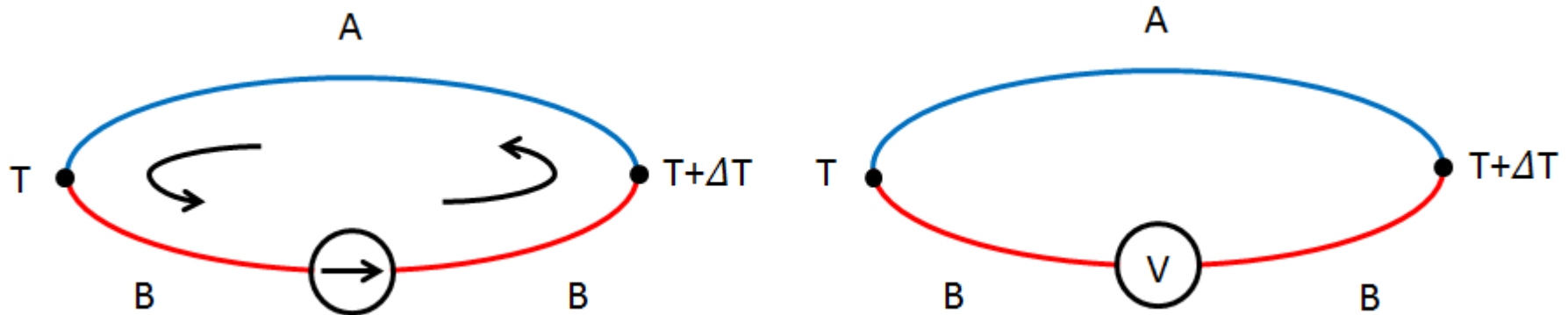
Open circuit voltage: $V = \alpha(T_h - T_c)$

$$\alpha = \frac{\Delta V}{\Delta T}$$

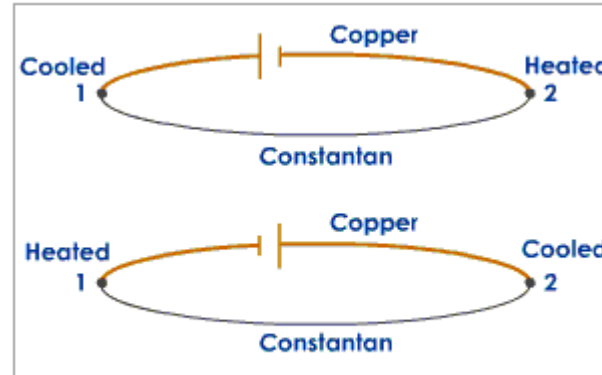
α : seebeck coefficient



- The electrons on a warmer end have a high average momentum as compared to the colder one. Therefore they will take energy with them (more in no.) as compared to the other one.
- This movement results in the more negative charge at colder part than warmer part, which Leads to the generation of electrical potential.



Peltier Effect

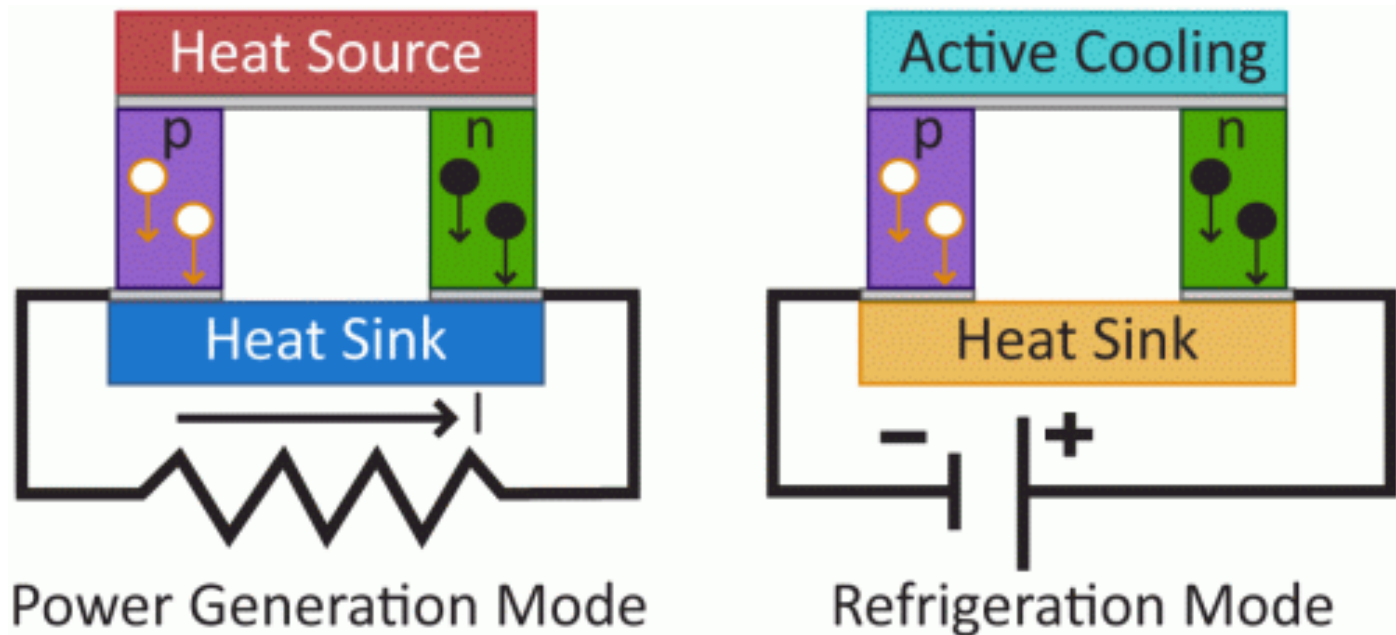


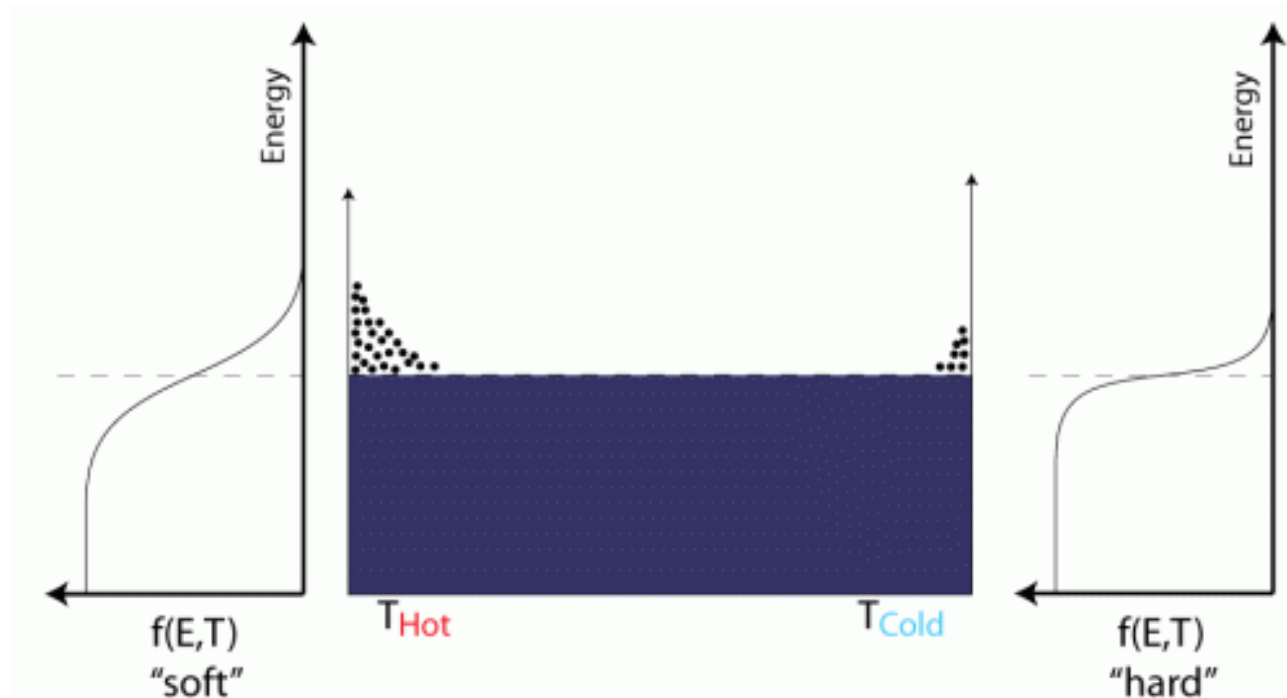
Peltier effect for copper-constantan thermocouple

- When a current is passed through a closed circuit of the two metals, heat is produced at one of the junctions and is absorbed at the other.
- Since the latter junction is cooled, this *Peltier effect* has been largely exploited for thermoelectric cooling applications (*e.g.*, auto seat coolers, computer component cooling).

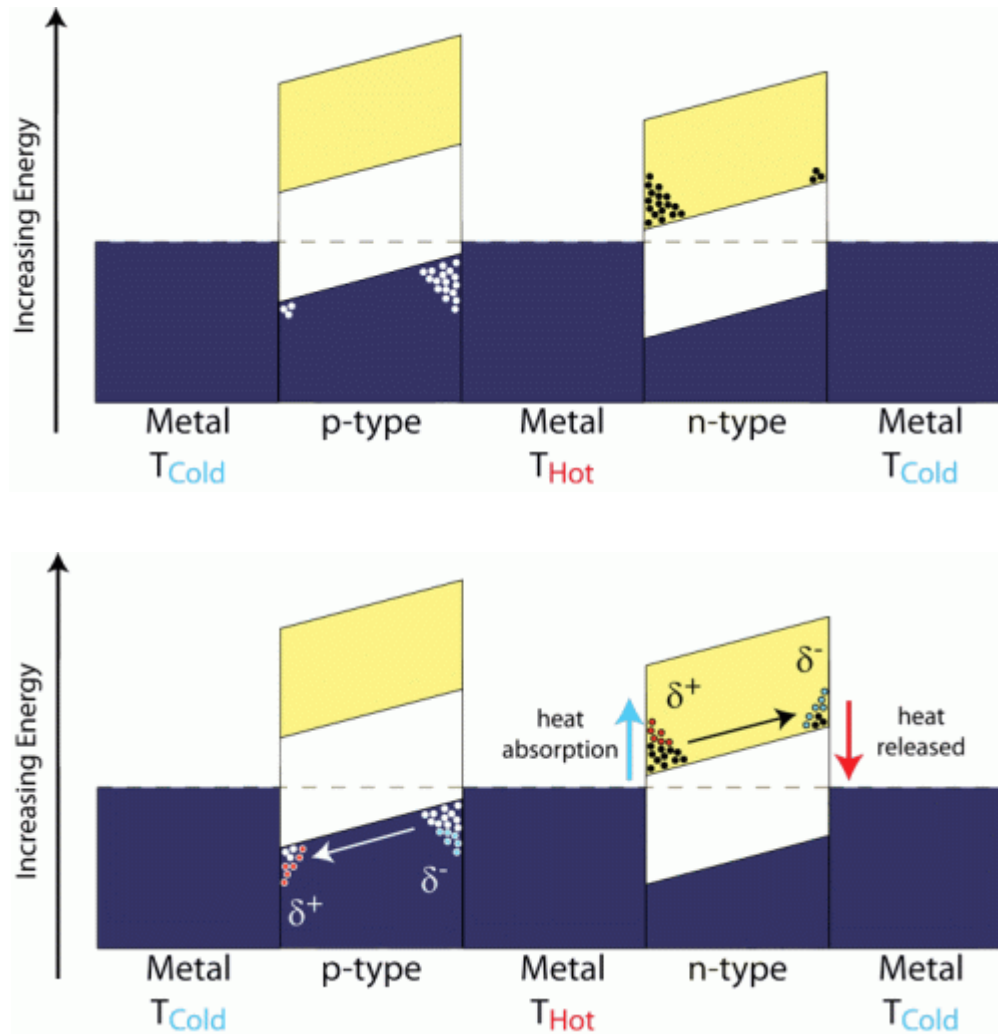


A TE device consists of a heat source and sink, joined together *via* n-type and p-type semiconductor materials

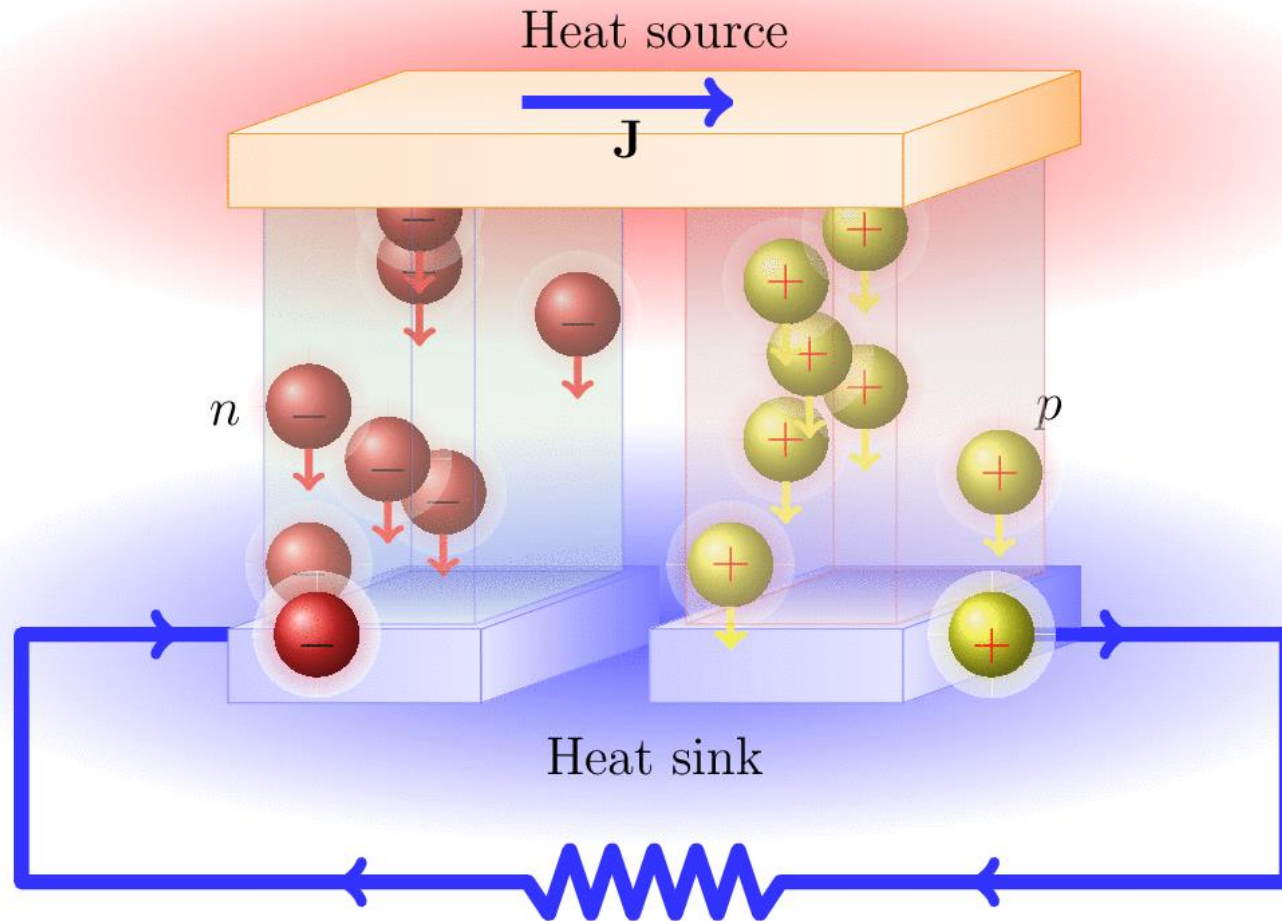


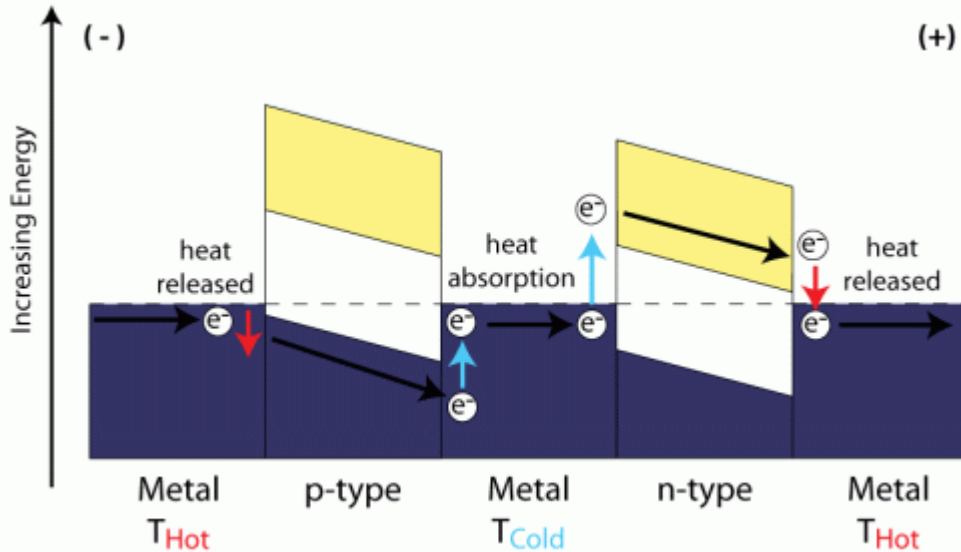


- At hot side Fermi distribution is soft i.e. the higher concentration of electrons above the Fermi energy but on cold side the Fermi distribution is sharp i.e. we have fewer electrons above Fermi energy.



The gradient in concentration drives diffusion of electrons and holes from hot to cold. Charges build up when electrons and holes migrate towards the cold side leaving behind charged donors/acceptors. This charge build up leads to an electric field causing backflow of current that will eventually cause the system to reach steady-state equilibrium.





In the case of thermoelectric cooling devices (Peltier coolers) a potential is applied across the device directing a current through the materials. As an electron moves from the metal to the p-type semiconductor it must release energy in the form of heat to enter the valence band (technically electrons do not transport through the p-type material- only holes, so a temporary electron-hole pair is assumed).

This released energy heats the metal. Conversely the electron must absorb energy as it passes back to the central metal region and again as it is promoted into the conduction band of the n-type semiconductor. The heat absorption results in active cooling in this metal region. Finally, the electron leaves the conduction band of the n-type material releasing heat into the last metal region.

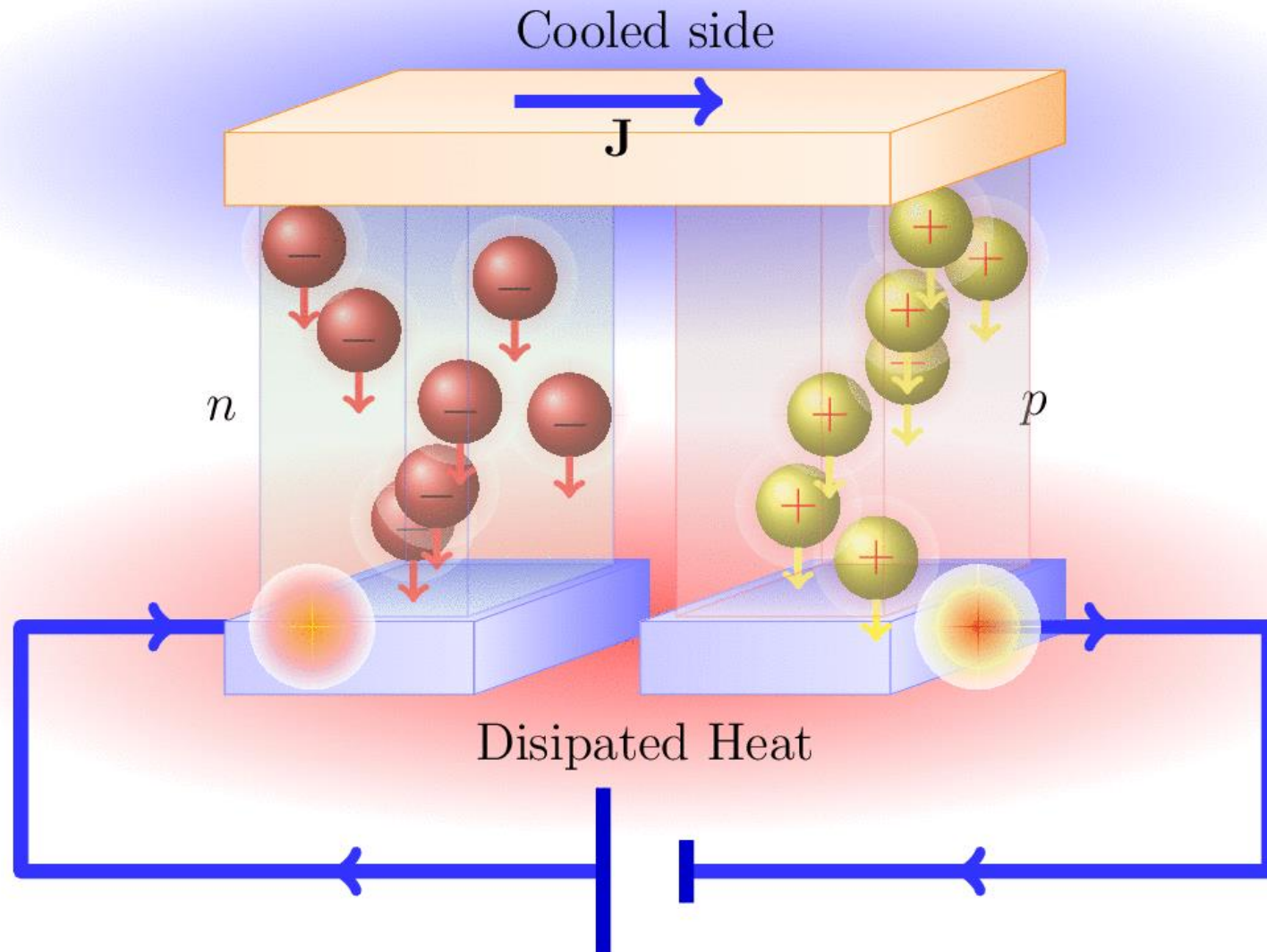




Figure of merit:

$$ZT = \frac{\alpha^2 \sigma}{\lambda}$$

ZT: the value to determine its suitability for power generation or refrigeration; a means to describe the transport properties of the TE material.

α : Seebeck coefficient ($\mu V/K$).

σ : Electrical conductivity.

λ : Thermal conductivity.



$$ZT = \frac{\alpha^2 \sigma}{\lambda}$$

A good TE material should have:

- A high Seebeck coefficient
- A high electrical conductivity
- A low thermal conductivity



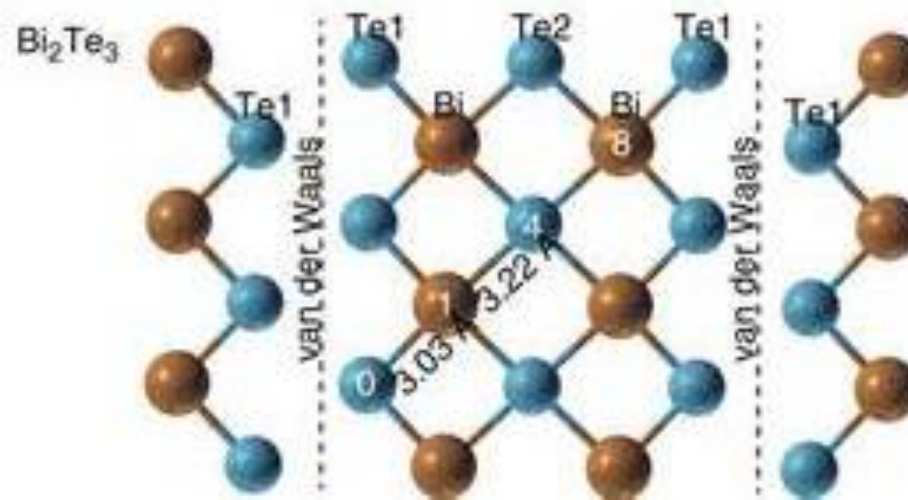
$\alpha = 1-10$ for metals

$\alpha = 150-250$ for semiconductors

- Semiconductors are much more effective TE materials (greater ZT values) than metals due to their significantly greater Seebeck coefficients and lower thermal conductivities.
- A barrier toward higher ZT values for semiconductors is their relatively small electrical conductivity, especially at low temperatures.
- In order to overcome this limitation, the chemical composition of the semiconductor may be fine-tuned to yield a small bandgap material, or one that is sufficiently doped with a low concentration (*ca.* $10^{18}-10^{19} \text{ cm}^{-3}$) of electronic/thermal carriers exhibiting high mobility through the lattice.

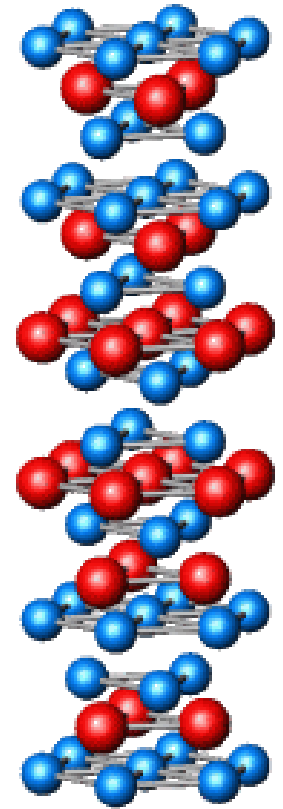
Bi_2Te_3 (Bismuth telluride)

- The most widely studied TE material is Bi_2Te_3 , consisting of a hexagonal unit cell with repeating. . . [Te–Bi–Te–Bi–Te. . . Te–Bi–Te–Bi–Te]. . . units.
- Whereas Te–Bi layers are bound by strong covalent interactions, the bonding between adjacent Te layers is through weak van der Waals interactions.





- This results in bulk anisotropic electrical and thermal conductivity, being most pronounced along planes that are perpendicular to the c -axis of the unit cell.
- To further improve ZT values, Bi_2Te_3 crystals may be doped with n- or p-type dopants, with most desired compositions of $\text{Bi}_2\text{Te}_{2.7}\text{Se}_{0.3}$ (n-type) and $\text{Bi}_{0.5}\text{Sb}_{1.5}\text{Te}_3$ (p-type), yielding $\text{ZT}=1$ at room temperature.
- This improvement results from a decrease in thermal conductivity of the lattice, brought about by phonon scattering by the dopant atoms (*i.e.*, perturbing the symmetry of the lattice, affecting the organized lattice vibration modes).





The following list represents the primary materials classes that have been designed in an effort to optimize ZT for thermoelectric applications. The general strategy is to dope the lattice with sufficient carriers (n- or p-dopants), while also interrupting the phonon transport through the solid through the introduction of large interstitial atoms.

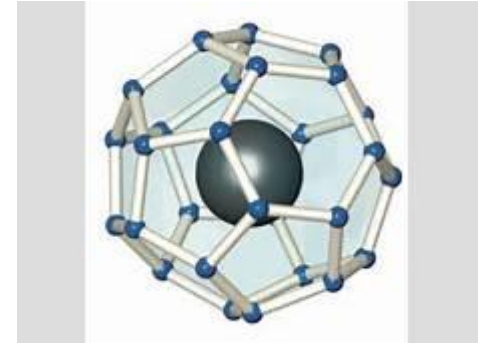


Complex Solid-State Inorganic Lattices

(*e.g.*, CsBi_4Te_6 , $\text{Bi}_{2-x}\text{Sb}_x\text{Te}_{3-y}\text{Se}_y$, ZrTe_5 , $\text{Ag}_n\text{Pb}_m\text{MnTe}_{m+2n}$ ($M = \text{Sb}, \text{Bi}$), $\text{AgPb}_{10}\text{SbTe}_{12}$, “Half-Heusler alloys” MNiSn ($M = \text{Zr}, \text{Hf}, \text{Ti}$), $\text{Zr}_{0.5}\text{Hf}_{0.5}\text{Ni}_{0.5}\text{Pd}_{0.5}\text{Sn}_{0.99}\text{Sb}_{0.01}$).

- For half-Heusler alloys, the unit cell is a combination of a NaCl lattice of two metals, with the third metal occupying tetrahedral interstitial sites.
- For instance, for TiNiSn , the Ti and Sn form the NaCl lattice, with Ni occupying 1/2 of the available tetrahedral interstitial sites (i.e., $\text{Ti:Ni:Sn} = 4:4:4$ atoms per unit cell).
- This combination offers a great deal of control over electronic/thermal conductivity of the solid.
- While the introduction of Sn (p-doping) increases electrical conductivity, heavy metal atoms such as Ti and Ni cause a decrease in thermal conductivity through phonon scattering.

Crystal Structures with “Rattlers”



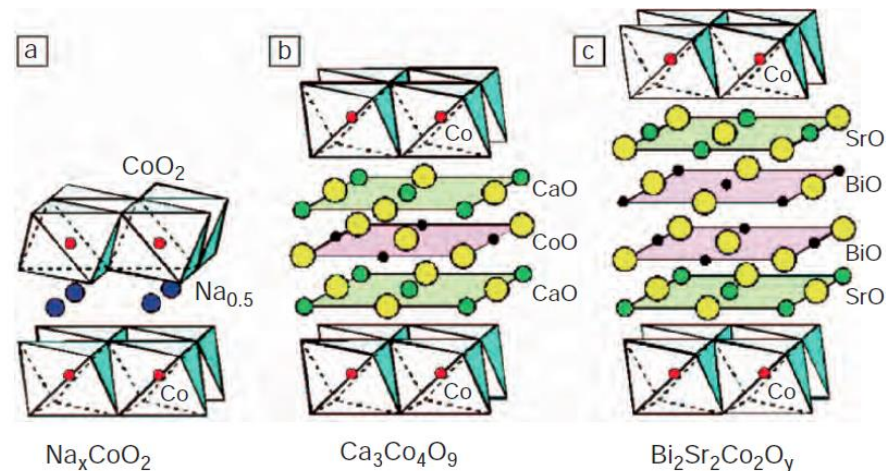
e.g., MSb₃ -based skutterudites, clathrate architectures.

- In these structures, the dopant atoms are weakly bound to the cage, and “rattle” in response to increasing temperature.
- As the atom within the cages becomes smaller/heavier, the amount of structural disorder will increase causing a larger decrease in lattice thermal conductivity.

Oxides

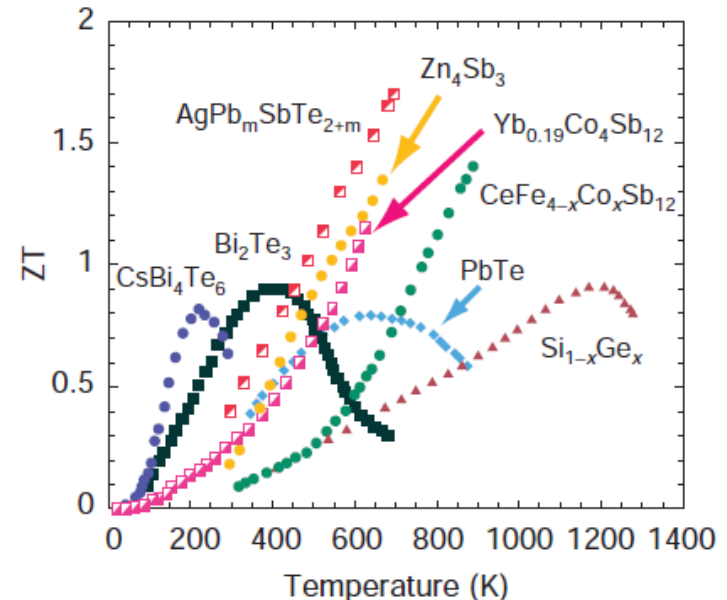
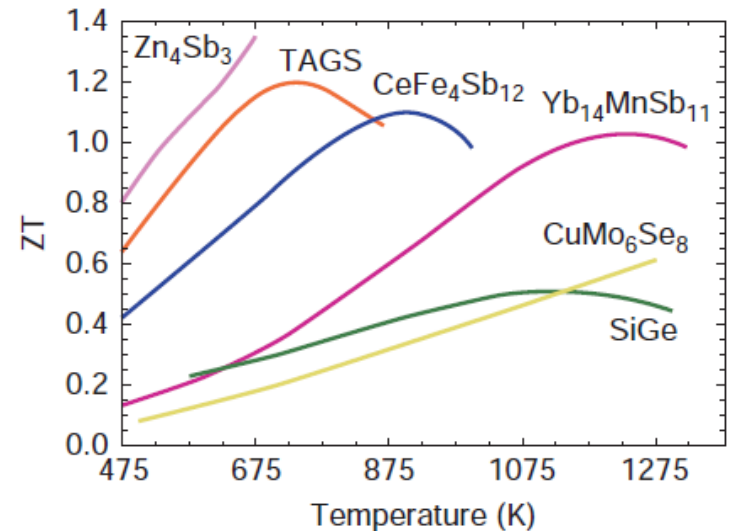
e.g., NaCo_2O_4 , $\text{Ca}_3\text{Co}_4\text{O}_9$, $\text{Al}_{0.02}\text{Zn}_{0.98}\text{O}$.

- These structures consist of CoO_2 layers, which serves as an effective electronic transport medium within the lattice.
- A variety of oxide layers (MO , $\text{M} = \text{Na}, \text{Ca}, \text{Sr}, \text{Bi}$) are placed in between the Co oxide layers to serve as phonon scatterers.





- The search for new TE materials continues to be an extremely active area of research;
- recent work has resulted in suitable TE components at virtually all temperature regimes.
- However, the search continues for materials with ZT values in excess of 2.0; most materials exhibit figure-of-merits ≤ 1.0 .





Summary

1. Semiconductors
 - a. definition
 - b. types (intrinsic and extrinsic)
 - c. direct and indirect bandgap
2. Silicon wafer production
 - a. MG-Si and EG-Si
 - b. CZ and FZ method
3. Integrated circuits
 - a. p-n junction and depletion region
 - b. diode
 - c. MOSFET
 - d. CMOS
 - e. ICs fabrication
(RCA clean; film fabrication; photolithography and ion implantation)



4. Thin-film deposition methodologies

PVD

CVD

5. LED

6. OLED

7. TE materials



For memory chips and removable flash drives, a gate stack structure known as SONOS (polysilicon/SiO₂/Si₃N₄/SiO₂/Si substrate) are typically used.

- Remembering: The device is programmed by applying a voltage to the gate to inject charge into the conduction band of the Si₃N₄ layer, resulting in a change in the threshold voltage. When the voltage is removed, the charge remains trapped within the Si₃N₄ layer.
- Erasing: To erase the device, a reverse bias voltage is applied that removes the trapped charge from the Si₃N₄ region.

High-k metal oxides such as ZrO₂ and HfO₂ are also capable of trapping injected electrons; if these replace Si₃N₄, the memory device is known as SOMOS (where M refers to metal oxide).



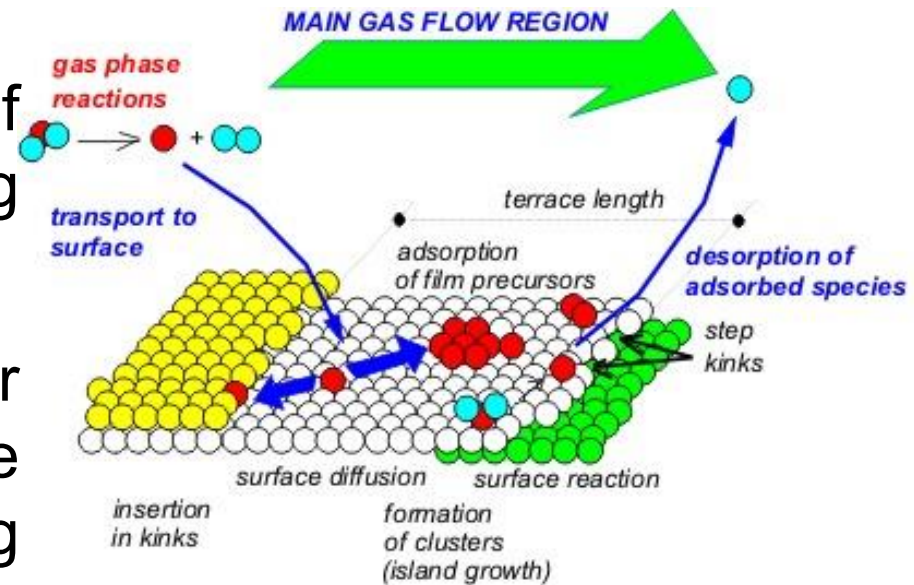
Step (d) : the deposition of a thin film (ca. 10–20 nm) of SiO_2 onto the trench sidewalls and bottom. This is used to establish an effective SiO_2/Si interface prior to the deposition of the thick (ca. 500 nm–1 μm) back-filled SiO_2 layer. That is, to ensure that the SiO_2 conforms to the entire channel surface, leaving no voidspaces that would decrease the effectiveness of the isolation trench.

Step (e): the thick oxide is deposited into the trench.

Step (f): chemical mechanical polishing (CMP) is used for planarization of the surface prior to subsequent layering steps. The silicon nitride layer acts a polishing stop, and is subsequently removed through plasma etching.

Thin-film growth

- the first stage is formation of subcritical embryos of varying sizes.
- These particles will further nucleate while taking on more precursor adsorption, forming a supercritical sized cluster.
- These clusters will coalesce into growth *islands*, which will expose new regions of the substrate that serve to adsorb additional precursor species.
- Isolated surface islands eventually grow together, leaving holes and channels that are filled by adsorbing precursor molecules to form a continuous thin film.





- The conductivity of ITO is due to both Sn dopants and O vacancies, as represented by the general formula $\text{In}_{2-x}\text{Sn}_x\text{O}_{3-2x}$.
- Since the organic hole-transport polymer is in direct contact with ITO in an OLED, the surface properties of ITO is important for effective hole injection. It has been shown that surface treatments such as UV ozone cleaning or Ar/O₂ plasma treatments, result in an increase in the work function of the ITO surface as the Sn:In ratio is decreased (and O concentration is increased).



Epitaxial Growth

Epitaxy means the growth of a single crystal film on top of a crystalline substrate.

Homoepitaxy

- The film and the substrate are the same material.
- Often used in Si on Si growth.
- Epitaxially grown layers are purer than the substrate and can be doped independently of it.

Heteroepitaxy

- Film and substrate are different materials.
- Such as Si on sapphire (α – *alumina*)
- Allows for optoelectronic structures and band gap engineered devices.